BK7259 Datasheet

DS-BK7259-E04 V1.3

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1 Features

Wi-Fi® 6 Single-band Connectivity

Low power design

- IEEE power save mode: 70 μA @ DTIM 10
- Active RX: 17.5 mA
- Deep sleep0: 2.5 μA

IEEE 802.11ax (Wi Fi 6)

- 1T1R operation in 2.4 GHz band
- 20 MHz and 40 MHz channel bandwidths
- 20 MHz and 40 MHz channels up to MCS9
- Uplink and downlink Orthogonal Frequency Division Multiple Access (OFDMA) for improved efficiency in dense IoT environments
- Uplink Multi-User Multiple-Input Multiple-Output (UL MU-MIMO) for increased total uplink throughput and reduced queuing latency in high-density environments
- Downlink Multi-User Multiple-Input Multiple-Output (DL MU-MIMO) with Beamformee support for higher capacity and better signal quality
- Target Wake Time (iTWT/bTWT) for optimized power saving and reduced collisions
- TX and RX Low-Density Parity Check (LDPC) support for extended range
- Spatial reuse and BSS coloring for higher throughput in crowded deployments
- Dual Carrier Modulation (DCM) and SU Extended Range (ER) for robust long range links
- Supports antenna diversity to combat multipath fading and improve reception

Backward compatibility with IEEE 802.11b/g/n

- Fully interoperable with legacy Wi Fi standards in 2.4 GHz band
- 20 MHz and 40 MHz channel bandwidths (HT20/HT40)
- Supports MCS 32 HT duplicate mode
- Short Guard Interval (SGI) support for improved throughput
- Greenfield and Mixed Mode preambles for compatibility and efficiency
- STBC (Space-Time Block Coding) for improved reliability in multipath environments
- Wi Fi Multimedia (WMM) for traffic prioritization

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- Block ACK, A-MPDU, A-MSDU aggregation for higher efficiency
- Automatic beacon monitoring with hardware TSF (Timing Synchronization Function)
- Up to 6 virtual Wi-Fi interfaces
- Antenna diversity for better reception in challenging RF conditions
- 802.11mc Fine Timing Measurement (FTM) for precise indoor positioning

Advanced Wi Fi features

- Channel State Information (CSI) for Wi Fi sensing applications
- Supports simultaneous operation of STA, SoftAP, Wi-Fi Direct (P2P), STA + SoftAP, and promiscuous modes
- PTA coexistence with integrated Bluetooth/802.15.4/2.4 GHz proprietary

Bluetooth®

- Bluetooth Core 6.0 certified
- Support for dual-mode Bluetooth 5.2 features
- Supports Basic Rate (BR), Enhanced Data Rate (EDR) 2 Mbps and 3 Mbps, Low Energy (LE) 1 Mbps, 2 Mbps, and long range (125 kbps and 500 kbps)
- LE Audio with isochronous channels (CIS/BIS)
- Angle of Arrival (AoA) and Angle of Departure (AoD) direction finding, antenna array supporting up to 16 antennas
- LE power control
- LE privacy 1.2 with resolvable private addresses
- LE secure connections pairing with ECDH for stronger link security
- Advertising extensions with multiple advertising sets
- Concurrent roles: Central, Peripheral, Broadcaster, Observer
- Bluetooth Mesh 1.1 support for large-scale IoT networks
- TX power up to +25 dBm
- Adaptive Frequency Hopping (AFH) for coexistence with Wi-Fi/802.15.4/2.4 GHz proprietary

IEEE 802.15.4

- 2.4 GHz IEEE 802.15.4 compliant
- Operating frequency range: 2400–2483.5 MHz
- O-QPSK with 250 kbps data rate
- Thread V1.4.0 certified
- Zigbee® 3.0 support

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- Matter over Thread
- Low latency channel access with CSMA-CA and guaranteed time slots (GTS) for time critical applications
- AES-128 link layer security for encrypted communication

2.4 GHz Proprietary

- Worldwide 2.4 GHz ISM band operation
- 1 Mbps air data rate
- GFSK modulation
- Automatic packet handling
- Auto acknowledgement
- Auto retransmission

Core

Wireless processor

- Dual 32-bit RISC cores, one up to 480 MHz and one up to 240 MHz
- Level 1 (L1) cache: 8 KB I-cache and 8 KB D-cache for each core
- 64 KB level 2 (L2) cache, frequency up to 240 MHz

Application processor

- Arm®v8.1-M Cortex®-M52 core at up to 480 MHz
  - Supports Arm TrustZone® technology
  - Hardware IEEE 754 compliant Floating-point Unit (FPU)
  - DSP extension
  - Built-in Memory Protection Unit (MPU)
  - Built-in Nested Vectored Interrupt Controller (NVIC)
  - Built-in Embedded Trace Macrocell (ETM)
  - L1 cache: 16 KB I-cache and 16 KB D-cache
- Armv8.1-M Cortex-M55 core at up to 480 MHz
  - Supports Arm TrustZone technology
  - Hardware IEEE 754 compliant Floating-point Unit (FPU)
  - DSP extension
  - MVE (Arm Helium™ technology), supporting Single Instruction Multiple Data (SIMD) 128-bit vector operations

  - Built-in Memory Protection Unit (MPU)
  - Built-in Nested Vectored Interrupt Controller (NVIC)
  - Built-in Embedded Trace Macrocell (ETM)
  - L1 cache: 16 KB I-cache and 16 KB D-cache
- Arm Ethos™-U65 NPU, frequency up to 480 MHz, 0.25 TOPS, 256 MACs/cycle
  - Works alongside Cortex-M55 core
  - Data types: Int-8 and Int-16
  - Heavy compute operators for CNN and RNN accelerated in hardware
  - Supports 8-bit weights
  - 55 KB internal SRAM
- 128 KB level 2 (L2) cache
- 64 KB TCM

Memories

- Flash
  - SiP flash up to 16 MB with XIP support
  - External flash expansion: up to 64 MB x 2 via QSPI with XIP support
- PSRAM
  - SiP PSRAM up to 64 MB (2 x 32 MB) with XIP support
  - External PSRAM expansion: up to 64 MB x 2 via QSPI with XIP support
- 128 KB ROM
- 1280 KB share SRAM
- 64 KB TCM
- 64 KB USB RAM dedicated to high-speed USB
- 4 KB OTP, with 3 KB available for customer applications
- 32-bit eFuse

Security

- Secure boot
- Secure debug
- Secure connection
- Firmware Over-The-Air (FOTA)
- Provisioning

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- TEE_M
- TrustEngine, which includes the following features:
  - Symmetric schemes, AES-ECB/CBC/CTR/CBC-MAC/CMAC/CCM/GCM (key size 128-bit, 192-bit and 256-bit)
  - Symmetric schemes, SM4-ECB/CBC/CTR/CBC-MAC/CMAC/CCM/GCM
  - Digest schemes, SHA1/224/256
  - Digest scheme, SM3
  - Asymmetric schemes, RSA 1024/2048/3072/4096 and ECCP 192/224/256/384/512/521
  - Asymmetric scheme, SM2
  - Key ladder for key management
  - Lifecycle management
  - True random number generator

Clock Management

- External oscillators: 26 MHz crystal oscillator (XTALH), 32.768 kHz crystal oscillator (X32K)
- Internal oscillators: 160–640 MHz digitally controlled oscillator (DCO), 32 kHz ring oscillator (ROSC)
- 960 MHz PLL (DPLL)
- 90–99 MHz audio PLL (APLL)

Power Management

- 2.3 to 3.6 V VBAT supply
- Embedded buck (DC-DC) converters and LDO regulators
- On-chip Power-On Reset (POR) and Brown-Out Detector (BOD)
- Low power consumption:
  - Active mode RX: 17.5 mA
  - Standby associated, DTIM = 10 without TCP keep-alive: 70 μA
  - Sleep mode: 70 μA
  - Deep sleep0 mode: 2.5 μA
  - Deep sleep1 mode: 25 μA
  - Shutdown mode: 1 μA

Peripherals

- GPIOs (up to 72 available): 62 in BK7259QN128A832 QFN128, 58 in BK7259QN128B832 QFN128
- 4x SPI interface (SPI)

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- 2x QSPI interface (QSPI)
- 6x UART interface (UART), 1 supports hardware flow control and flash download support
- 1x Smart Card controller (SC)
- 2x SD/SDIO and eMMC host controller (SDMMC)
- 2x I2C interface (I2C)
- 1x I3C interface (I3C)
- 1x full-speed USB 2.0 (USB_FS) (BK7259QN128A832 only)
- 1x high-speed USB 2.0 (USB_HS)
- 2x CAN controller with CAN FD (CAN)
- 1x LIN controller (LIN)
- 1x general-purpose DMA controller (GDMA) with 8 channels
- 1x high-performance DMA controller (HPDMA) with 4 channels
- 1x Ethernet MAC interface (ENET)
- 12x 32-bit PWM channel
- Video
  - 1x 24-bit display controller (DISPLAY)
  - 1x MIPI® DSI, 4 data lanes, up to 1.5 Gbps each
  - 1x 10-bit DVP interface
  - 1x MIPI® CSI, 2 data lanes, up to 1.5 Gbps each
  - 1x image signal processor (ISP)
  - 1x video processing unit (VPU)
  - 1x graphics processing unit (GPU)
  - 1x display processor unit (DPU)
- Audio
  - 1x 20-band digital hardware equalizer (EQ)
  - Up to 3x audio ADC
  - Up to 2x audio DAC
  - 2x DMIC interface (DMIC)
  - 1x voice activity detector (VAD)
  - 2x SPDIF interface (SPDIF)
  - 1x HDMI CEC controller (HDMI CEC)
  - 5x I2S interface (I2S)

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- 14-bit AUX ADC, up to 10 channels
- 2x 12-bit AUX DAC
- 2x analog comparator (ACOMP)
- 18x 32-bit general-purpose timer
- Watchdog timers
  - 1x AON watchdog timer (AON WDT)
  - 4x CPU window watchdog timer (CPU WWDT)
  - 1x AHB window watchdog timer (AHB WWDT)
- Real-time counters (RTC)
  - 1x 36-bit high-voltage RTC (HV RTC)
  - 1x 64-bit AON RTC
- 4x IrDA Interface (IrDA)
- 1x temperature sensor (TEMP)
- 1x touch sensor (TOUCH)
- Serial-wire debug port interface (SW-DP)

Packaging

- BK7259QN128A832/BK7259QN128B832: QFN128 package, 12.3 x 12.3 x 0.85 mm
- Operating temperature range: -40 to +85 °C

Applications

The BK7259 is designed for the next generation of smart, connected, and battery-efficient devices. With Wi-Fi 6, dual-mode Bluetooth 6.0, IEEE 802.15.4 (Thread/Zigbee/Matter), and 2.4 GHz proprietary all integrated into a single chip, it enables seamless interoperability across ecosystems while delivering ultra-low power consumption for long battery life.

- Smart home and appliances: door locks, sensors, lighting, and connected appliances that run reliably for years on batteries.
- Consumer electronics: wearables, audio devices with LE Audio, toys, and drones that need responsive, low latency wireless.
- Industrial IoT: factory sensors, smart meters, and automation systems that demand secure, always-on connectivity with minimal energy use.
- Healthcare and medical: portable monitors and medical devices that combine secure data transfer with extended operating time.

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- Edge devices: Matter-enabled hubs, controllers, and gateways that unify Wi-Fi, Bluetooth, Thread and 2.4 GHz proprietary in one secure platform.

With its deep sleep0 current as low as 2.5 μA, the BK7259 makes it possible to design smaller, longer lasting, and more secure IoT products across consumer, industrial, and healthcare markets.

2 Overview

The BK7259 is a highly integrated, low-power multi-protocol wireless MCU that combines an advanced MCU subsystem with Wi-Fi 6 (802.11b/g/n/ax), dual-mode Bluetooth 6.0, IEEE 802.15.4, and 2.4 GHz proprietary connectivity. It is designed for a wide range of connected applications, including smart home and appliances, industrial automation, smart energy, healthcare, and secure payment systems.

The MCU subsystem is built around a wireless processor and an application processor, delivering high performance for protocol stacks, edge computing, and real-time control. The device integrates 1280 KB SRAM, 128 KB ROM, and flexible memory options including up to 64 MB SiP PSRAM and up to 16 MB SiP flash, enabling efficient execution of complex applications. The Quad SPI interface supports high-bandwidth external memory access, while secure on-the-fly decryption ensures protection of code and data stored in off-chip XIP flash or PSRAM.

The wireless subsystem includes a 1×1 single-band Wi-Fi 6 (802.11b/g/n/ax) radio supporting 20 MHz/40 MHz channels in 2.4 GHz band. The Bluetooth LE 6.0 radio supports 1 Mbps, 2 Mbps, and long-range PHYs, as well as advanced features such as LE Audio and AoA/AoD for high-precision positioning. The integrated 802.15.4 radio enables support for Zigbee, Thread, and Matter, making the BK7259 a versatile platform for multi-protocol IoT devices.

The BK7259 is well-suited for Matter applications across Wi-Fi and Thread, and can operate as a Matter end device within major ecosystems.

A comprehensive Secure Element (SE) is embedded, providing secure boot, secure debug, secure connection, FOTA, provisioning, TEE_M, and TrustEngine. The integrated TrustEngine accelerates cryptographic operations with support for AES, SM4, SHA, SM3, RSA, ECC, and SM2, while a true random number generator and key ladder ensure robust key management. Together, these features establish a strong hardware root of trust and lifecycle security management.

The BK7259 is optimized for low-power operation, with active receive current as low as 17.5 mA and deep sleep0 current down to 2.5 μA, enabling long battery life in portable and energy-constrained devices. Power management is simplified with three buck converters and multiple LDO regulators, supporting a wide supply range of 2.3 V to 3.6 V.

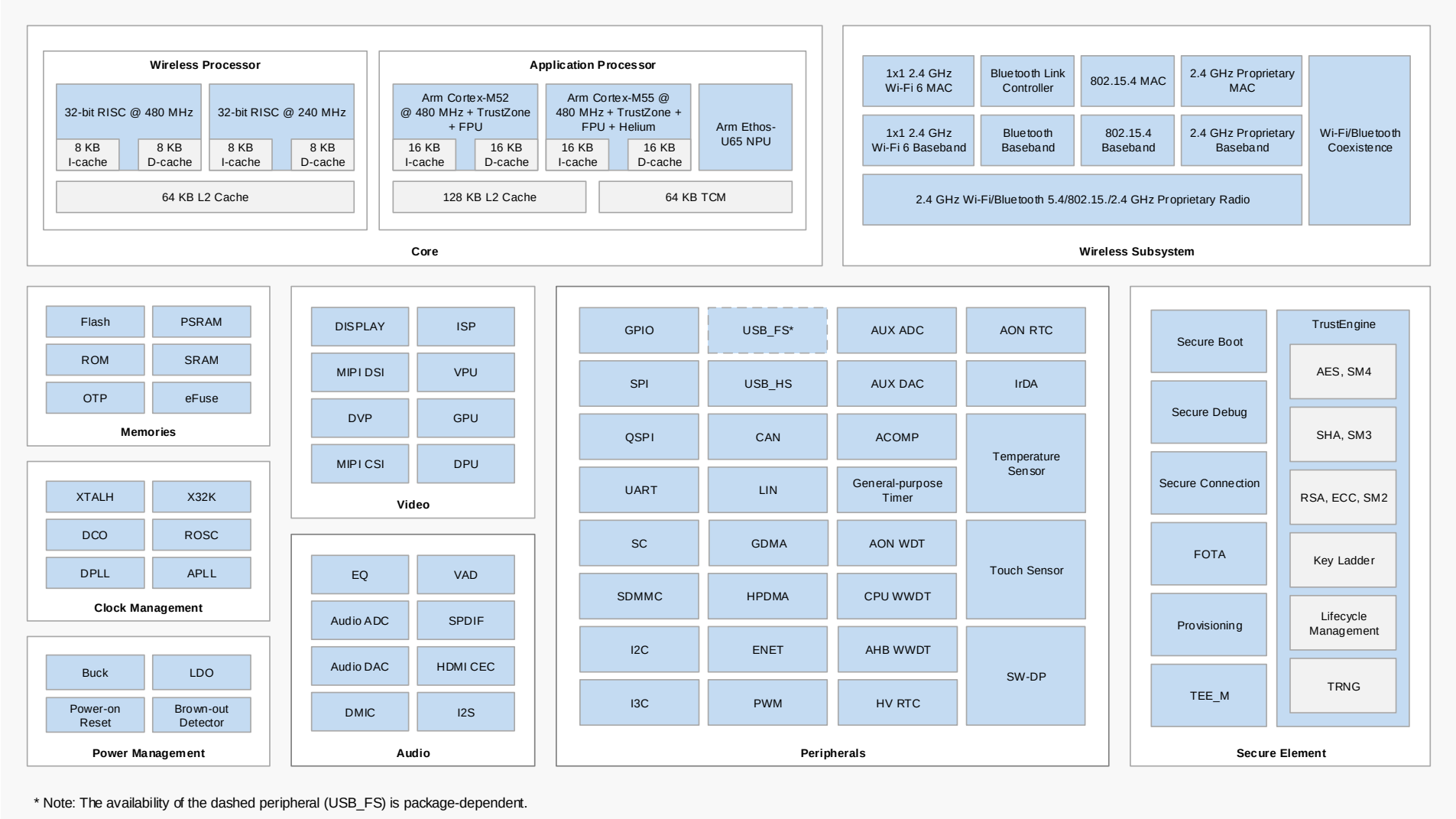
Available in two QFN128 (12.3 × 12.3 mm) package options, the BK7259 supports an extended operating temperature range of -40 °C to +85 °C, making it suitable for both consumer and industrial environments.

The advanced design of the BK7259 delivers tight integration, low power, and strong security in a cost- and space-efficient wireless MCU platform, enabling developers to accelerate the deployment of next-generation IoT products.

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Figure 2-1 shows the general block diagram of the BK7259.

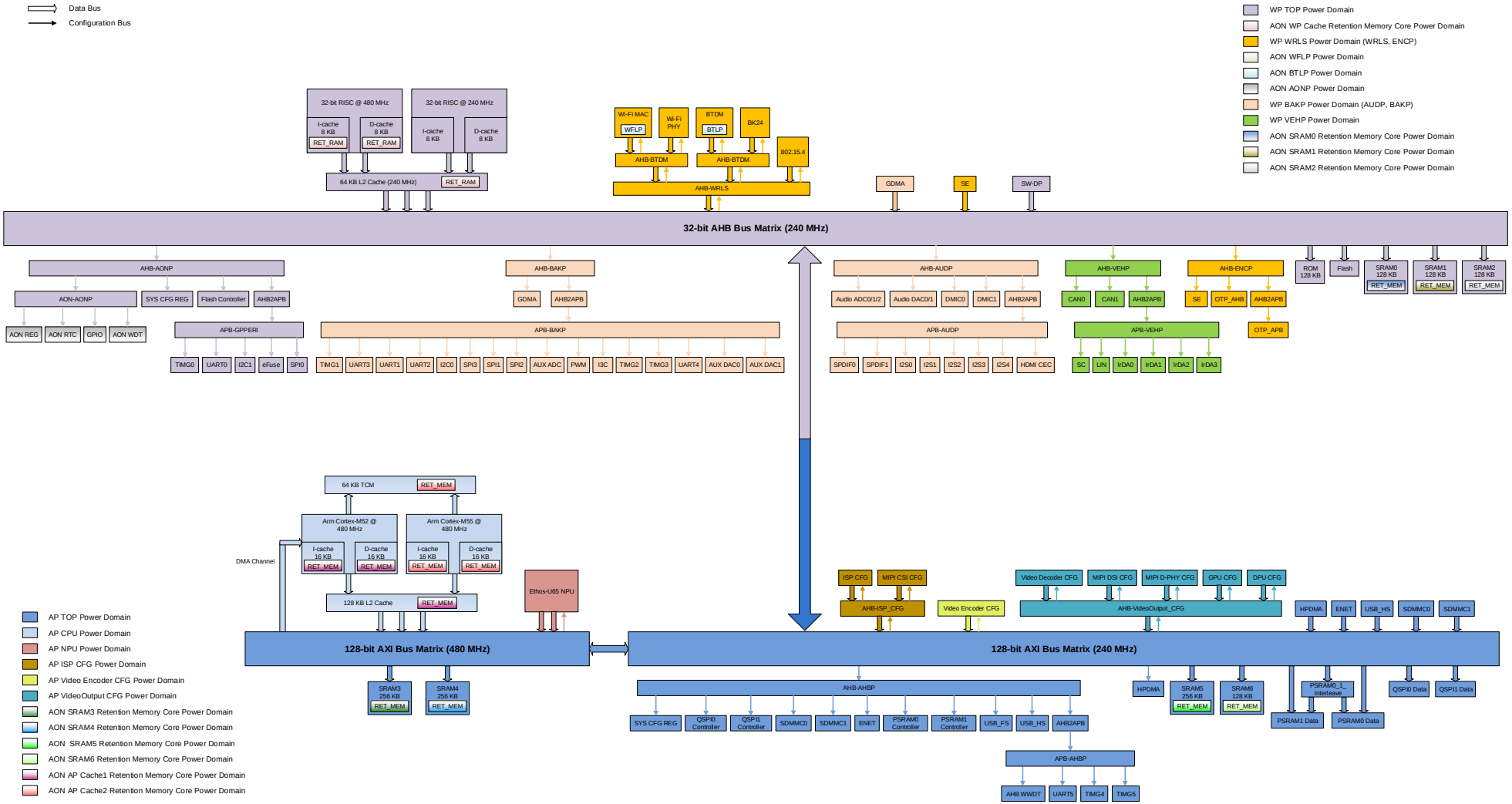
Figure 2-1 BK7259 Block Diagram



* Note: The availability of the dashed peripheral (USB_FS) is package-dependent.

Figure 2-2 shows the bus interconnect of the BK7259.

Figure 2-2 BK7259 Bus Interconnect



The digital domain of the BK7259 is divided into three primary power domains: the WP subsystem, the AP subsystem, and the AON (Always-On) domain. The WP and AP subsystems are further partitioned into multiple independently switchable sub-power domains.

- WP subsystem power domain: supplied by VDDDIG (adjustable LDO output voltage: 0.45–0.975 V)
- AP subsystem power domain: supplied by VDDCORE_HS (adjustable LDO output voltage: 0.6–0.975 V)
- AON power domain: supplied by VDDAON (adjustable LDO output voltage: 0.6–0.95 V)
  - The AON cache/SRAM retention memory core, WFLP, and BTLP power domains can be individually configured to remain powered during deep sleep1 mode.
  - The AON AONP power domain always remains powered during deep sleep1 mode.

The BK7259 devices are offered in two QFN128 packages. Table 2-1 provides a comparison of the features available for each device option.

Table 2-1 Device Options and Features

Feature		BK7259QN128A832 QFN128	BK7259QN128B832 QFN128
SiP flash		8 MB	8 MB
SiP PSRAM		32 MB (2 x 16 MB)	32 MB (2 x 16 MB)
GPIO		62	58
WKUP pin		31	30
SPI	Master/Slave	4	4
QSPI		2	2
UART		6	6
Smart Card controller		1	1
SDMMC		2	2
I2C	Master/Slave	2	2
I3C		1	1
Full-speed USB 2.0		1	-
High-speed USB 2.0		1	1
CAN		2	2
LIN		1	1
GDMA		1	1

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Feature		BK7259QN128A832 QFN128	BK7259QN128B832 QFN128
HPDMA		1	1
Ethernet MAC interface		1	1
PWM	Number of channels	12	12
Display controller		1	1
MIPI DSI		1	1
DVP interface		1	1
MIPI CSI		1	1
ISP		Yes	Yes
VPU		Yes	Yes
GPU		Yes	Yes
DPU		Yes	Yes
Digital equalizer		1	1
Audio ADC		1	3
Audio DAC		Mono	Stereo
DMIC		2	2
Voice activity detector		1	1
SPDIF		2	2
HDMI CEC		1	1
I2S	Master/Slave	5	5
AUX ADC	14 bits	1	1
	Number of channels	10	9
AUX DAC	12 bits	2	2
Analog comparator		2 (10 input channels per comparator)	2 (9 input channels per comparator)
General-purpose timer		18	18
AON WDT		1	1
CPU WWDT		4	4
AHB WWDT		1	1

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Feature	BK7259QN128A832 QFN128	BK7259QN128B832 QFN128
HV RTC	1	1
AON RTC	1	1
IrDA	4	4
Temperature sensor	1	1
Touch sensor	1	1
SW-DP	Yes	Yes
Package	12.3 x 12.3 x 0.85 mm QFN128	12.3 x 12.3 x 0.85 mm QFN128
Operating voltage	2.3 to 3.6 V	
Operating temperature	-40 to +85 °C	

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3 Pin Descriptions

The BK7259 provides Wi-Fi, dual-mode Bluetooth, IEEE 802.15.4, and 2.4 GHz proprietary functionality in two QFN128 package options with different pin assignments.

3.1 BK7259QN128A832 QFN128 Pin Descriptions

Figure 3-1 shows the pin assignments of the BK7259QN128A832’s 12.3 x 12.3 mm, 128-pin QFN package.

Figure 3-1 BK7259QN128A832 QFN128 Pin Assignments

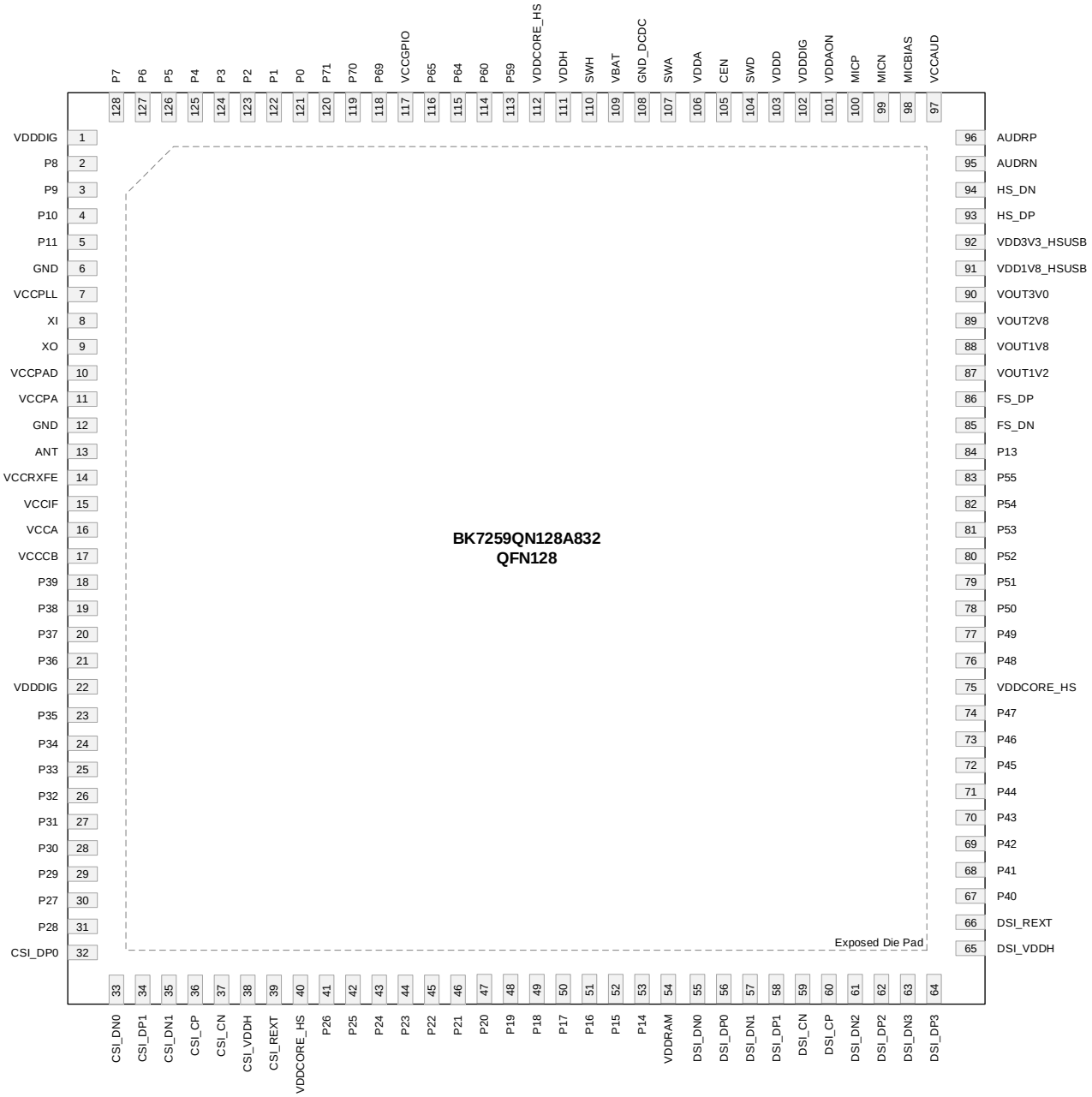


Table 3-1 shows the pin descriptions of the BK7259QN128A832 QFN128 package.

Note: Due to power-up glitches, the following I/O pins are not recommended for applications sensitive to power-up fluctuations, such as LEDs or motor control: GPIO10, GPIO20, GPIO21, and GPIO22.

Table 3-1 BK7259QN128A832 QFN128 Pin Descriptions

Pin #	Name	I/O	Type	Description
1	VDDDIG	-	Analog output	Digital core LDO output
2	P8	I/O	Digital/Analog	• GPIO8: general-purpose I/O • SD0_DATA6: data • 32K_XO: 32.768 kHz crystal output • ACOMP_CH8: non-inverting input channel • WKUP8: wake-up from deep sleep0 mode • ADC10: analog input channel
3	P9	I/O	Digital/Analog	• GPIO9: general-purpose I/O • SD0_DATA7: data • 32K_XI: 32.768 kHz crystal input • WKUP9: wake-up from deep sleep0 mode
4	P10	I/O	Digital/Analog	• GPIO10: general-purpose I/O • DL_UART_RX: UART flash download receive data input (UART0) • SD0_DATA2: data • CLK_AUXS: reference clock, derived from CLK_DCO/CLK_APLL/ WP_CLK_320M/WP_CLK_480M (divide by 1 to 64) • WKUP10: wake-up from deep sleep0 mode
5	P11	I/O	Digital/Analog	• GPIO11: general-purpose I/O • DL_UART_TX: UART flash download transmit data output (UART0) • SD0_DATA3: data • WKUP11: wake-up from deep sleep0 mode
6	GND	-	GND	Ground
7	VCCPLL	-	Analog input	RF PLL power supply
8	XI	-	Analog input	26 MHz crystal input
9	XO	-	Analog output	26 MHz crystal output

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Pin #	Name	I/O	Type	Description
10	VCCPAD	-	Analog input	RF PA driver power supply
11	VCCPA	-	Analog input	RF PA power supply
12	GND	-	GND	Ground
13	ANT	-	RF	2.4 GHz RF signal port
14	VCCRXFE	-	Analog input	RF receiver power supply
15	VCCIF	-	Analog input	IF power supply
16	VCCA	-	Analog input	Analog power supply
17	VCCCB	-	Analog input	Power supply for bandgap and analog peripherals
18	P39	I/O	Digital/Analog	• GPIO39: general-purpose I/O • DVP_PXD7: data • DAC0: output channel
19	P38	I/O	Digital/Analog	• GPIO38: general-purpose I/O • DVP_PXD6: data • DAC1: output channel
20	P37	I/O	Digital	• GPIO37: general-purpose I/O • DVP_PXD5: data
21	P36	I/O	Digital	• GPIO36: general-purpose I/O • DVP_PXD4: data
22	VDDDIG	-	Analog output	Digital core LDO output
23	P35	I/O	Digital	• GPIO35: general-purpose I/O • DVP_PXD3: data
24	P34	I/O	Digital	• GPIO34: general-purpose I/O • DVP_PXD2: data
25	P33	I/O	Digital	• GPIO33: general-purpose I/O • DVP_PXD1: data
26	P32	I/O	Digital	• GPIO32: general-purpose I/O • DVP_PXD0: data
27	P31	I/O	Digital/Analog	• GPIO31: general-purpose I/O • DVP_VSYNC: vertical synchronization

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Pin #	Name	I/O	Type	Description
				• WKUP31: wake-up from deep sleep0 mode
28	P30	I/O	Digital/Analog	• GPIO30: general-purpose I/O • DVP_HSYNC: horizontal synchronization • WKUP30: wake-up from deep sleep0 mode
29	P29	I/O	Digital/Analog	• GPIO29: general-purpose I/O • DVP_PCLK: pixel clock • WKUP29: wake-up from deep sleep0 mode
30	P27	I/O	Digital/Analog	• GPIO27: general-purpose I/O • FEM_LNA_EN: LNA bypass • CLK_AUXS_DVP: DVP master clock, derived from CLK_XTALH/ AP_CLK_240M/AP_CLK_320M/ AP_CLK_480M (divide by 1 to 16) • QSPI0_IO3: data • DVP_MCLK: DVP master clock, derived from CLK_XTALH/AP_CLK_240M/ AP_CLK_320M/AP_CLK_480M (divide by 1 to 16) • WKUP27: wake-up from deep sleep0 mode
31	P28	I/O	Digital/Analog	• GPIO28: general-purpose I/O • WIFI_RX_EN: receive enable • CLK_AUXS: reference clock, derived from CLK_DCO/CLK_APLL/ WP_CLK_320M/WP_CLK_480M (divide by 1 to 64) • ENET_TXEN: transmit data enable • ACOMP_CH2: non-inverting input channel • WKUP28: wake-up from deep sleep0 mode • ADC4: analog input channel
32	CSI_DP0	-	Analog input	MIPI CSI differential input lane 0 positive
33	CSI_DN0	-	Analog input	MIPI CSI differential input lane 0 negative

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Pin #	Name	I/O	Type	Description
34	CSI_DP1	-	Analog input	MIPI CSI differential input lane 1 positive
35	CSI_DN1	-	Analog input	MIPI CSI differential input lane 1 negative
36	CSI_CP	-	Analog input	MIPI CSI differential input clock positive
37	CSI_CN	-	Analog input	MIPI CSI differential input clock negative
38	CSI_VDDH	-	Analog output	Power supply for MIPI CSI
39	CSI_REXT	-	Analog output	MIPI CSI reference resistor. This analog signal connects to the ground through an external resistor.
40	VDDCORE_HS	-	Analog output	High-power digital core LDO output
41	P26	I/O	Digital/Analog	• GPIO26: general-purpose I/O • WIFI_TX_EN: transmit enable • QSPI0_IO2: data • RGB_G5: green data • ENET_RXD3: receive data • WKUP26: wake-up from deep sleep0 mode
42	P25	I/O	Digital/Analog	• GPIO25: general-purpose I/O • QSPI0_IO1: data • RGB_G6: green data • ENET_RXD2: receive data • ACOMP_CH3: non-inverting input channel • WKUP25: wake-up from deep sleep0 mode • ADC1: analog input channel
43	P24	I/O	Digital/Analog	• GPIO24: general-purpose I/O • LPO_CLK: clock derived from 32 kHz CLK_LPO1 • SD1_STB: data strobe • QSPI0_IO0: data • RGB_G7: green data • ENET_RXD1: receive data • ACOMP_CH4: non-inverting input channel • WKUP24: wake-up from deep sleep0 mode

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Pin #	Name	I/O	Type	Description
				• ADC2: analog input channel
44	P23	I/O	Digital/Analog	• GPIO23: general-purpose I/O • SD1_DATA7: data • QSPI0_CS: chip select • RGB_B3: blue data • ENET_RXD0: receive data • ACOMP_CH5: non-inverting input channel • WKUP23: wake-up from deep sleep0 mode • ADC3: analog input channel
45	P22	I/O	Digital/Analog	• GPIO22: general-purpose I/O • CLK26M: clock derived from 26 MHz CLK_XTALH • SD1_DATA6: data • QSPI0_SCK: serial clock • RGB_B4: blue data • ENET_RXDV: receive data valid • ACOMP_CH6: non-inverting input channel • WKUP22: wake-up from deep sleep0 mode • ADC5: analog input channel
46	P21	I/O	Digital/Analog	• GPIO21: general-purpose I/O • SWDIO: serial wire data • SD1_DATA5: data • ENET_REF_CLK: RMII reference clock • RGB_B5: blue data • ENET_GRCLK: RGMII timing reference for RX data transfers • ACOMP_CH7: non-inverting input channel • WKUP21: wake-up from deep sleep0 mode • ADC6: analog input channel
47	P20	I/O	Digital/Analog	• GPIO20: general-purpose I/O • SWCLK: serial wire clock

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Pin #	Name	I/O	Type	Description
				• SD1_DATA4: data • RGB_B6: blue data • ENET_MDIO: management data • WKUP20: wake-up from deep sleep0 mode
48	P19	I/O	Digital/Analog	• GPIO19: general-purpose I/O • SD1_DATA3: data • RGB_B7: blue data • ENET_MDC: management data clock • TOUCH0: touch sensing I/O • WKUP19: wake-up from deep sleep0 mode
49	P18	I/O	Digital/Analog	• GPIO18: general-purpose I/O • SD1_DATA2: data • RGB_DCLK: clock output • CLK_AUXS_ENET: reference clock for external Ethernet PHY, derived from CLK_DCO/CLK_APLL (divide by 1 to 16) • WKUP18: wake-up from deep sleep0 mode
50	P17	I/O	Digital/Analog	• GPIO17: general-purpose I/O • SD1_DATA1: data • RGB_DISP: display on enable • ENET_TXD0: transmit data • WKUP17: wake-up from deep sleep0 mode
51	P16	I/O	Digital/Analog	• GPIO16: general-purpose I/O • SD1_DATA0: data • RGB_HSYNC: horizontal synchronization • ENET_TXD1: transmit data • WKUP16: wake-up from deep sleep0 mode
52	P15	I/O	Digital/Analog	• GPIO15: general-purpose I/O • SD1_CMD: command/response • RGB_VSYNC: vertical synchronization

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Pin #	Name	I/O	Type	Description
				• ENET_TXD2: transmit data • WKUP15: wake-up from deep sleep0 mode
53	P14	I/O	Digital/Analog	• GPIO14: general-purpose I/O • SD1_CLK: clock • RGB_DE: data enable • ENET_TXD3: transmit data • WKUP14: wake-up from deep sleep0 mode
54	VDDRAM	-	Analog output	EXMEM LDO output
55	DSI_DN0	-	Analog output	MIPI DSI differential output lane 0 negative
56	DSI_DP0	-	Analog output	MIPI DSI differential output lane 0 positive
57	DSI_DN1	-	Analog output	MIPI DSI differential output lane 1 negative
58	DSI_DP1	-	Analog output	MIPI DSI differential output lane 1 positive
59	DSI_CN	-	Analog output	MIPI DSI differential output clock negative
60	DSI_CP	-	Analog output	MIPI DSI differential output clock positive
61	DSI_DN2	-	Analog output	MIPI DSI differential output lane 2 negative
62	DSI_DP2	-	Analog output	MIPI DSI differential output lane 2 positive
63	DSI_DN3	-	Analog output	MIPI DSI differential output lane 3 negative
64	DSI_DP3	-	Analog output	MIPI DSI differential output lane 3 positive
65	DSI_VDDH	-	Analog output	Power supply for MIPI DSI
66	DSI_REXT	-	Analog output	MIPI DSI reference resistor. This analog signal connects to the ground through an external resistor.
67	P40	I/O	Digital	• GPIO40: general-purpose I/O • DVP_PXD8: data • ENET_GTCLK: RGMII timing reference for TX data transfers • RGB_G4: green data
68	P41	I/O	Digital	• GPIO41: general-purpose I/O • DVP_PXD9: data • ENET_TXD3: transmit data • RGB_G3: green data

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Pin #	Name	I/O	Type	Description
69	P42	I/O	Digital	• GPIO42: general-purpose I/O • ENET_TXD2: transmit data • RGB_G2: green data
70	P43	I/O	Digital	• GPIO43: general-purpose I/O • ENET_TXD1: transmit data • RGB_R7: red data
71	P44	I/O	Digital	• GPIO44: general-purpose I/O • ENET_TXD0: transmit data • RGB_R6: red data
72	P45	I/O	Digital	• GPIO45: general-purpose I/O • ENET_TXEN: transmit data enable • RGB_R5: red data
73	P46	I/O	Digital	• GPIO46: general-purpose I/O • CLK_AUXS_ENET: reference clock for external Ethernet PHY, derived from CLK_DCO/CLK_APLL (divide by 1 to 16) • RGB_R4: red data
74	P47	I/O	Digital	• GPIO47: general-purpose I/O • ENET_MDC: management data clock • RGB_R3: red data
75	VDDCORE_HS	-	Analog output	High-power digital core LDO output
76	P48	I/O	Digital	• GPIO48: general-purpose I/O • ENET_MDIO: management data • RGB_B2: blue data
77	P49	I/O	Digital	• GPIO49: general-purpose I/O • ENET_REF_CLK: RMII reference clock • ENET_GRCLK: RGMII timing reference for RX data transfers • RGB_B1: blue data
78	P50	I/O	Digital	• GPIO50: general-purpose I/O • ENET_RXDV: receive data valid • RGB_B0: blue data
79	P51	I/O	Digital	• GPIO51: general-purpose I/O

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Pin #	Name	I/O	Type	Description
				• ENET_RXD0: receive data • RGB_G1: green data
80	P52	I/O	Digital	• GPIO52: general-purpose I/O • ENET_RXD1: receive data • RGB_G0: green data
81	P53	I/O	Digital	• GPIO53: general-purpose I/O • ENET_RXD2: receive data • RGB_R2: red data
82	P54	I/O	Digital	• GPIO54: general-purpose I/O • ENET_RXD3: receive data • RGB_R1: red data
83	P55	I/O	Digital	• GPIO55: general-purpose I/O • ENET_PHY_INT: PHY interrupt input • RGB_R0: red data
84	P13	I/O	Digital/Analog	• GPIO13: general-purpose I/O • ENET_GTCLK: RGMII timing reference for TX data transfers • ACOMP_CH0: non-inverting input channel • WKUP13: wake-up from deep sleep0 mode • ADC15: analog input channel
85	FS_DN	I/O	Digital	Full-speed USB D-
86	FS_DP	I/O	Digital	Full-speed USB D+
87	VOUT1V2	-	Analog output	1.2 V LDO output
88	VOUT1V8	-	Analog output	1.8 V LDO output
89	VOUT2V8	-	Analog output	2.8 V LDO output
90	VOUT3V0	-	Analog output	3.0 V LDO output
91	VDD1V8_HSUSB	-	Analog output	1.8 V power supply for high-speed USB
92	VDD3V3_HSUSB	-	Analog output	3.3 V power supply for high-speed USB
93	HS_DP	I/O	Digital	High-speed USB D+
94	HS_DN	I/O	Digital	High-speed USB D-

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Pin #	Name	I/O	Type	Description
95	AUDRN	-	Analog output	Audio right channel negative output
96	AUDRP	-	Analog output	Audio right channel positive output
97	VCCAUD	-	Analog input	Power supply for audio ADCs/DACs
98	MICBIAS	-	Analog output	Microphone bias output
99	MICN	-	Analog input	Microphone negative input
100	MICP	-	Analog input	Microphone positive input
101	VDDAON	-	Analog output	AON LDO output
102	VDDDIG	-	Analog output	Digital core LDO output
103	VDDD	-	Analog output	Digital buck/LDO output
104	SWD	-	Analog output	Digital buck switch output
105	CEN	-	Analog input	Chip enable, active high
106	VDDA	-	Analog output	Analog buck/LDO output
107	SWA	-	Analog output	Analog buck switch output
108	GND_DCDC	-	GND	Buck ground
109	VBAT	-	Power	Chip power supply
110	SWH	-	Analog output	High-power digital buck switch output
111	VDDH	-	Analog output	High-power digital buck/LDO output
112	VDDCORE_HS	-	Analog output	High-power digital core LDO output
113	P59	I/O	Digital	• GPIO59: general-purpose I/O • CLK_AUXS_DVP: DVP master clock, derived from CLK_XTALH/ AP_CLK_240M/AP_CLK_320M/ AP_CLK_480M (divide by 1 to 16)
114	P60	I/O	Digital	• GPIO60: general-purpose I/O • ENET_PHY_INT: PHY interrupt input
115	P64	I/O	Digital	• GPIO64: general-purpose I/O • SD0_DATA5: data • UART5_RX: receive data input
116	P65	I/O	Digital	• GPIO65: general-purpose I/O • SD0_DATA4: data

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Pin #	Name	I/O	Type	Description
				• UART5_TX: transmit data output
117	VCCGPIO	-	Analog output	Power supply for GPIOs
118	P69	I/O	Digital	• GPIO69: general-purpose I/O • SD0_DATA0: data
119	P70	I/O	Digital	• GPIO70: general-purpose I/O • SD0_CMD: command/response
120	P71	I/O	Digital	• GPIO71: general-purpose I/O • SD0_CLK: clock
121	P0	I/O	Digital/Analog	• GPIO0: general-purpose I/O • ACOMP_CH9: non-inverting input channel • WKUP0: wake-up from deep sleep0 mode • ADC12: analog input channel
122	P1	I/O	Digital/Analog	• GPIO1: general-purpose I/O • ACOMP_CH10: non-inverting input channel • WKUP1: wake-up from deep sleep0 mode • ADC13: analog input channel
123	P2	I/O	Digital/Analog	• GPIO2: general-purpose I/O • SD0_CLK: clock • QSPI1_SCK: serial clock • WKUP2: wake-up from deep sleep0 mode
124	P3	I/O	Digital/Analog	• GPIO3: general-purpose I/O • SD0_CMD: command/response • QSPI1_CS: chip select • WKUP3: wake-up from deep sleep0 mode
125	P4	I/O	Digital/Analog	• GPIO4: general-purpose I/O • SD0_DATA0: data • QSPI1_IO0: data • WKUP4: wake-up from deep sleep0 mode

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Pin #	Name	I/O	Type	Description
126	P5	I/O	Digital/Analog	• GPIO5: general-purpose I/O • SD0_DATA1: data • QSPI1_IO1: data • WKUP5: wake-up from deep sleep0 mode
127	P6	I/O	Digital/Analog	• GPIO6: general-purpose I/O • CLK13M: clock derived from 26 MHz CLK_XTALH (divide by 1 to 8) • SD0_DATA4: data • QSPI1_IO2: data • WKUP6: wake-up from deep sleep0 mode
128	P7	I/O	Digital/Analog	• GPIO7: general-purpose I/O • SD0_DATA5: data • QSPI1_IO3: data • WKUP7: wake-up from deep sleep0 mode
Die pad	GND_SLUG	-	GND	Ground

3.2 BK7259QN128B832 QFN128 Pin Descriptions

Figure 3-2 shows the pin assignments of the BK7259QN128B832's 12.3 x 12.3 mm, 128-pin QFN package.

Figure 3-2 BK7259QN128B832 QFN128 Pin Assignments

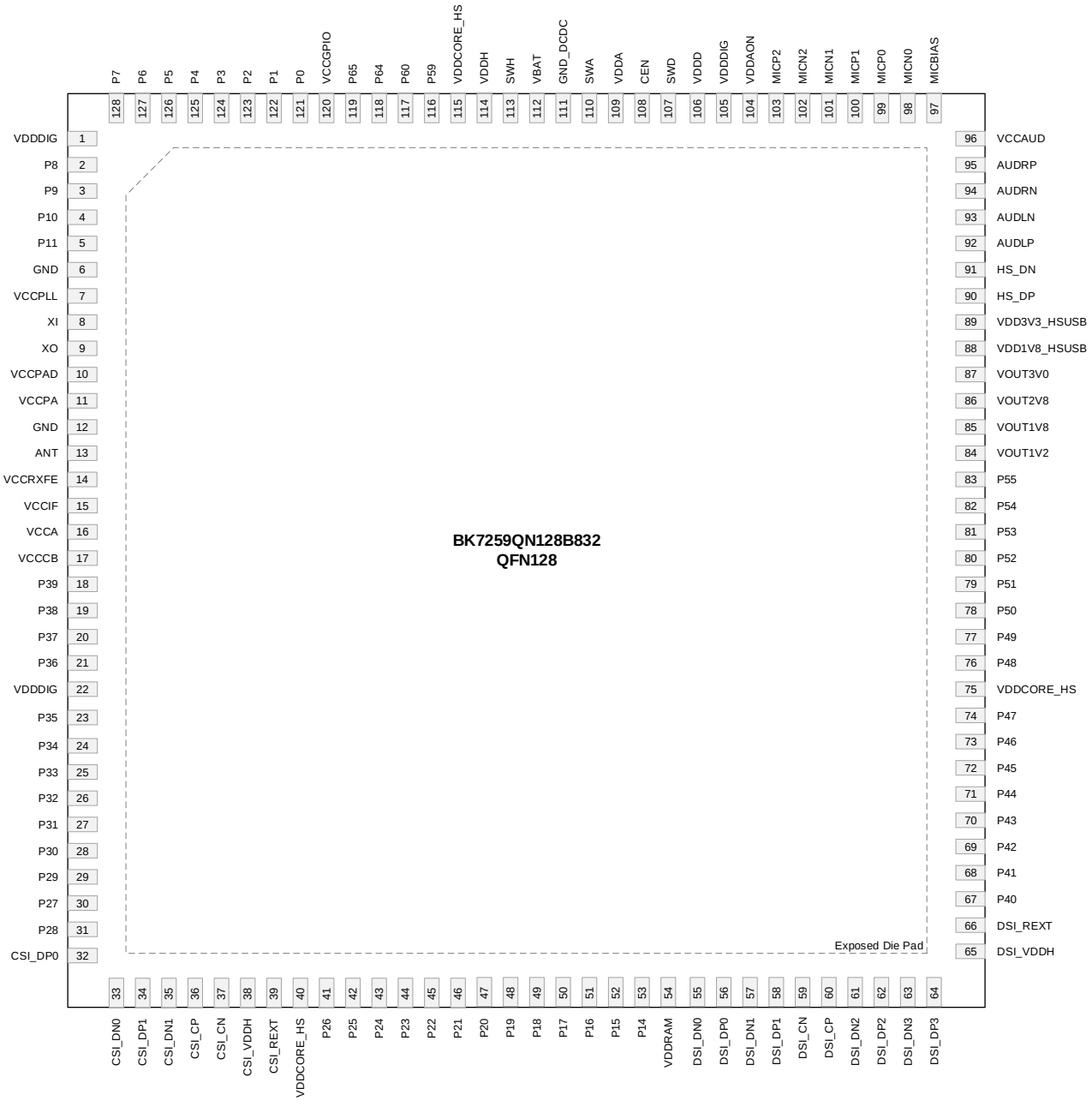


Table 3-2 shows the pin descriptions of the BK7259QN128B832 QFN128 package.

Note: Due to power-up glitches, the following I/O pins are not recommended for applications sensitive to power-up fluctuations, such as LEDs or motor control: GPIO10, GPIO20, GPIO21, and GPIO22.

Table 3-2 BK7259QN128B832 QFN128 Pin Descriptions

Pin #	Name	I/O	Type	Description
1	VDDDIG	-	Analog output	Digital core LDO output
2	P8	I/O	Digital/Analog	• GPIO8: general-purpose I/O • SD0_DATA6: data • 32K_XO: 32.768 kHz crystal output • ACOMP_CH8: non-inverting input channel • WKUP8: wake-up from deep sleep0 mode • ADC10: analog input channel
3	P9	I/O	Digital/Analog	• GPIO9: general-purpose I/O • SD0_DATA7: data • 32K_XI: 32.768 kHz crystal input • WKUP9: wake-up from deep sleep0 mode
4	P10	I/O	Digital/Analog	• GPIO10: general-purpose I/O • DL_UART_RX: UART flash download receive data input (UART0) • SD0_DATA2: data • CLK_AUXS: reference clock, derived from CLK_DCO/CLK_APLL/ WP_CLK_320M/WP_CLK_480M (divide by 1 to 64) • WKUP10: wake-up from deep sleep0 mode
5	P11	I/O	Digital/Analog	• GPIO11: general-purpose I/O • DL_UART_TX: UART flash download transmit data output (UART0) • SD0_DATA3: data • WKUP11: wake-up from deep sleep0 mode
6	GND	-	GND	Ground
7	VCCPLL	-	Analog input	RF PLL power supply
8	XI	-	Analog input	26 MHz crystal input
9	XO	-	Analog output	26 MHz crystal output

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Pin #	Name	I/O	Type	Description
10	VCCPAD	-	Analog input	RF PA driver power supply
11	VCCPA	-	Analog input	RF PA power supply
12	GND	-	GND	Ground
13	ANT	-	RF	2.4 GHz RF signal port
14	VCCRXFE	-	Analog input	RF receiver power supply
15	VCCIF	-	Analog input	IF power supply
16	VCCA	-	Analog input	Analog power supply
17	VCCCB	-	Analog input	Power supply for bandgap and analog peripherals
18	P39	I/O	Digital/Analog	• GPIO39: general-purpose I/O • DVP_PXD7: data • DAC0: output channel
19	P38	I/O	Digital/Analog	• GPIO38: general-purpose I/O • DVP_PXD6: data • DAC1: output channel
20	P37	I/O	Digital	• GPIO37: general-purpose I/O • DVP_PXD5: data
21	P36	I/O	Digital	• GPIO36: general-purpose I/O • DVP_PXD4: data
22	VDDDIG	-	Analog output	Digital core LDO output
23	P35	I/O	Digital	• GPIO35: general-purpose I/O • DVP_PXD3: data
24	P34	I/O	Digital	• GPIO34: general-purpose I/O • DVP_PXD2: data
25	P33	I/O	Digital	• GPIO33: general-purpose I/O • DVP_PXD1: data
26	P32	I/O	Digital	• GPIO32: general-purpose I/O • DVP_PXD0: data
27	P31	I/O	Digital/Analog	• GPIO31: general-purpose I/O • DVP_VSYNC: vertical synchronization

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Pin #	Name	I/O	Type	Description
				• WKUP31: wake-up from deep sleep0 mode
28	P30	I/O	Digital/Analog	• GPIO30: general-purpose I/O • DVP_HSYNC: horizontal synchronization • WKUP30: wake-up from deep sleep0 mode
29	P29	I/O	Digital/Analog	• GPIO29: general-purpose I/O • DVP_PCLK: pixel clock • WKUP29: wake-up from deep sleep0 mode
30	P27	I/O	Digital/Analog	• GPIO27: general-purpose I/O • FEM_LNA_EN: LNA bypass • CLK_AUXS_DVP: DVP master clock, derived from CLK_XTALH/ AP_CLK_240M/AP_CLK_320M/ AP_CLK_480M (divide by 1 to 16) • QSPI0_IO3: data • DVP_MCLK: DVP master clock, derived from CLK_XTALH/AP_CLK_240M/ AP_CLK_320M/AP_CLK_480M (divide by 1 to 16) • WKUP27: wake-up from deep sleep0 mode
31	P28	I/O	Digital/Analog	• GPIO28: general-purpose I/O • WIFI_RX_EN: receive enable • CLK_AUXS: reference clock, derived from CLK_DCO/CLK_APLL/ WP_CLK_320M/WP_CLK_480M (divide by 1 to 64) • ENET_TXEN: transmit data enable • ACOMP_CH2: non-inverting input channel • WKUP28: wake-up from deep sleep0 mode • ADC4: analog input channel
32	CSI_DP0	-	Analog input	MIPI CSI differential input lane 0 positive
33	CSI_DN0	-	Analog input	MIPI CSI differential input lane 0 negative

Pin #	Name	I/O	Type	Description
34	CSI_DP1	-	Analog input	MIPI CSI differential input lane 1 positive
35	CSI_DN1	-	Analog input	MIPI CSI differential input lane 1 negative
36	CSI_CP	-	Analog input	MIPI CSI differential input clock positive
37	CSI_CN	-	Analog input	MIPI CSI differential input clock negative
38	CSI_VDDH	-	Analog output	Power supply for MIPI CSI
39	CSI_REXT	-	Analog output	MIPI CSI reference resistor. This analog signal connects to the ground through an external resistor.
40	VDDCORE_HS	-	Analog output	High-power digital core LDO output
41	P26	I/O	Digital/Analog	• GPIO26: general-purpose I/O • WIFI_TX_EN: transmit enable • QSPI0_IO2: data • RGB_G5: green data • ENET_RXD3: receive data • WKUP26: wake-up from deep sleep0 mode
42	P25	I/O	Digital/Analog	• GPIO25: general-purpose I/O • QSPI0_IO1: data • RGB_G6: green data • ENET_RXD2: receive data • ACOMP_CH3: non-inverting input channel • WKUP25: wake-up from deep sleep0 mode • ADC1: analog input channel
43	P24	I/O	Digital/Analog	• GPIO24: general-purpose I/O • LPO_CLK: clock derived from 32 kHz CLK_LPO1 • SD1_STB: data strobe • QSPI0_IO0: data • RGB_G7: green data • ENET_RXD1: receive data • ACOMP_CH4: non-inverting input channel • WKUP24: wake-up from deep sleep0 mode

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Pin #	Name	I/O	Type	Description
				• ADC2: analog input channel
44	P23	I/O	Digital/Analog	• GPIO23: general-purpose I/O • SD1_DATA7: data • QSPI0_CS: chip select • RGB_B3: blue data • ENET_RXD0: receive data • ACOMP_CH5: non-inverting input channel • WKUP23: wake-up from deep sleep0 mode • ADC3: analog input channel
45	P22	I/O	Digital/Analog	• GPIO22: general-purpose I/O • CLK26M: clock derived from 26 MHz CLK_XTALH • SD1_DATA6: data • QSPI0_SCK: serial clock • RGB_B4: blue data • ENET_RXDV: receive data valid • ACOMP_CH6: non-inverting input channel • WKUP22: wake-up from deep sleep0 mode • ADC5: analog input channel
46	P21	I/O	Digital/Analog	• GPIO21: general-purpose I/O • SWDIO: serial wire data • SD1_DATA5: data • ENET_REF_CLK: RMII reference clock • RGB_B5: blue data • ENET_GRCLK: RGMII timing reference for RX data transfers • ACOMP_CH7: non-inverting input channel • WKUP21: wake-up from deep sleep0 mode • ADC6: analog input channel
47	P20	I/O	Digital/Analog	• GPIO20: general-purpose I/O • SWCLK: serial wire clock

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Pin #	Name	I/O	Type	Description
				• SD1_DATA4: data • RGB_B6: blue data • ENET_MDIO: management data • WKUP20: wake-up from deep sleep0 mode
48	P19	I/O	Digital/Analog	• GPIO19: general-purpose I/O • SD1_DATA3: data • RGB_B7: blue data • ENET_MDC: management data clock • TOUCH0: touch sensing I/O • WKUP19: wake-up from deep sleep0 mode
49	P18	I/O	Digital/Analog	• GPIO18: general-purpose I/O • SD1_DATA2: data • RGB_DCLK: clock output • CLK_AUXS_ENET: reference clock for external Ethernet PHY, derived from CLK_DCO/CLK_APLL (divide by 1 to 16) • WKUP18: wake-up from deep sleep0 mode
50	P17	I/O	Digital/Analog	• GPIO17: general-purpose I/O • SD1_DATA1: data • RGB_DISP: display on enable • ENET_TXD0: transmit data • WKUP17: wake-up from deep sleep0 mode
51	P16	I/O	Digital/Analog	• GPIO16: general-purpose I/O • SD1_DATA0: data • RGB_HSYNC: horizontal synchronization • ENET_TXD1: transmit data • WKUP16: wake-up from deep sleep0 mode
52	P15	I/O	Digital/Analog	• GPIO15: general-purpose I/O • SD1_CMD: command/response • RGB_VSYNC: vertical synchronization

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Pin #	Name	I/O	Type	Description
				• ENET_TXD2: transmit data • WKUP15: wake-up from deep sleep0 mode
53	P14	I/O	Digital/Analog	• GPIO14: general-purpose I/O • SD1_CLK: clock • RGB_DE: data enable • ENET_TXD3: transmit data • WKUP14: wake-up from deep sleep0 mode
54	VDDRAM	-	Analog output	EXMEM LDO output
55	DSI_DN0	-	Analog output	MIPI DSI differential output lane 0 negative
56	DSI_DP0	-	Analog output	MIPI DSI differential output lane 0 positive
57	DSI_DN1	-	Analog output	MIPI DSI differential output lane 1 negative
58	DSI_DP1	-	Analog output	MIPI DSI differential output lane 1 positive
59	DSI_CN	-	Analog output	MIPI DSI differential output clock negative
60	DSI_CP	-	Analog output	MIPI DSI differential output clock positive
61	DSI_DN2	-	Analog output	MIPI DSI differential output lane 2 negative
62	DSI_DP2	-	Analog output	MIPI DSI differential output lane 2 positive
63	DSI_DN3	-	Analog output	MIPI DSI differential output lane 3 negative
64	DSI_DP3	-	Analog output	MIPI DSI differential output lane 3 positive
65	DSI_VDDH	-	Analog output	Power supply for MIPI DSI
66	DSI_REXT	-	Analog output	MIPI DSI reference resistor. This analog signal connects to the ground through an external resistor.
67	P40	I/O	Digital	• GPIO40: general-purpose I/O • DVP_PXD8: data • ENET_GTCLK: RGMII timing reference for TX data transfers • RGB_G4: green data
68	P41	I/O	Digital	• GPIO41: general-purpose I/O • DVP_PXD9: data • ENET_TXD3: transmit data • RGB_G3: green data

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Pin #	Name	I/O	Type	Description
69	P42	I/O	Digital	• GPIO42: general-purpose I/O • ENET_TXD2: transmit data • RGB_G2: green data
70	P43	I/O	Digital	• GPIO43: general-purpose I/O • ENET_TXD1: transmit data • RGB_R7: red data
71	P44	I/O	Digital	• GPIO44: general-purpose I/O • ENET_TXD0: transmit data • RGB_R6: red data
72	P45	I/O	Digital	• GPIO45: general-purpose I/O • ENET_TXEN: transmit data enable • RGB_R5: red data
73	P46	I/O	Digital	• GPIO46: general-purpose I/O • CLK_AUXS_ENET: reference clock for external Ethernet PHY, derived from CLK_DCO/CLK_APLL (divide by 1 to 16) • RGB_R4: red data
74	P47	I/O	Digital	• GPIO47: general-purpose I/O • ENET_MDC: management data clock • RGB_R3: red data
75	VDDCORE_HS	-	Analog output	High-power digital core LDO output
76	P48	I/O	Digital	• GPIO48: general-purpose I/O • ENET_MDIO: management data • RGB_B2: blue data
77	P49	I/O	Digital	• GPIO49: general-purpose I/O • ENET_REF_CLK: RMII reference clock • ENET_GRCLK: RGMII timing reference for RX data transfers • RGB_B1: blue data
78	P50	I/O	Digital	• GPIO50: general-purpose I/O • ENET_RXDV: receive data valid • RGB_B0: blue data
79	P51	I/O	Digital	• GPIO51: general-purpose I/O

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Pin #	Name	I/O	Type	Description
				• ENET_RXD0: receive data • RGB_G1: green data
80	P52	I/O	Digital	• GPIO52: general-purpose I/O • ENET_RXD1: receive data • RGB_G0: green data
81	P53	I/O	Digital	• GPIO53: general-purpose I/O • ENET_RXD2: receive data • RGB_R2: red data
82	P54	I/O	Digital	• GPIO54: general-purpose I/O • ENET_RXD3: receive data • RGB_R1: red data
83	P55	I/O	Digital	• GPIO55: general-purpose I/O • ENET_PHY_INT: PHY interrupt input • RGB_R0: red data
84	VOUT1V2	-	Analog output	1.2 V LDO output
85	VOUT1V8	-	Analog output	1.8 V LDO output
86	VOUT2V8	-	Analog output	2.8 V LDO output
87	VOUT3V0	-	Analog output	3.0 V LDO output
88	VDD1V8_HSUSB	-	Analog output	1.8 V power supply for high-speed USB
89	VDD3V3_HSUSB	-	Analog output	3.3 V power supply for high-speed USB
90	HS_DP	I/O	Digital	High-speed USB D+
91	HS_DN	I/O	Digital	High-speed USB D-
92	AUDLP	-	Analog output	Audio left channel positive output
93	AUDLN	-	Analog output	Audio left channel negative output
94	AUDRN	-	Analog output	Audio right channel negative output
95	AUDRP	-	Analog output	Audio right channel positive output
96	VCCAUD	-	Analog input	Power supply for audio ADCs/DACs
97	MICBIAS	-	Analog output	Microphone bias output
98	MICN0	-	Analog input	Microphone 0 negative input
99	MICP0	-	Analog input	Microphone 0 positive input

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Pin #	Name	I/O	Type	Description
100	MICP1	-	Analog input	Microphone 1 positive input
101	MICN1	-	Analog input	Microphone 1 negative input
102	MICN2	-	Analog input	Microphone 2 negative input
103	MICP2	-	Analog input	Microphone 2 positive input
104	VDDAON	-	Analog output	AON LDO output
105	VDDDIG	-	Analog output	Digital core LDO output
106	VDDD	-	Analog output	Digital buck/LDO output
107	SWD	-	Analog output	Digital buck switch output
108	CEN	-	Analog input	Chip enable, active high
109	VDDA	-	Analog output	Analog buck/LDO output
110	SWA	-	Analog output	Analog buck switch output
111	GND_DCDC	-	GND	Buck ground
112	VBAT	-	Power	Chip power supply
113	SWH	-	Analog output	High-power digital buck switch output
114	VDDH	-	Analog output	High-power digital buck/LDO output
115	VDDCORE_HS	-	Analog output	High-power digital core LDO output
116	P59	I/O	Digital	• GPIO59: general-purpose I/O • CLK_AUXS_DVP: DVP master clock, derived from CLK_XTALH/ AP_CLK_240M/AP_CLK_320M/ AP_CLK_480M (divide by 1 to 16)
117	P60	I/O	Digital	• GPIO60: general-purpose I/O • ENET_PHY_INT: PHY interrupt input
118	P64	I/O	Digital	• GPIO64: general-purpose I/O • SD0_DATA5: data • UART5_RX: receive data input
119	P65	I/O	Digital	• GPIO65: general-purpose I/O • SD0_DATA4: data • UART5_TX: transmit data output
120	VCCGPIO	-	Analog output	Power supply for GPIOs

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Pin #	Name	I/O	Type	Description
121	P0	I/O	Digital/Analog	• GPIO0: general-purpose I/O • ACOMP_CH9: non-inverting input channel • WKUP0: wake-up from deep sleep0 mode • ADC12: analog input channel
122	P1	I/O	Digital/Analog	• GPIO1: general-purpose I/O • ACOMP_CH10: non-inverting input channel • WKUP1: wake-up from deep sleep0 mode • ADC13: analog input channel
123	P2	I/O	Digital/Analog	• GPIO2: general-purpose I/O • SD0_CLK: clock • QSPI1_SCK: serial clock • WKUP2: wake-up from deep sleep0 mode
124	P3	I/O	Digital/Analog	• GPIO3: general-purpose I/O • SD0_CMD: command/response • QSPI1_CS: chip select • WKUP3: wake-up from deep sleep0 mode
125	P4	I/O	Digital/Analog	• GPIO4: general-purpose I/O • SD0_DATA0: data • QSPI1_IO0: data • WKUP4: wake-up from deep sleep0 mode
126	P5	I/O	Digital/Analog	• GPIO5: general-purpose I/O • SD0_DATA1: data • QSPI1_IO1: data • WKUP5: wake-up from deep sleep0 mode
127	P6	I/O	Digital/Analog	• GPIO6: general-purpose I/O • CLK13M: clock derived from 26 MHz CLK_XTALH (divide by 1 to 8) • SD0_DATA4: data • QSPI1_IO2: data

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Pin #	Name	I/O	Type	Description
				• WKUP6: wake-up from deep sleep0 mode
128	P7	I/O	Digital/Analog	• GPIO7: general-purpose I/O • SD0_DATA5: data • QSPI1_IO3: data • WKUP7: wake-up from deep sleep0 mode
Die pad	GND_SLUG	-	GND	Ground

3.3 Pin Multiplexing

Each I/O pin of the BK7259 can be configured to perform multiple functions: GPIO function, peripheral functions, and analog functions.

High-performance peripherals have alternate functions mapped onto fixed I/O pins. For normal peripherals, through the I/O multiplexer (IOMUX), their input signals can be from any I/O pin, and their output signals can be routed to any I/O pin. Each I/O pin can be configured as a GPIO or an I/O for peripherals through the GPIOx_FUN_SEL field of the GPIO configuration register.

The I/O pins can be configured as the following states:

- High impedance
- Input high-impedance
- Input pull-up
- Input pull-down
- Output push-pull with pull-up or pull-down
- Input/output
- Peripheral function push-pull with pull-up or pull-down
- Peripheral function open-drain with pull-up or pull-down
- Analog function

The definitions of I/O types are listed below:

- I: input
- O: output
- IO: input/output
- Z: high impedance

- IOZ: input/output/high impedance

3.3.1 I/O Pin Predefined States at Reset and After Reset

Table 3-3 shows the predefined states at reset and after reset for I/Os.

Table 3-3 I/O Pin Predefined States at Reset and After Reset

I/O	Predefined State	
	At Reset	After Reset
GPIO0	High impedance	High impedance
GPIO1	High impedance	High impedance
GPIO2	High impedance	High impedance
GPIO3	High impedance	High impedance
GPIO4	High impedance	High impedance
GPIO5	High impedance	High impedance
GPIO6	High impedance	High impedance
GPIO7	High impedance	High impedance
GPIO8	High impedance	High impedance
GPIO9	High impedance	High impedance
GPIO10	High impedance	Peripheral function (DL_UART_RX)
GPIO11	High impedance	High impedance
GPIO13	High impedance	High impedance
GPIO14	High impedance	High impedance
GPIO15	High impedance	High impedance
GPIO16	High impedance	High impedance
GPIO17	High impedance	High impedance
GPIO18	High impedance	High impedance
GPIO19	High impedance	High impedance
GPIO20	High impedance	Peripheral function (pull-down) (SWCLK)
GPIO21	High impedance	Peripheral function (pull-up) (SWDIO)
GPIO22	Input pull-down	High impedance

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I/O	Predefined State	
	At Reset	After Reset
GPIO23	High impedance	High impedance
GPIO24–71	High impedance	High impedance

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3.3.2 Pin Multiplexing of High-performance Peripherals

Table 3-4 shows the pin multiplexing of high-performance peripherals.

Table 3-4 Pin Multiplexing of High-performance Peripherals

GPIO	Flash Download	Debug	Alternate Functions (GPIOx_FUNC_SEL)				
			8'd126	8'd129	8'd130	8'd131	8'd132
	UART	SW-DP (Default)[1]	Clock FEM Support	SDMMC0/1 Clock DVP ENET	QSPI0/1 Clock ENET UART5	DISPLAY DVP	ENET
GPIO0							
GPIO1							
GPIO2				SD0_CLK (IOZ)	QSPI1_SCK (IOZ)		
GPIO3				SD0_CMD (IOZ)	QSPI1_CS (O)		
GPIO4				SD0_DATA0 (IO)	QSPI1_IO0 (IO)		
GPIO5				SD0_DATA1 (IO)	QSPI1_IO1 (IO)		
GPIO6			CLK13M (O)	SD0_DATA4 (IO)	QSPI1_IO2 (IO)		
GPIO7				SD0_DATA5 (IO)	QSPI1_IO3 (IO)		
GPIO8				SD0_DATA6 (IO)			
GPIO9				SD0_DATA7 (IO)			
GPIO10	DL_UART_RX (I)			SD0_DATA2 (IO)	CLK_AUXS (O)		
GPIO11	DL_UART_TX (O)			SD0_DATA3 (IO)			

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GPIO	Flash Download	Debug	Alternate Functions (GPIOx_FUNC_SEL)				
			8'd126	8'd129	8'd130	8'd131	8'd132
	UART	SW-DP (Default)[1]	Clock FEM Support	SDMMC0/1 Clock DVP ENET	QSPI0/1 Clock ENET UART5	DISPLAY DVP	ENET
GPIO13							ENET_GTCLK (O)
GPIO14				SD1_CLK (IOZ)		RGB_DE (O)	ENET_TXD3 (O)
GPIO15				SD1_CMD (IOZ)		RGB_VSYNC (O)	ENET_TXD2 (O)
GPIO16				SD1_DATA0 (IO)		RGB_HSYNC (O)	ENET_TXD1 (O)
GPIO17				SD1_DATA1 (IO)		RGB_DISP (O)	ENET_TXD0 (O)
GPIO18				SD1_DATA2 (IO)		RGB_DCLK (O)	CLK_AUXS_ENET (O)
GPIO19				SD1_DATA3 (IO)		RGB_B7 (O)	ENET_MDC (O)
GPIO20		SWCLK (I)		SD1_DATA4 (IO)		RGB_B6 (O)	ENET_MDIO (IO)
GPIO21		SWDIO (IO)		SD1_DATA5 (IO)	ENET_REF_CLK (I)	RGB_B5 (O)	ENET_GRCLK (I)
GPIO22			CLK26M (O)	SD1_DATA6 (IO)	QSPI0_SCK (IOZ)	RGB_B4 (O)	ENET_RXDV (I)
GPIO23				SD1_DATA7 (IO)	QSPI0_CS (O)	RGB_B3 (O)	ENET_RXD0 (I)
GPIO24			LPO_CLK (O)	SD1_STB (I)	QSPI0_IO0 (IO)	RGB_G7 (O)	ENET_RXD1 (I)
GPIO25					QSPI0_IO1 (IO)	RGB_G6 (O)	ENET_RXD2 (I)
GPIO26			WIFI_TX_EN (O)		QSPI0_IO2 (IO)	RGB_G5 (O)	ENET_RXD3 (I)
GPIO27			FEM_LNA_EN (O)	CLK_AUXS_DVP (O)	QSPI0_IO3 (IO)	DVP_MCLK (O)	
GPIO28			WIFI_RX_EN (O)	CLK_AUXS (O)			ENET_TXEN (O)

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GPIO	Flash Download	Debug	Alternate Functions (GPIOx_FUNC_SEL)				
			8'd126	8'd129	8'd130	8'd131	8'd132
	UART	SW-DP (Default)[1]	Clock FEM Support	SDMMC0/1 Clock DVP ENET	QSPI0/1 Clock ENET UART5	DISPLAY DVP	ENET
GPIO29				DVP_PCLK (I)			
GPIO30				DVP_HSYNC (I)			
GPIO31				DVP_VSYNC (I)			
GPIO32				DVP_PXD0 (I)			
GPIO33				DVP_PXD1 (I)			
GPIO34				DVP_PXD2 (I)			
GPIO35				DVP_PXD3 (I)			
GPIO36				DVP_PXD4 (I)			
GPIO37				DVP_PXD5 (I)			
GPIO38				DVP_PXD6 (I)			
GPIO39				DVP_PXD7 (I)			
GPIO40				DVP_PXD8 (I)	ENET_GTCLK (O)	RGB_G4 (O)	
GPIO41				DVP_PXD9 (I)	ENET_TXD3 (O)	RGB_G3 (O)	
GPIO42					ENET_TXD2 (O)	RGB_G2 (O)	
GPIO43					ENET_TXD1 (O)	RGB_R7 (O)	
GPIO44					ENET_TXD0 (O)	RGB_R6 (O)	

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GPIO	Flash Download	Debug	Alternate Functions (GPIOx_FUNC_SEL)				
			8'd126	8'd129	8'd130	8'd131	8'd132
	UART	SW-DP (Default)[1]	Clock FEM Support	SDMMC0/1 Clock DVP ENET	QSPI0/1 Clock ENET UART5	DISPLAY DVP	ENET
GPIO45					ENET_TXEN (O)	RGB_R5 (O)	
GPIO46					CLK_AUXS_ENET (O)	RGB_R4 (O)	
GPIO47					ENET_MDC (O)	RGB_R3 (O)	
GPIO48					ENET_MDIO (IO)	RGB_B2 (O)	
GPIO49				ENET_REF_CLK (I)	ENET_GRCLK (I)	RGB_B1 (O)	
GPIO50					ENET_RXDV (I)	RGB_B0 (O)	
GPIO51					ENET_RXD0 (I)	RGB_G1 (O)	
GPIO52					ENET_RXD1 (I)	RGB_G0 (O)	
GPIO53					ENET_RXD2 (I)	RGB_R2 (O)	
GPIO54					ENET_RXD3 (I)	RGB_R1 (O)	
GPIO55					ENET_PHY_INT (I)	RGB_R0 (O)	
GPIO59				CLK_AUXS_DVP (O)			
GPIO60					ENET_PHY_INT (I)		
GPIO64				SD0_DATA5 (IO)	UART5_RX (I)		
GPIO65				SD0_DATA4 (IO)	UART5_TX (O)		
GPIO69				SD0_DATA0 (IO)			

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GPIO	Flash Download	Debug	Alternate Functions (GPIOx_FUNC_SEL)				
			8'd126	8'd129	8'd130	8'd131	8'd132
	UART	SW-DP (Default)(1)	Clock FEM Support	SDMMC0/1 Clock DVP ENET	QSPI0/1 Clock ENET UART5	DISPLAY DVP	ENET
GPIO70				SD0_CMD (IOZ)			
GPIO71				SD0_CLK (IOZ)			

(1) Upon a reset, the SW-DP is selected by default. The two I/O pins, SWDIO and SWCLK, are configured by hardware in debug mode.

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3.3.3 I/O Matrix of Normal Peripherals

Table 3-5 shows the I/O matrix of normal peripherals.

Table 3-5 I/O Matrix of Normal Peripherals

Function	GPIOx_FUNC_SEL Value	GPIO Function Name	I/O Type	Description
GPIO	8'd0	GPIO_HIGH_Z	Z	General-purpose I/O
	8'd1	GPIO_I	I	General-purpose I/O
	8'd2	GPIO_O	O	General-purpose I/O
	8'd3	GPIO_IO	IO	General-purpose I/O
DMIC0	8'd4	DMIC0_CLK	O	Clock
	8'd5	DMIC0_DAT	I	Data
DMIC1	8'd6	DMIC1_CLK	O	Clock
	8'd7	DMIC1_DAT	I	Data
HDMI CEC	8'd8	HDMI_CEC	IO	Data
I2S0	8'd9	I2S0_DIN	I	Serial data input
	8'd10	I2S0_DOUT	O	Serial data output
	8'd11	I2S0_SCK	IO	Serial clock
	8'd12	I2S0_SYNC	IO	Frame synchronization
I2S1	8'd13	I2S1_DIN	I	Serial data input
	8'd14	I2S1_DOUT	O	Serial data output
	8'd15	I2S1_SCK	IO	Serial clock
	8'd16	I2S1_SYNC	IO	Frame synchronization
I2S2	8'd17	I2S2_DIN	I	Serial data input
	8'd18	I2S2_DOUT	O	Serial data output
	8'd19	I2S2_SCK	IO	Serial clock
	8'd20	I2S2_SYNC	IO	Frame synchronization
I2S3	8'd21	I2S3_DIN	I	Serial data input
	8'd22	I2S3_DOUT	O	Serial data output

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Function	GPIOx_FUNC_SEL Value	GPIO Function Name	I/O Type	Description
	8'd23	I2S3_SCK	IO	Serial clock
	8'd24	I2S3_SYNC	IO	Frame synchronization
I2S4	8'd25	I2S4_DIN	I	Serial data input
	8'd26	I2S4_DOUT	O	Serial data output
	8'd27	I2S4_SCK	IO	Serial clock
	8'd28	I2S4_SYNC	IO	Frame synchronization
I2S	8'd29	I2S_MCLK	O	Master clock
SW-DP	8'd30	SWCLK	I	Serial wire clock
	8'd31	SWDIO	IO	Serial wire data
PWM	8'd32	PWM0	IO	PWM0 channel
	8'd33	PWM1	IO	PWM1 channel
	8'd34	PWM2	IO	PWM2 channel
	8'd35	PWM3	IO	PWM3 channel
	8'd36	PWM4	IO	PWM4 channel
	8'd37	PWM5	IO	PWM5 channel
	8'd38	PWM6	IO	PWM6 channel
	8'd39	PWM7	IO	PWM7 channel
	8'd40	PWM8	IO	PWM8 channel
	8'd41	PWM9	IO	PWM9 channel
	8'd42	PWM10	IO	PWM10 channel
	8'd43	PWM11	IO	PWM11 channel
SPI0	8'd44	SPI0_MISO	IO	Master in slave out
	8'd45	SPI0_MOSI	IO	Master out slave in
	8'd46	SPI0_CSN	IO	Chip select
	8'd47	SPI0_SCK	IO	Serial clock
SPI1	8'd48	SPI1_MISO	IO	Master in slave out
	8'd49	SPI1_MOSI	IO	Master out slave in

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Function	GPIOx_FUNC_SEL Value	GPIO Function Name	I/O Type	Description
	8'd50	SPI1_CSN	IO	Chip select
	8'd51	SPI1_SCK	IO	Serial clock
SPI2	8'd52	SPI2_MISO	IO	Master in slave out
	8'd53	SPI2_MOSI	IO	Master out slave in
	8'd54	SPI2_CSN	IO	Chip select
	8'd55	SPI2_SCK	IO	Serial clock
SPI3	8'd56	SPI3_MISO	IO	Master in slave out
	8'd57	SPI3_MOSI	IO	Master out slave in
	8'd58	SPI3_NSS	IO	Chip select
	8'd59	SPI3_SCK	IO	Serial clock
I3C	8'd60	I3C_SCL	IOZ	Serial clock
	8'd61	I3C_SDA	IOZ	Serial data
	8'd62	I3C_SDA_PURN	O	Serial data pull-up enable, active low
N/A	8'd63	N/A	-	N/A
Bluetooth AoA/AoD	8'd64	BT_ANT0	I	Bluetooth antenna select
	8'd65	BT_ANT1	I	Bluetooth antenna select
	8'd66	BT_ANT2	I	Bluetooth antenna select
	8'd67	BT_ANT3	I	Bluetooth antenna select
CAN0	8'd68	CAN0_RX	I	Receive
	8'd69	CAN0_STBY	O	Transceiver standby mode, active high
	8'd70	CAN0_TX	O	Transmit
CAN1	8'd71	CAN1_RX	I	Receive
	8'd72	CAN1_STBY	O	Transceiver standby mode, active high
	8'd73	CAN1_TX	O	Transmit
I2C0	8'd74	I2C0_SCL	IOZ	Serial clock

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Function	GPIOx_FUNC_SEL Value	GPIO Function Name	I/O Type	Description
	8'd75	I2C0_SDA	IOZ	Serial data
I2C1	8'd76	I2C1_SCL	IOZ	Serial clock
	8'd77	I2C1_SDA	IOZ	Serial data
IrDA0	8'd78	IRDA0	IO	Data
IrDA1	8'd79	IRDA1	IO	Data
IrDA2	8'd80	IRDA2	IO	Data
IrDA3	8'd81	IRDA3	IO	Data
LIN	8'd82	LIN_RXD	I	Receive data input
	8'd83	LIN_SLEEP	O	Transceiver sleep mode, active low
	8'd84	LIN_TXD	O	Transmit data output
SC	8'd85	SC_CLK	O	Clock
	8'd86	SC_IO	IOZ	Data input/output
	8'd87	SC_RSTN	O	Reset
	8'd88	SC_VCC	O	Power supply to the smart card
SPDIF0	8'd89	SPDIF0_RX	I	Receive
	8'd90	SPDIF0_TX	O	Transmit
SPDIF1	8'd91	SPDIF1_RX	I	Receive
	8'd92	SPDIF1_TX	O	Transmit
UART0	8'd93	UART0_RTS	O	Request to send
	8'd94	UART0_CTS	I	Clear to send
	8'd95	UART0_RX	I	Receive data input
	8'd96	UART0_TX	O	Transmit data output
UART1	8'd97	UART1_RX	I	Receive data input
	8'd98	UART1_TX	O	Transmit data output
UART2	8'd99	UART2_RX	I	Receive data input
	8'd100	UART2_TX	O	Transmit data output

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Function	GPIOx_FUNC_SEL Value	GPIO Function Name	I/O Type	Description
UART3	8'd101	UART3_RX	I	Receive data input
	8'd102	UART3_TX	O	Transmit data output
UART4	8'd103	UART4_RX	I	Receive data input
	8'd104	UART4_TX	O	Transmit data output

3.3.4 Pin Multiplexing of Analog Functions

Table 3-6 shows the pin multiplexing of analog functions. The I/O pins must be set to input pull-up or input pull-down to enable analog functions.

Table 3-6 Pin Multiplexing of Analog Functions

GPIO	Analog Functions			
	Analog1	Analog2	Analog3	Analog4
	X32K TOUCH	ACOMP	WKUP	AUX ADC AUX DAC
GPIO0		ACOMP_CH9 (I)	WKUP0 (I)	ADC12 (I)
GPIO1		ACOMP_CH10 (I)	WKUP1 (I)	ADC13 (I)
GPIO2			WKUP2 (I)	
GPIO3			WKUP3 (I)	
GPIO4			WKUP4 (I)	
GPIO5			WKUP5 (I)	
GPIO6			WKUP6 (I)	
GPIO7			WKUP7 (I)	
GPIO8	32K_XO (O)	ACOMP_CH8 (I)	WKUP8 (I)	ADC10 (I)
GPIO9	32K_XI (I)		WKUP9 (I)	
GPIO10			WKUP10 (I)	
GPIO11			WKUP11 (I)	
GPIO13		ACOMP_CH0 (I)	WKUP13 (I)	ADC15 (I)
GPIO14			WKUP14 (I)	

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GPIO	Analog Functions			
	Analog1	Analog2	Analog3	Analog4
	X32K TOUCH	ACOMP	WKUP	AUX ADC AUX DAC
GPIO15			WKUP15 (I)	
GPIO16			WKUP16 (I)	
GPIO17			WKUP17 (I)	
GPIO18			WKUP18 (I)	
GPIO19	TOUCH0 (I)		WKUP19 (I)	
GPIO20			WKUP20 (I)	
GPIO21		ACOMP_CH7 (I)	WKUP21 (I)	ADC6 (I)
GPIO22		ACOMP_CH6 (I)	WKUP22 (I)	ADC5 (I)
GPIO23		ACOMP_CH5 (I)	WKUP23 (I)	ADC3 (I)
GPIO24		ACOMP_CH4 (I)	WKUP24 (I)	ADC2 (I)
GPIO25		ACOMP_CH3 (I)	WKUP25 (I)	ADC1 (I)
GPIO26			WKUP26 (I)	
GPIO27			WKUP27 (I)	
GPIO28		ACOMP_CH2 (I)	WKUP28 (I)	ADC4 (I)
GPIO29			WKUP29 (I)	
GPIO30			WKUP30 (I)	
GPIO31			WKUP31 (I)	
GPIO32				
GPIO33				
GPIO34				
GPIO35				
GPIO36				
GPIO37				
GPIO38				DAC1 (O)
GPIO39				DAC0 (O)

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GPIO	Analog Functions			
	Analog1	Analog2	Analog3	Analog4
	X32K TOUCH	ACOMP	WKUP	AUX ADC AUX DAC
GPIO40				
GPIO41				
GPIO42				
GPIO43				
GPIO44				
GPIO45				
GPIO46				
GPIO47				
GPIO48				
GPIO49				
GPIO50				
GPIO51				
GPIO52				
GPIO53				
GPIO54				
GPIO55				
GPIO59				
GPIO60				
GPIO64				
GPIO65				
GPIO69				
GPIO70				
GPIO71				

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4 Functional Description

4.1 Wireless Processor (WP)

The wireless processor is based on dual 32-bit RISC cores, one operating at up to 480 MHz and the other at up to 240 MHz. The wireless processor features the following caches:

- Level 1 (L1) cache: 8 KB I-cache and 8 KB D-cache for each core
- 64 KB level 2 (L2) cache, frequency up to 240 MHz

4.2 Application Processor (AP)

The application processor includes:

- An Armv8.1-M Cortex-M52 core
- An Arm Ethos-U65 neural processing unit (NPU) that works in concert with an Armv8.1-M Cortex-M55 core

Both the Cortex-M52 and Cortex-M55 cores operate at up to 480 MHz.

The application processor features the following caches and TCM:

- Level 1 (L1) cache:
  - Cortex-M52 core: 16 KB I-cache and 16 KB D-cache
  - Cortex-M55 core: 16 KB I-cache and 16 KB D-cache
- 128 KB level 2 (L2) cache
- 64 KB TCM
  - When the wireless processor subsystem accesses it, the address range is 0x28200000–0x2820FFFF (NSC) and 0x38200000–0x3820FFFF (NS)
  - When the application processor subsystem, including the HPDMA (except the Cortex-M52 and Cortex-M55 cores), accesses it, the address range is 0x20000000–0x2000FFFF (NSC) and 0x30000000–0x3900FFFF (NS)

4.2.1 Cortex-M52 Processor

4.2.1.1 Cortex-M52 Processor Overview

The Cortex-M52 processor implements the Armv8.1-M mainline architecture. It supports computing applications, such as digital signal processing (DSP) and machine learning. The Cortex-M52 processor is energy efficient and

achieves high compute performance across scalar and vector operations while maintaining low power consumption.

4.2.1.2 Cortex-M52 Features

The Cortex-M52 processor implements the Armv8.1-M Mainline architecture and also supports previous Armv8-M architectural features.

Cortex-M52 Processor Architectural Features

- Arm PMSAv8 memory system architecture with memory protection
- Arm FPv5 hardware supporting scalar half- and single-precision floating-point operation that is compliant with IEEE 754-2008
- Digital Signal Processing (DSP) extension
- DSP debug extension
- Exception model
- External Implementation Defined Attribution Unit (IDAU)
- L1 cache: 16 KB I-cache and 16 KB D-cache
- Main extension
- Memory Protection Unit (MPU) with up to 16 non-secure regions and 16 secure regions
- Low Overhead Branch (LOB) extension
- Support for Data Independent Timing (DIT) operation
- Nested Vector Interrupt Controller (NVIC) with 64 interrupts and 8 levels of priority
- Reliability, Availability, and Serviceability (RAS) extension
- Security Attribution Unit (SAU) with up to 8 non-secure or non-secure callable memory regions
- Security extension, an implementation of Arm TrustZone
- Unprivileged Debug Extension (UDE)

Debug and Trace Features

- BreakPoint Unit (BPU) with 8 comparators
- Data Watchpoint and Trace (DWT) unit with 4 comparators
- Embedded Trace Macrocell (ETM)
- Performance Monitoring Unit (PMU)

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4.2.2 Cortex-M55 Processor

4.2.2.1 Cortex-M55 Processor Overview

The Cortex-M55 processor is a processor that implements the Armv8.1-M Mainline architecture which includes support for the M-profile vector extension (MVE, also referred to as Arm Helium™ technology). The processor also supports previous Armv8-M architectural features.

The design is focused on compute applications such as digital signal processing and machine learning. The Cortex-M55 processor is energy efficient and achieves high compute performance across scalar and vector operations while maintaining low power consumption.

4.2.2.2 Cortex-M55 Features

The Cortex-M55 processor implements the Armv8.1-M Mainline architecture and also supports previous Armv8-M architectural features.

Cortex-M55 Processor Architectural Features

- Arm PMSAv8 memory system architecture with memory protection
- Arm FPv5 hardware supporting scalar half, single, and double-precision floating-point operation that is compliant with IEEE 754-2008
- Digital Signal Processing (DSP) extension
- DSP debug extension
- Exception model
- External Implementation Defined Attribution Unit (IDAU)
- L1 cache: 16 KB I-cache and 16 KB D-cache
- Main extension
- Memory Protection Unit (MPU) with up to 16 non-secure regions and 16 secure regions
- MVE, supporting Single Instruction Multiple Data (SIMD) 128-bit vector operations
- Support for Data Independent Timing (DIT) operation
- Nested Vector Interrupt Controller (NVIC) with 64 interrupts and 8 levels of priority
- Reliability, Availability, and Serviceability (RAS) extension
- Security Attribution Unit (SAU) with up to 8 non-secure or non-secure callable memory regions
- Security extension, an implementation of Arm TrustZone
- Unprivileged Debug Extension (UDE)

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Debug and Trace Features

- BreakPoint Unit (BPU) with 8 comparators
- Data Watchpoint and Trace (DWT) unit with 4 comparators
- Embedded Trace Macrocell (ETM)
- Performance Monitoring Unit (PMU)

4.2.3 Ethos-U65 NPU

The Neural Processing Unit (NPU) improves the inference performance of neural networks. The NPU targets quantized Convolutional Neural Networks (CNN) and 8-bit and 16-bit integer Recurrent Neural Networks (RNN). The NPU supports 8-bit weights.

At 480 MHz, the NPU achieves a theoretical performance of 0.25 TOPS (8-bit integer).

The NPU utilizes TensorFlow Lite Micro as its primary neural framework, supported by a comprehensive suite of software components including the TensorFlow Lite Micro Runtime, CMSIS-NN, Optimizer, and Driver.

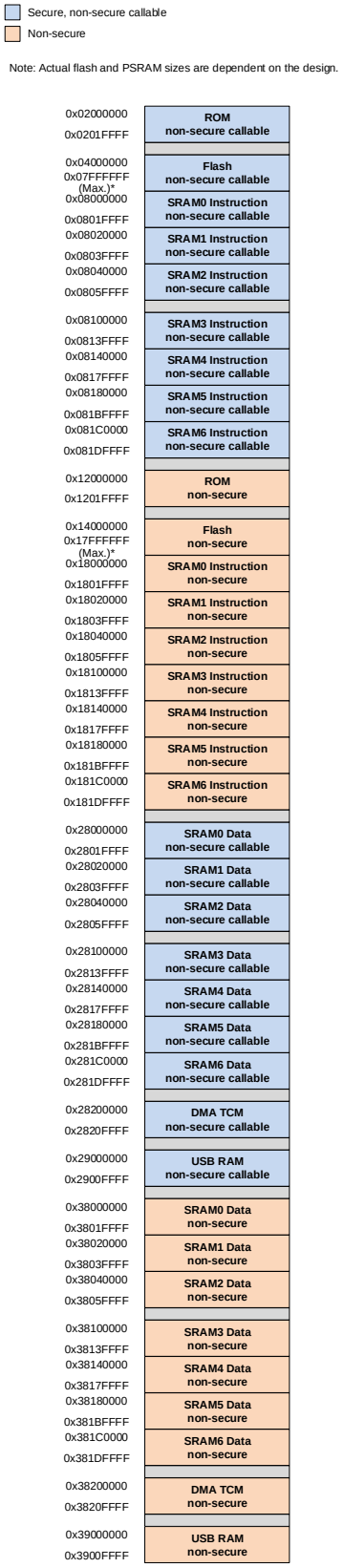
4.3 Memory Organization

When the TrustZone is enabled, the SAU and IDAU define the access permissions based on secure and non-secure states. According to IDAU security attribution, all memory spaces of the BK7259 are mapped twice for secure and non-secure states.

Both instruction and data are in little endian format.

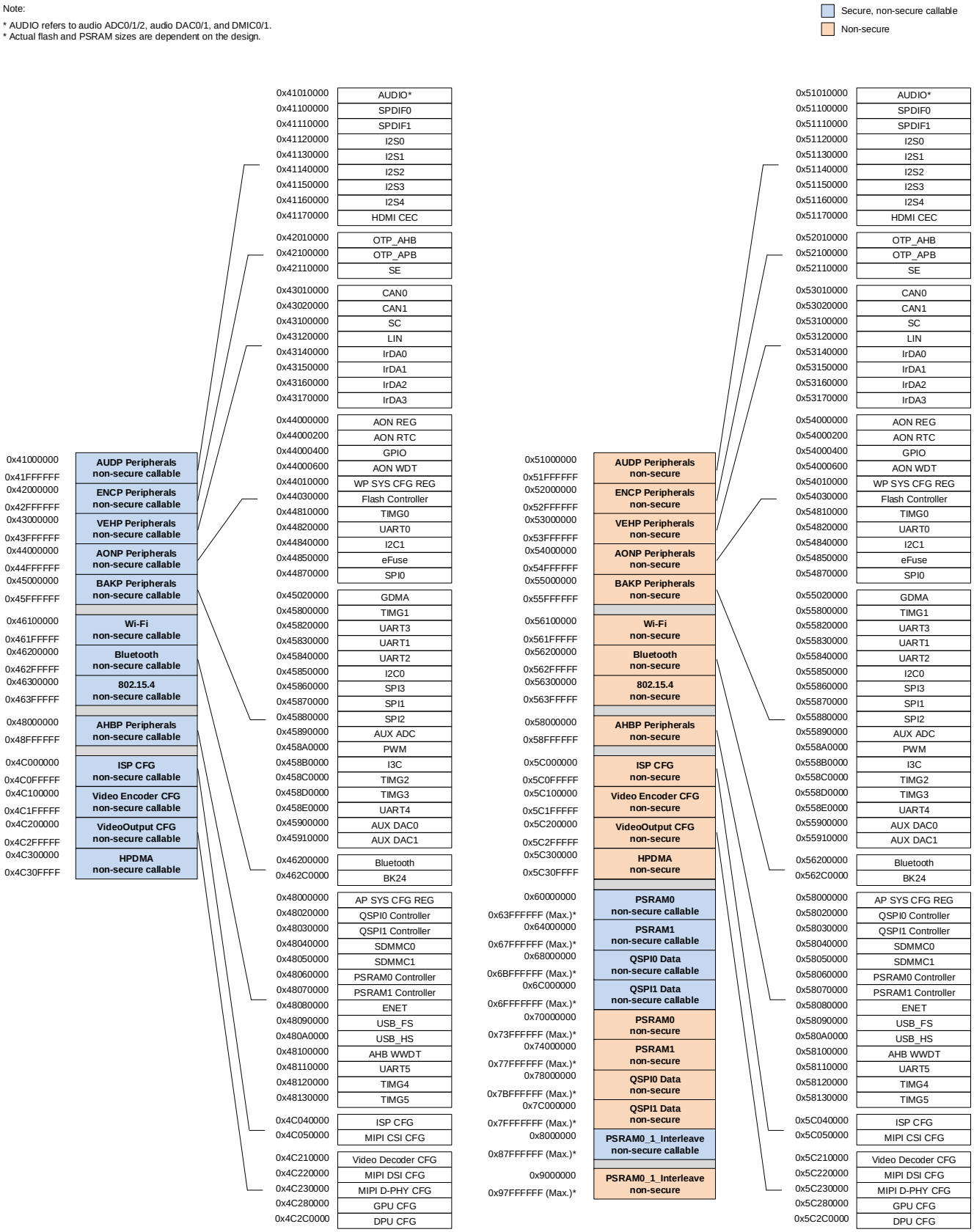
Figure 4-1 illustrates the memory map based on IDAU mapping.

Figure 4-1 Memory Map Based on IDAU Mapping



Note:

* AUDIO refers to audio ADC0/1/2, audio DAC0/1, and DMIC0/1.
* Actual flash and PSRAM sizes are dependent on the design.



The internal memory of the BK7259 refers to the memory integrated onto the chip die or co-packaged within the chip, including:

- 128 KB ROM
- 1280 KB share SRAM
  - 128 KB SRAM0
  - 128 KB SRAM1
  - 128 KB SRAM2
  - 256 KB SRAM3
  - 256 KB SRAM4
  - 256 KB SRAM5
  - 128 KB SRAM6
- 64 KB TCM
- 64 KB USB RAM dedicated to high-speed USB
- 4 KB OTP, with 3 KB available for customer applications
- 32-bit eFuse
- SiP flash: up to 16 MB, dependent on the package configuration
- SiP PSRAM: up to 64 MB (2 x 32 MB) in total, dependent on the package configuration
  - PSRAM0 and PSRAM1 can be accessed separately.
  - PSRAM0 and PSRAM1 can be accessed as a whole PSRAM.
  - The BK7259 supports PSRAM low-power mode, which allows for data retention.

The RISC cores, Cortex-M52 core, and Cortex-M55 core have direct access to these internal memories.

The RISC cores, Cortex-M52 core, and Cortex-M55 core can access internal SRAMs (SRAM0, SRAM1, SRAM2, SRAM3, SRAM4, SRAM5, and SRAM6) via both the instruction bus and data bus with bytes, half-words, or full words.

4.4 RF Transceiver

The BK7259 integrates a high-performance Wi-Fi/Bluetooth/802.15.4/2.4 GHz proprietary transceiver, along with a dedicated low-power Bluetooth transmitter. The Bluetooth transmitter features a polar power amplifier (PA), enabling low-power Bluetooth operation. The transceiver integrates on-chip baluns. On the receive side, the balun converts the single-ended RF signal from the antenna into a differential signal and the low noise amplifier (LNA) amplifies the differential signal to achieve a better noise and linearity trade-off. On the transmit side, the PA amplifies the differential signal and the balun converts the differential signal to a single-ended signal for feeding

the antenna. This enables transmit and receive operations with only one ANT pin connected to the antenna. The frequency synthesizer is fully integrated, eliminating the need for any external components.

4.5 Wi-Fi Subsystem

4.5.1 Wi-Fi Baseband

The BK7259 Wi-Fi baseband supports the following features:

- Standards and compatibility
  - 802.11ax (HE), backward compatible with 802.11n (HT), 802.11g/b
- Channel bandwidths
  - 20 MHz and 40 MHz
- Modulation and coding
  - 802.11ax: MCS 0–9 (up to 256-QAM)
  - 802.11n: MCS 0–7 (up to 64-QAM)
  - Legacy: BPSK, QPSK, 16-QAM, 64-QAM
  - Dual Carrier Modulation (DCM)
  - BCC coding
  - LDPC coding for Forward Error Correction
- Frame formats
  - HE_SU, HE_MU, HE_ER_SU, HE_TB
  - HT (n), OFDM (g), DSSS/CCK (b)
  - Channel State Information (CSI) feedback
- Multi-user features
  - Uplink MU-MIMO and OFDMA TX (STA)
  - Downlink MU-MIMO and OFDMA RX (STA)
- Performance enhancements
  - Guard intervals: 0.8 μs, 1.6 μs, 3.2 μs (HE-LTF variants)
  - Short GI (0.4 μs, legacy n)
  - RIFS (Reduced Inter-Frame Spacing, 802.11n)
  - Aggressive packet extension
  - Extended Range (ER SU)

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- Optional PHY features
  - 20 MHz coexistence with mid-packet detection (enhanced CCA)
  - Receiver beam change
- Spectral intelligence
  - Spectrum monitoring
  - Interference identification/classification
- Power efficiency
  - Legacy PS, U-APSD, TWT
  - Dynamic bandwidth management

4.5.2 Wi-Fi MAC

The BK7259 Wi-Fi MAC supports the following features:

- Core MAC functions
  - CSMA/CA with backoff, NAV, dual-NAV (HE dual-NAV)
  - Fragmentation/reassembly
  - Block Acknowledgment (BA): M-BA, C-BA
  - QoS support (802.11e/WMM)
  - Power save modes: legacy PS, U-APSD, Target Wake Time (TWT)
  - Management Information Base (MIB) counters
  - Agile Multiband (MBO) and Optimized Connectivity Experience (OCE)
- Frame types and control
  - Trigger frames (basic, MU-BAR, MU-RTS, BFRP, BSRP, padding)
  - HE variants of HT Control (UL Power Headroom, Receive Operation Mode)
  - Operating Mode Indication (OMI)
  - Buffer Status Reports (BSR)
  - MU frame exchange sequences
  - MU acknowledgment (ACK)
  - A-MPDU aggregation/de-aggregation (including single-MPDU A-MPDU)
- Multi-user and scheduling
  - MU-MIMO MAC (UL/DL)
  - OFDMA scheduling (UL/DL)
  - Target Wake Time scheduling

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  - Multiple BSS/Station operation
- Security and management
  - WPA2/WPA3 (802.11i/802.11w)
  - Protected Management Frames (PMF)
  - Association, authentication, roaming support

4.5.3 Wi-Fi Radio

The BK7259 Wi-Fi radio supports the following features:

- 802.11ax 1x1 operation in 2.4 GHz band
- Integrated 2.4 GHz PAs and LNA
- Integrated 2.4 GHz T/R switch
- Integrated 2.4 GHz baluns
- Supports both internal and external PAs
- Adjustable TX power
- Antenna diversity

4.6 Bluetooth Subsystem

4.6.1 Bluetooth Baseband

The Bluetooth baseband supports the features below:

- Bluetooth Core Specification compliant, targeting BR/EDR/LE features
- BR/EDR features:
  - All ACL, CSB, SCO, and eSCO packet types (1, 3, and 5 slots packets), including EDR packets
  - Encryption/decryption (E0 and/or AES-CCM)
  - Bit stream processing (HEC, CRC, whitening, FEC 1/3, FEC 2/3)
  - Hopping frequency calculation (1600 and 3200 hops/s)
  - Adaptive Frequency Hopping (AFH)
  - Bluetooth clock and offsets
  - TDMA/TDD frames formatting and synchronization
  - Audio path including CVSD, a/μ-Law, PCM codecs, and VoHCI support
- Bluetooth LE Features:

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- All packet types (Broadcasting/Advertising/Data/Control)
- Advertising extensions
- Isochronous channel/audio over Bluetooth LE operations support
- Encryption/decryption (AES-CCM)
- Bit stream processing (CRC, whitening)
- Frequency hopping calculation
- FDMA/TDMA/events formatting and synchronization
- All device roles support (Broadcaster, Central, Observer, Peripheral)

4.6.2 Bluetooth Interfaces

- UART HCI interface, up to 10 Mbps
- PCM/I2S audio interface

4.6.3 Bluetooth Link Controller

The Bluetooth link controller has the following features:

- Classic Bluetooth
  - Device discovery (inquiry and inquiry scan)
  - Connection establishment (page and page scan)
  - Up to 7 connections as master
  - Up to 3 connections as slave
  - Asynchronous data reception and transmission
  - Up to 2 synchronous links (SCO/eSCO)
  - Master/slave switch
  - Adaptive Frequency Hopping (AFH)
  - Adaptive frequency hopping and channel assessment
  - Data broadcasting
  - Authentication and encryption
  - Secure simple pairing
  - Multi-point and scatternet management
  - Sniff mode
  - Connectionless slave broadcast (transmitter and receiver)

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- Bluetooth LE
  - Advertising
  - Scanning
  - Up to 10 connections
  - Asynchronous data reception and transmission
  - Adaptive frequency hopping and channel assessment
  - Encryption
  - Connection parameter update
  - Data length extension up to 251 bytes
  - Privacy 1.2
  - LE ping

4.7 IEEE 802.15.4 PHY and MAC

The BK7259 implements IEEE 802.15.4 specification and supports multiple PHY and MAC functionalities.

The 802.15.4 PHY supports Energy detection(ED), Link quality indication(LQI), channel selection, clear channel assessment (CCA), and transmitting as well as receiving packets across the physical medium.

The 802.15.4 MAC supports beacon management, channel access, GTS management, frame validation, acknowledged frame delivery, association, and disassociation. In addition, the MAC sublayer provides hooks for implementing application-appropriate security mechanisms.

802.15.4 Personal Area Networks (PANs) are of two kinds, Beacon Enabled (Slotted) and Non-Beacon Enabled (Unslotted).

In Beacon Enabled PAN, channel time is defined by superframe structure. Superframe have an active and inactive portion, bounded by beacon transmission. Active portion is divided into aNumSuperframeSlots (=16) equally spaced slots and composed of a Contention Access Period (CAP) and a Contention Free Period (CFP). Beacon frame is transmitted as part of the first slot in CAP. GTS slots are located within CFP. Beacon transmission and GTS transmission/reception happens at slot boundary. PAN coordinator only transmits beacon frame. Device only need to receive/track beacon frame. Coordinator tracks beacon of its coordinator and transmits beacon frame for its network. MAC Command Frames and Data (not marked by Higher Layer to be transmitted only during GTS) are transmitted during CAP. For transmission during CAP, CSMA-CA channel access mechanism is used. Two types of CSMA-CA algorithm used. For the beacon enabled PAN, slotted version of CSMA-CA. For the non-beacon enabled PAN, un-slotted version of CSMA-CA.

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Non-beacon enabled PAN can be viewed as special case of beacon enabled PAN, where Beacon Interval (BI) is infinite, there is no CFP (GTS) and no inactive portion. Entire period is CAP. And during this CAP, un-slotted version of CSMA-CA is used. Device transmits Command Frame or Data to its coordinator during CAP. In beacon enabled PAN, Coordinator informs the availability of Command Frame or Data in its Beacon and devices requests for the Data based on this information. In non-beacon enabled PAN, higher layer will initiate MLME-POLL request for devices to send data request to coordinator and see if the ACK response suggests data available for the device, based on the Frame Pending field.

Higher layer sets up a PAN, by configuring a device as PAN coordinator. Following that for any device to participate in the PAN operation, need to Associate with one of the coordinator in the PAN. Association is applicable in both beacon enabled and non-beacon enabled PAN.

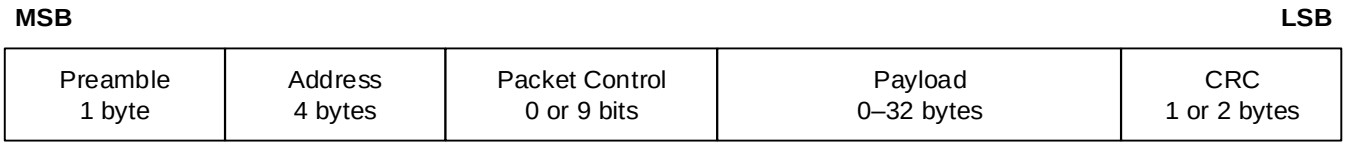
4.8 2.4 GHz Proprietary Baseband

Beken's 2.4 GHz proprietary baseband supports the following features:

- 1 Mbps air data rate
- GFSK modulation
- Automatic packet handling
- Auto acknowledgement
- Auto retransmission

The 2.4 GHz proprietary packet contains a preamble field, address field, packet control field, payload field, and a CRC field. Figure 4-2 shows the packet format with MSB to the left.

Figure 4-2 2.4 GHz Proprietary Packet Format



4.9 FEM Support

The BK7259 supports the use of an external Front-End Module (FEM). The external FEM is controlled by three signals WIFI_TX_EN, WIFI_RX_EN, and FEM_LNA_EN.

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Table 4-1 FEM Operating Modes

Operating Mode	WIFI_TX_EN (GPIO26)	WIFI_RX_EN (GPIO28)	FEM_LNA_EN (GPIO27)
Transmit mode	1	0	0
Receive mode	0	1	1
LNA bypass mode	0	1	0
	1	1	X
Standby mode	0	0	0
Not supported	All other states		

4.10 Clock Management

The primary clock sources available in the BK7259 are as follows:

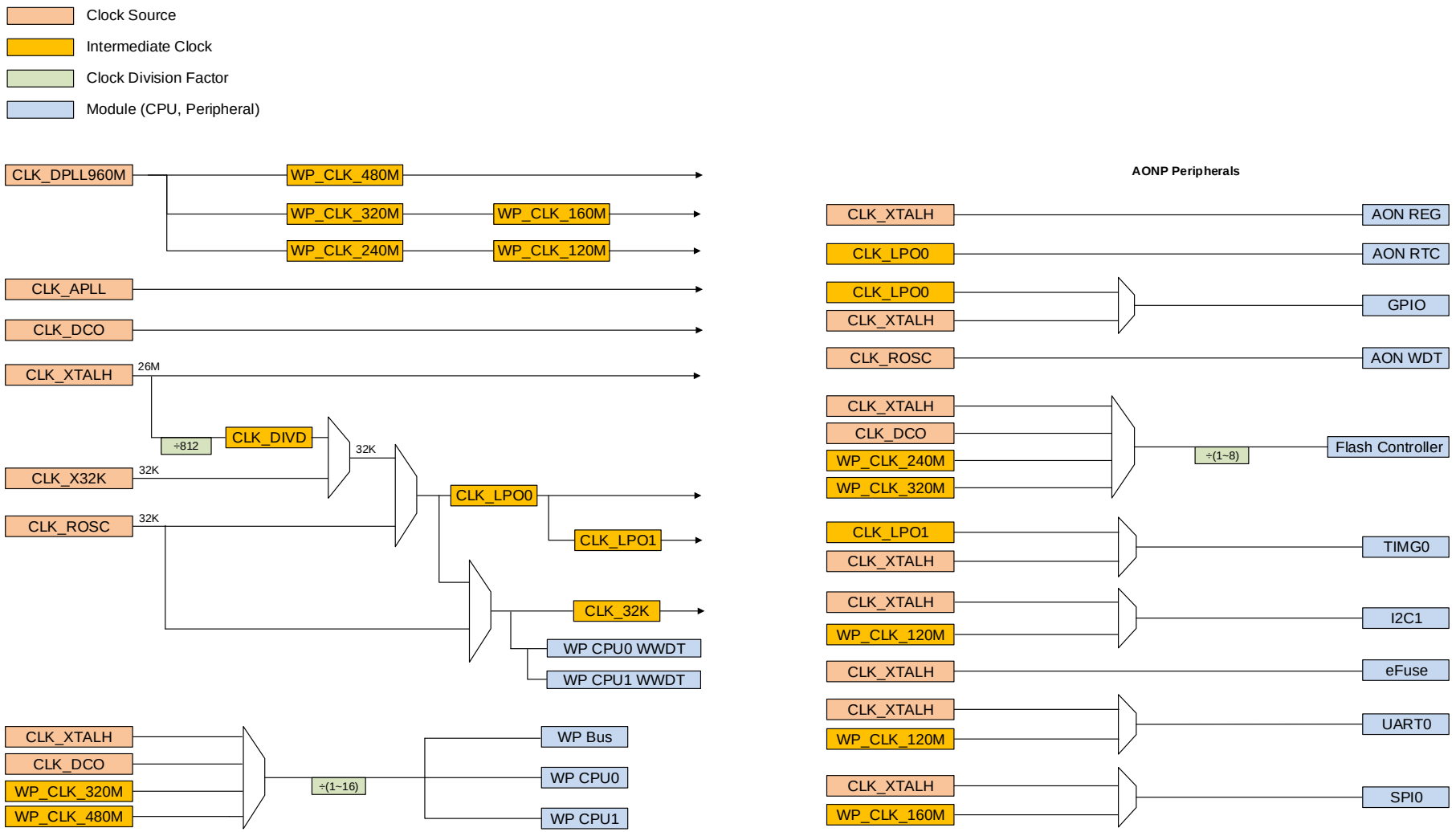
- High-frequency clocks
  - 26 MHz crystal oscillator: it outputs clock signal CLK_XTALH
  - 160–640 MHz internal digitally controlled oscillator (DCO): it outputs clock signal CLK_DCO
  - 960 MHz digital PLL (DPLL): it outputs clock signal CLK_DPLL960M
- Low-frequency clocks
  - 32 kHz (32.768 kHz) crystal oscillator: it outputs clock signal CLK_X32K
  - 32 kHz internal ring oscillator (ROSC): it outputs clock signal CLK_ROSC
- Audio clock
  - 90–99 MHz audio PLL (APLL): it outputs clock signal CLK_APLL

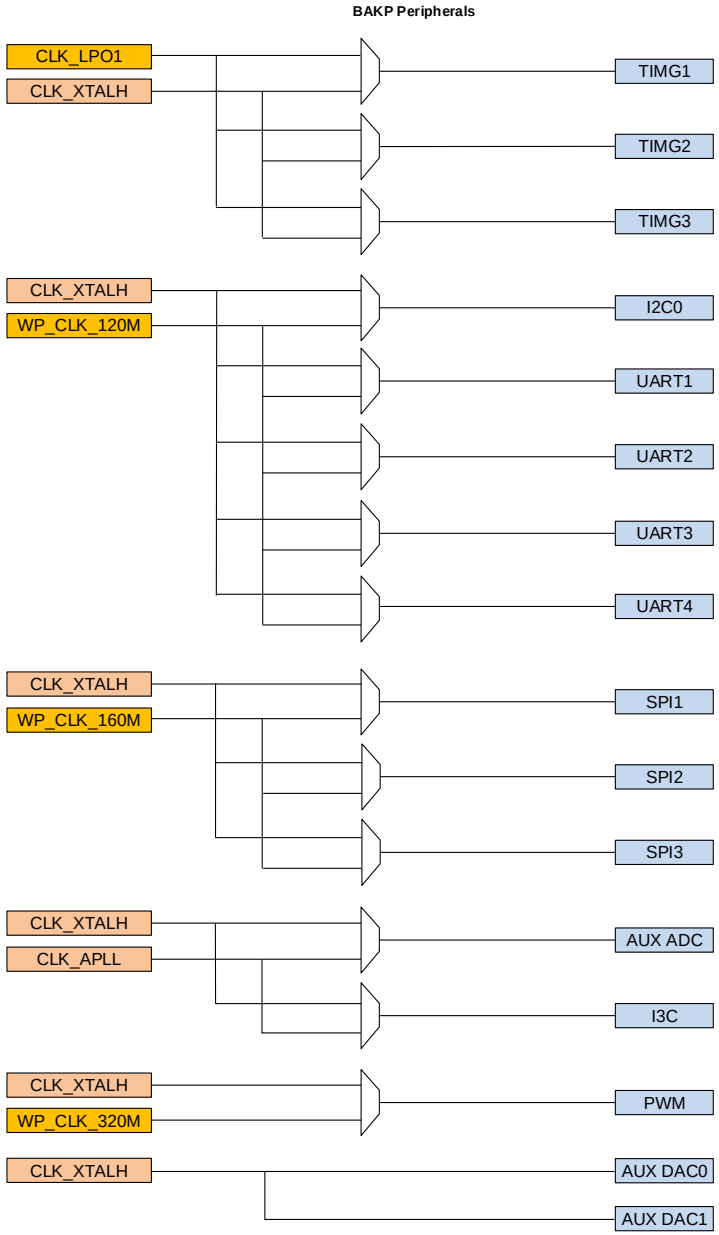
The system generates three low-power clock sources for standby: CLK_LPO0, CLK_LPO1, and CLK_32K. The three clocks can be selected from CLK_DIVD, CLK_X32K, or CLK_ROSC.

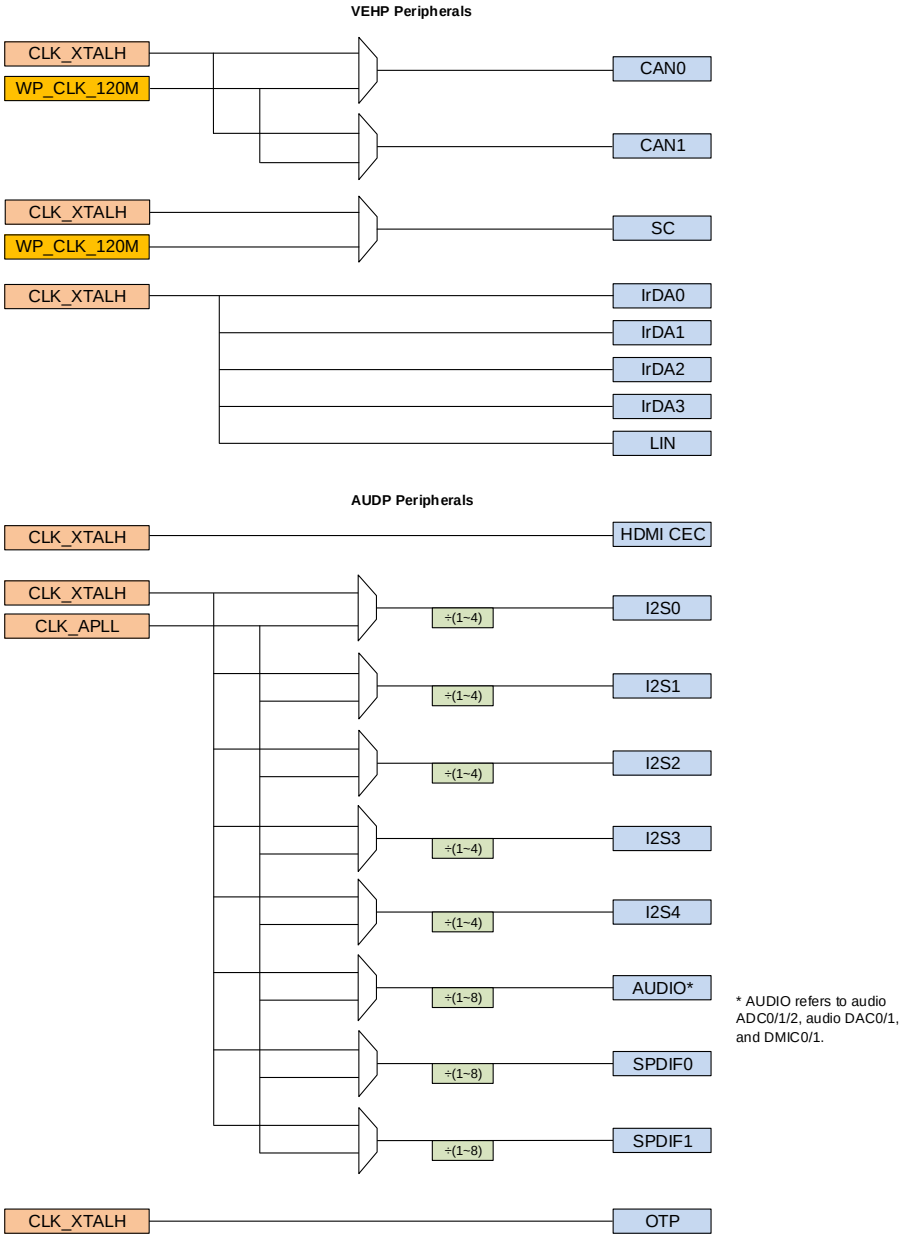
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Figure 4-3 shows the clock tree for the wireless processor subsystem.

Figure 4-3 Clock Tree for Wireless Processor Subsystem







* AUDIO refers to audio ADC0/1/2, audio DAC0/1, and DMIC0/1.

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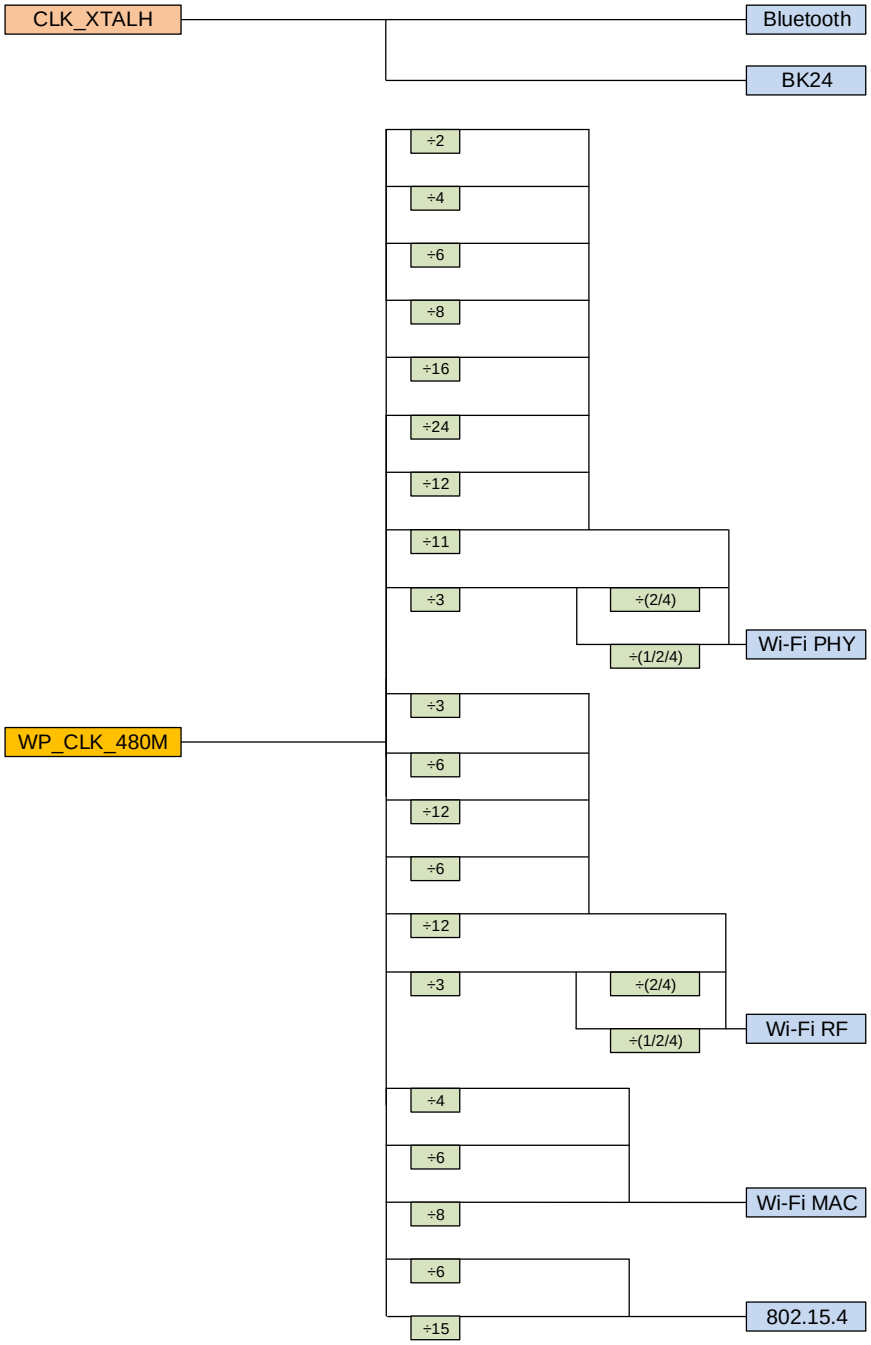
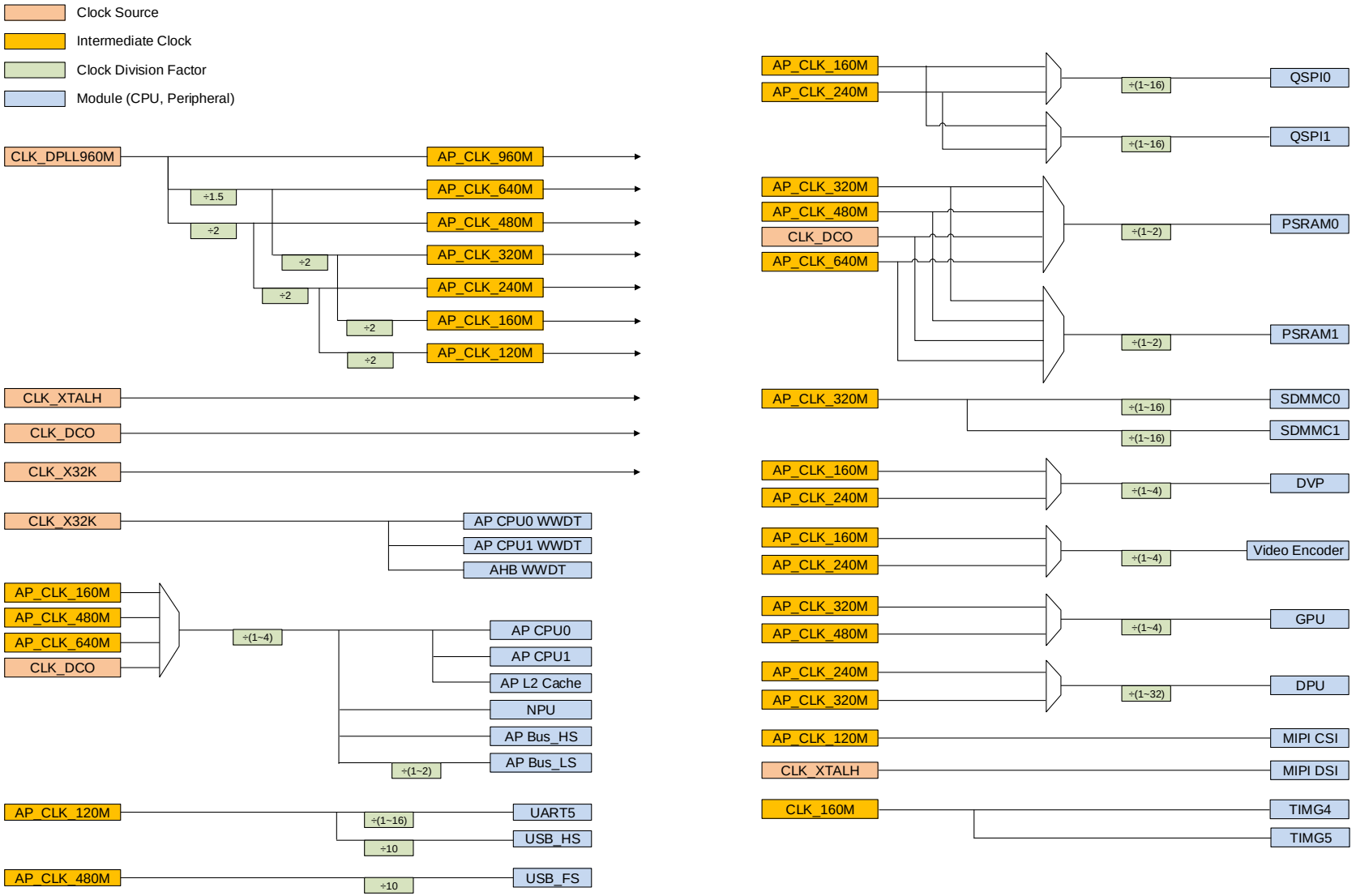

CLK_XTALH
Bluetooth
BK24
÷2
÷4
÷6
÷8
÷16
÷24
÷12
÷11
÷3
÷(2/4)
÷(1/2/4)
Wi-Fi PHY
WP_CLK_480M
÷3
÷6
÷12
÷6
÷12
÷3
÷(2/4)
÷(1/2/4)
Wi-Fi RF
÷4
÷6
÷8
Wi-Fi MAC
÷6
÷15
802.15.4

Figure 4-4 shows the clock tree for the application processor subsystem.

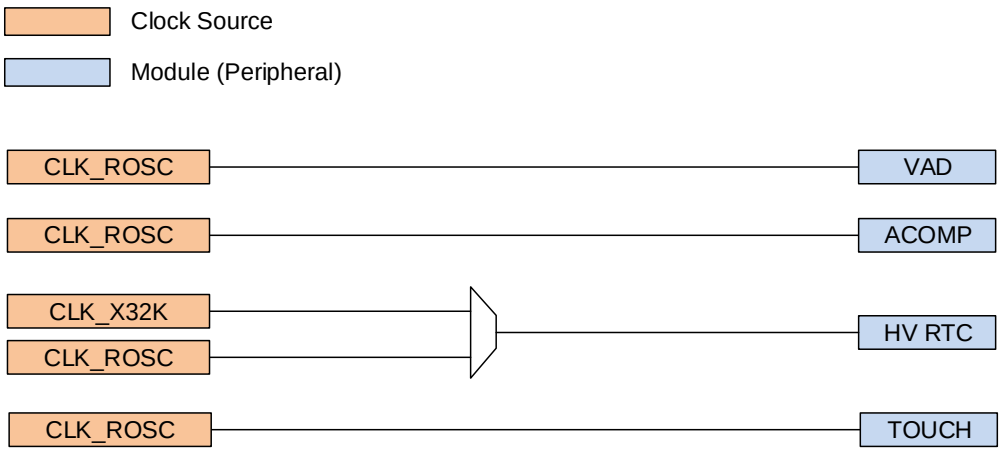
Figure 4-4 Clock Tree for Application Processor Subsystem



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Figure 4-5 shows the clock tree for analog peripherals.

Figure 4-5 Clock Tree for Analog Peripherals



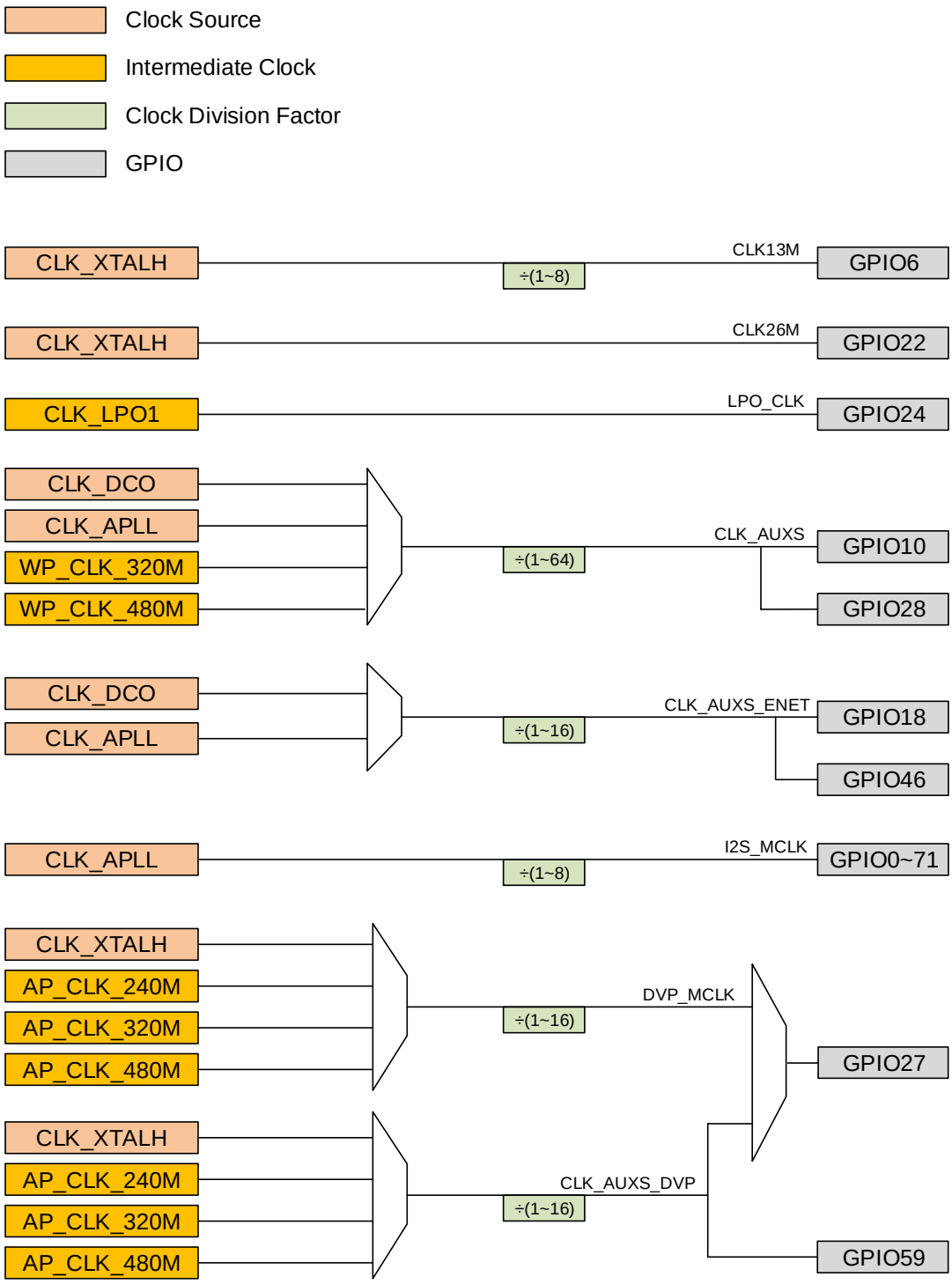
The BK7259 also has a clock output capability, which allows clock signals to be output to external components through GPIOs. GPIOs can output the following clock signals:

- CLK13M: clock derived from 26 MHz CLK_XTALH (division factor 1 to 8)
- CLK26M: clock derived from 26 MHz CLK_XTALH
- LPO_CLK: clock derived from 32 kHz CLK_LPO1 clock
- CLK_AUXS: reference clock, derived from CLK_DCO/CLK_APLL/WP_CLK_320M/WP_CLK_480M (division factor 1 to 64)
- CLK_AUXS_ENET: reference clock for external Ethernet PHY, derived from CLK_DCO/CLK_APLL (division factor 1 to 16)
- I2S_MCLK: reference clock for external audio codec, derived from CLK_APLL (division factor 1 to 8)
- DVP_MCLK: reference clock for external DVP sensor, derived from CLK_XTALH/AP_CLK_240M/ AP_CLK_320M/AP_CLK_480M (division factor 1 to 16)
- CLK_AUXS_DVP: reference clock for external DVP sensor, derived from CLK_XTALH/AP_CLK_240M/ AP_CLK_320M/AP_CLK_480M (division factor 1 to 16)

Figure 4-6 shows the clock tree for clock output.

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Figure 4-6 Clock Tree for Clock Output



4.11 Reset

A reset can be triggered by the following sources: power-on reset, brown-out reset, watchdog reset, and wake-up from shutdown, deep sleep0, or deep sleep1 mode.

Power-on reset, brown-out reset, and AON WDT reset reset the whole chip to its initial state. The CPU core reset (CPU WWDT reset) resets each respective CPU core. The scope of the AHB WWDT reset is configurable, allowing for the reset of either the entire application processor subsystem or specific modules within it.

Wake-up from shutdown or deep sleep0 mode triggers the whole system reset, while wake-up from deep sleep1 mode triggers the reset of digital blocks.

4.12 Power Management

4.12.1 Power Scheme

The power management system on the BK7259 includes three buck converters and several internal LDO regulators to provide voltage and noise isolation to various parts of the chip.

VBAT is the primary external supply for the chip, with a voltage range of 2.3 to 3.6 V. It serves as the input supply for internal buck converters and LDO regulators. The buck converters and LDO regulators generate the following main power supplies:

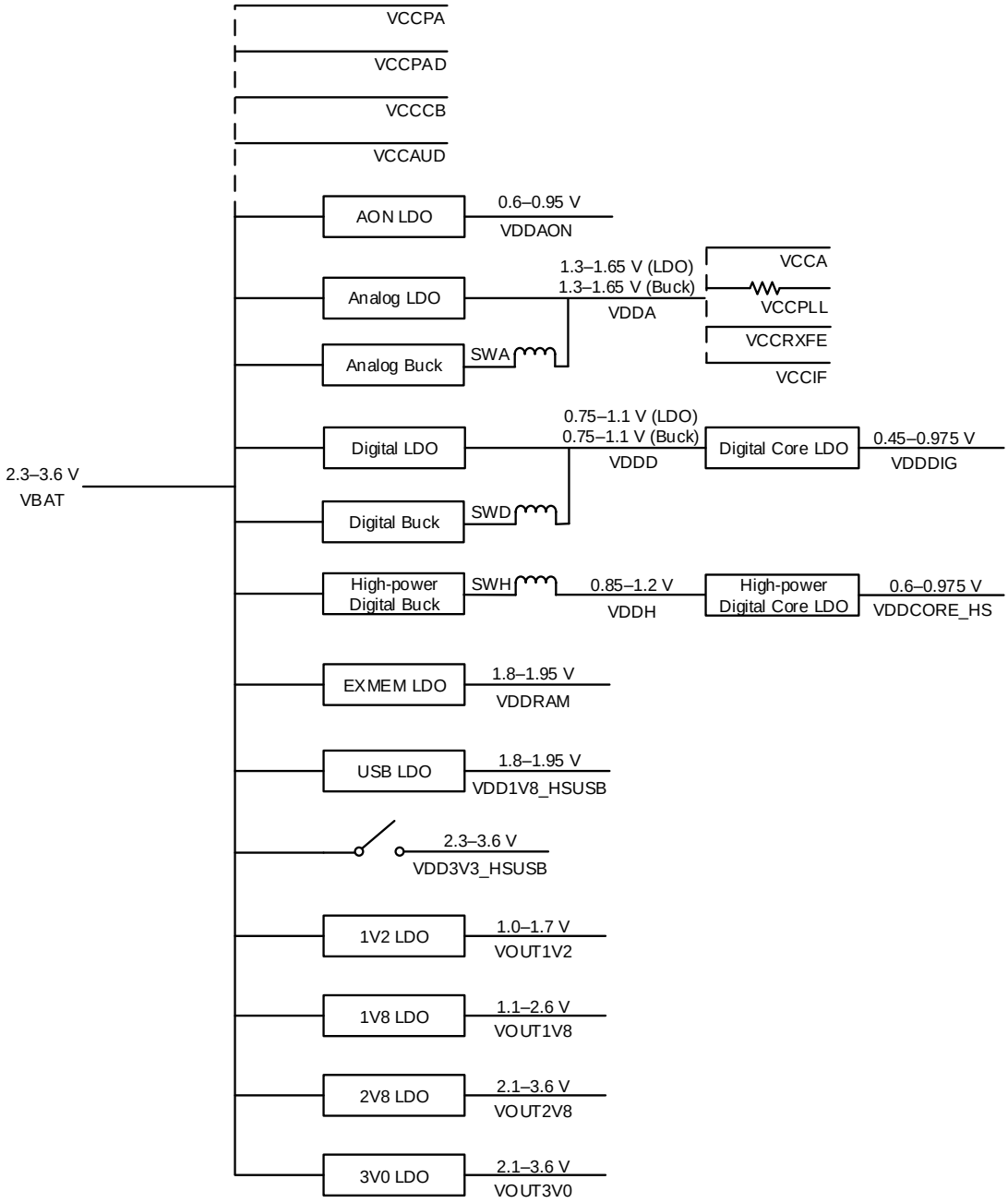
- VDDAON: power supply for AON power domain.
- VDDA: power supply for RF/analog blocks. It is externally connected to VCCA, VCCPLL, VCCRXFE, and VCCIF to supply power to the RF transceiver and Bluetooth transmitter. Internally, it provides power to XTALH, DCO, DPLL, APLL, and audio ADCs/DACs directly.
- VDDDIG: power supply for WP subsystem digital domain.
- VDDCORE_HS: power supply for AP subsystem digital domain.
- VDDRAM: power supply for PSRAM.
- VDD1V8_USB: power supply for high-speed USB.
- VOUT1V2/VOUT1V8/VOUT2V8/VOUT3V0: power supply for external components.

GPIO0–GPIO55 are supplied by VBAT, while GPIO56–GPIO71 are powered by an external supply via the VCCGPIO pin.

Figure 4-7 shows the power distribution of the BK7259.

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Figure 4-7 Internal Power Distribution



Note: Outputs from the buck converters and LDO regulators require proper bypass capacitors to reduce supply noise. Please refer to the hardware schematic for details on selecting bypass capacitors.

Figure 4-8 shows the power-up sequence of the BK7259.

Figure 4-8 BK7259 Power-up Sequence

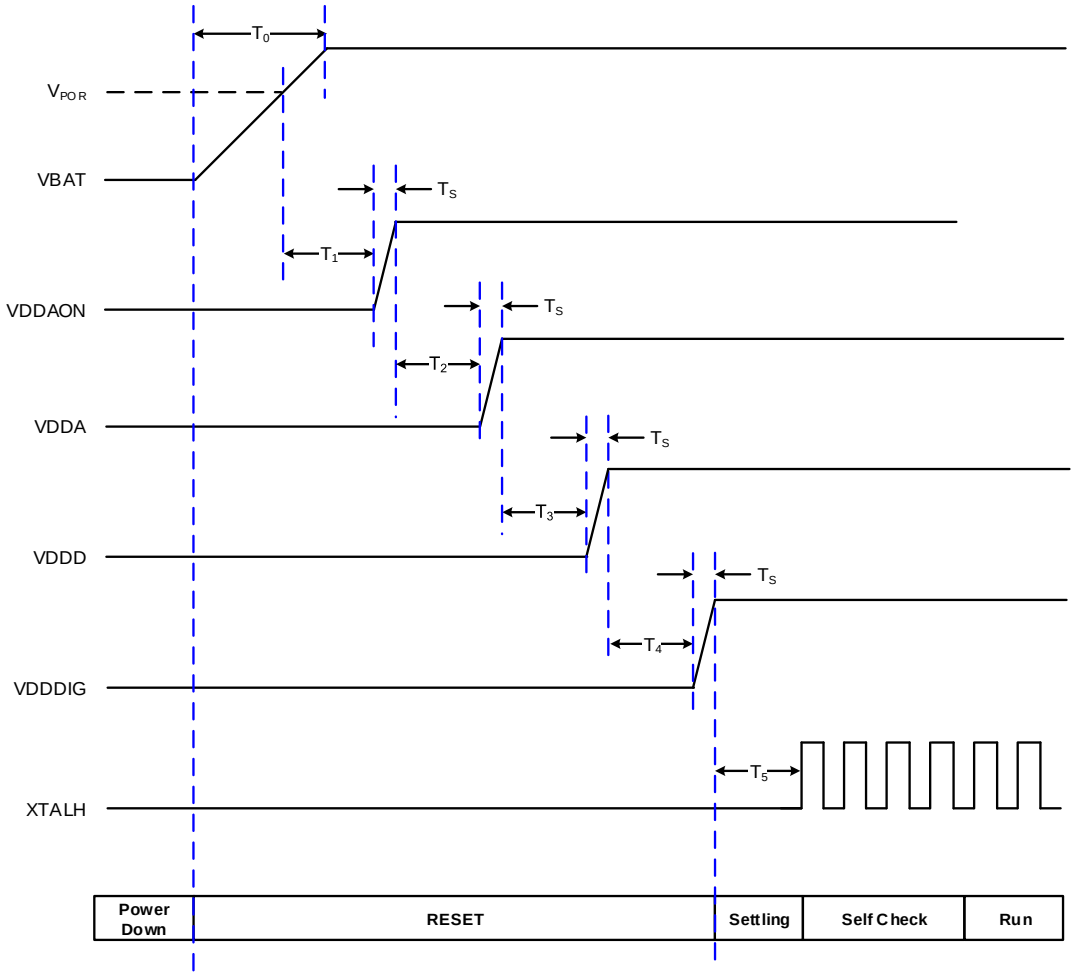


Table 4-2 Timing Parameters of Power-up Sequence

Parameter	Description	Min.	Typ.	Max.	Unit
V_{POR}	VBAT POR threshold	-	TBD	-	V
T_0	VBAT power-up time	TBD	-	-	μs
T_1	AON LDO ready time	-	TBD	-	ms
T_2	Analog buck/LDO ready time	-	TBD	-	μs
T_3	Digital buck/LDO ready time	-	TBD	-	μs
T_4	Digital core LDO ready time	-	TBD	-	μs
T_5	XTALH settle time	TBD	-	-	μs
T_S	LDO settle time	TBD	-	-	μs

Parameter	Description	Min.	Typ.	Max.	Unit
SR_{VBAT_RISE}	VBAT power-up rise slew rate	TBD	-	-	mV/ms

Figure 4-9 shows the power-down sequence of the BK7259.

Figure 4-9 BK7259 Power-down Sequence

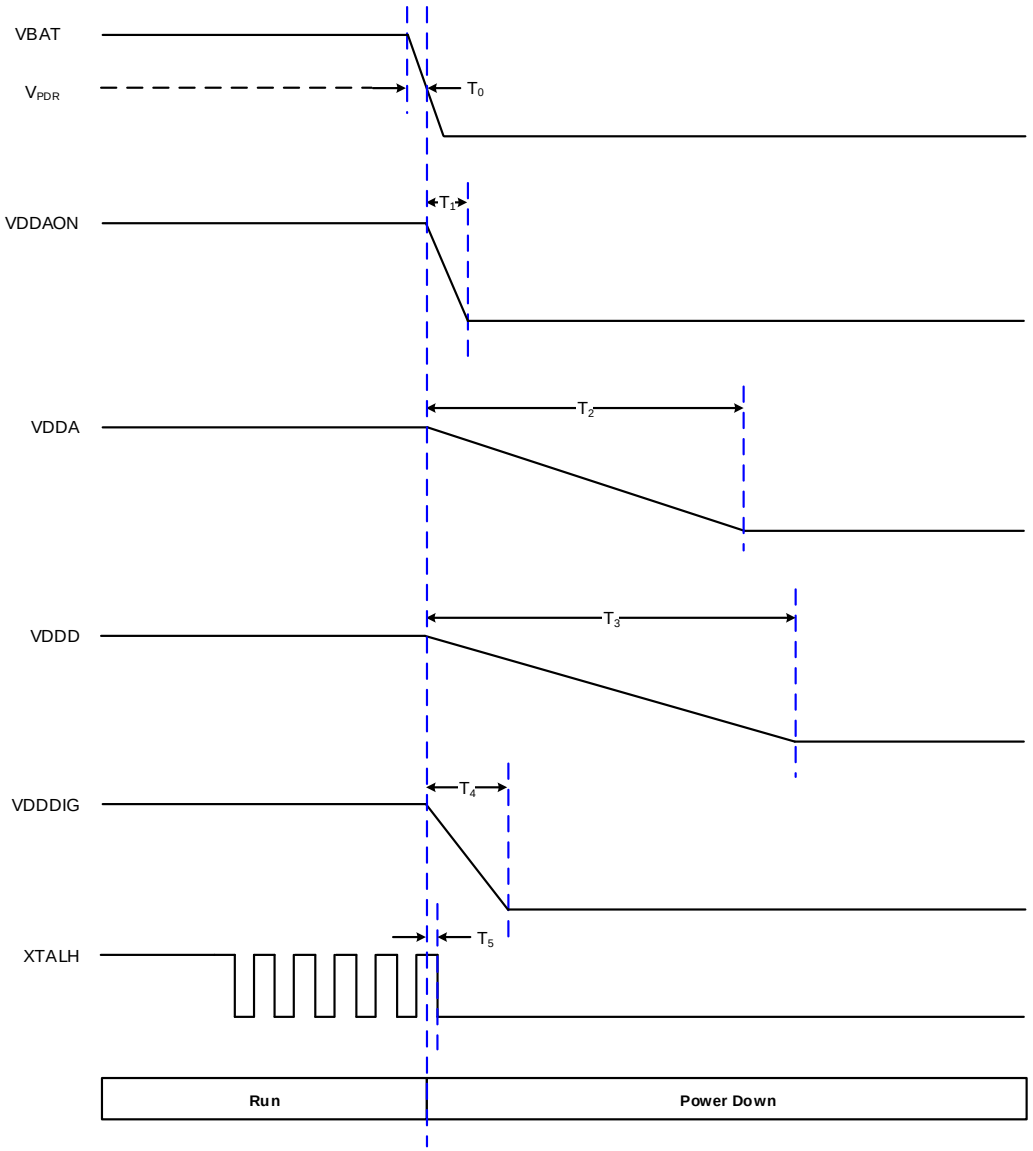


Table 4-3 Timing Parameters of Power-down Sequence

Parameter	Description	Min.	Typ.	Max.	Unit
V_{PDR}	VBAT PDR threshold	-	TBD	-	V

Parameter	Description	Min.	Typ.	Max.	Unit
T_0	VBAT power-down time	-	TBD	-	μs
T_1	AON LDO power-down time	-	TBD	-	μs
T_2	Analog buck/LDO power-down time	-	TBD	-	ms
T_3	Digital buck/LDO power-down time	-	TBD	-	ms
T_4	Digital core LDO power-down time	-	TBD	-	ms
T_5	XTALH power-down time	-	TBD	-	μs
SR_{VBAT_FALL}	VBAT power-down fall slew rate	TBD	-	-	mV/ms

4.12.2 Power Supply Supervisor

The BK7259 integrates a power-on reset (POR)/power-down reset (PDR)/brown-out detector (BOD) circuit. The POR, PDR, and BOD all monitor VBAT power supply.

Power-on Reset (POR)

The POR supervisor monitors VBAT power supply and compares it to a fixed threshold. The system remains in reset mode when VBAT is below this threshold.

Power-down Reset (PDR)

The PDR supervisor monitors VBAT power supply. A reset is generated when VBAT drops below a fixed threshold.

Brown-out Detector (BOD)

The BOD supervisor monitors VBAT power supply and compares it to programmable threshold levels. The system remains in reset mode when VBAT drops below the selected threshold. During reset:

- All circuits are powered down except the ROSC and high-voltage RTC (HV RTC).
- The HV RTC provides continuous timekeeping.

4.12.3 Power Modes

The BK7259 supports four low-power modes except active mode, namely shutdown mode, deep sleep0 mode, deep sleep1 mode, and sleep mode, among which the shutdown mode has the lowest power consumption.

Shutdown Mode: All circuits are turned off. A high level on the CEN pin will bring the system to active mode.

Deep Sleep0 Mode: All circuits are powered down except the HV RTC. An edge transition on GPIO0–31 (WKUP), an HV RTC event, or a touch sensor event can power up the system again. The 32-bit retention register retains its contents.

Deep Sleep1 Mode: All circuits are powered down except the AON power domain. A GPIO event, an AON RTC event, a VAD event, or a touch sensor event can power up the system again.

Sleep Mode: The MCU and all digital blocks stop their clocks. All digital blocks are in retention state. A GPIO event, an AON RTC event, a Wi-Fi MAC counter event, a Bluetooth MAC counter event, a VAD event, or a touch sensor event can bring the system back to active mode with normal voltage.

Active Mode: The MCU is active, and all peripherals are available.

4.13 I/O Multiplexer (IOMUX)

The BK7259 has an I/O multiplexer (IOMUX) that allows the use of I/O pins as either GPIOs or I/Os for normal peripherals. The GPIOx_FUNC_SEL field of the GPIO configuration register selects I/O functions:

- When GPIOx_FUNC_SEL is set to 0 to 3, the I/O is configured as a GPIO. The GPIO type can be configured as:
  - High impedance
  - Input
  - Output
  - Input/output
- When GPIOx_FUNC_SEL is set to 4 to 94 (except 63), the I/O is configured as an I/O of a normal peripheral. It's should be noted that multiple I/Os cannot be configured as the same peripheral I/O.

4.14 General-purpose I/Os (GPIO)

The BK7259 has up to 62 GPIOs, which can be configured as high-impedance, input, output, or input/output. All GPIOs are shared with peripheral alternate functions and/or analog functions. Section 3.3 Pin Multiplexing provides the mux functions of I/Os.

The main features of GPIOs include:

- Push-pull
- Internal pull-up/down resistors
- Configurable drive strength
- Alternate function

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- Analog function
- Interrupt generation:
  - High or low level
  - Rising or falling edge

4.15 SPI Interfaces (SPI)

The BK7259 integrates four SPI interfaces that can operate in master or slave mode. The SPI interfaces allow a clock frequency up to 40 MHz in both master and slave modes.

The SPI interfaces support the following features:

- 4-wire or 3-wire full-duplex synchronous communication
- Configurable 8-bit or 16-bit data width
- Programmable clock polarity and phase
- Programmable data order with MSB-first or LSB-first shifting
- Embedded 64-depth RX FIFO and 64-depth TX FIFO with DMA capability

4.16 Quad SPI Interfaces (QSPI)

The BK7259 embeds two Quad SPI interfaces that provide support for communicating with external flash, PSRAM, or AMOLED displays. The QSPI interface allows communicating up to 80 MHz.

The features of the QSPI interface are listed below:

- Single, dual, or quad SPI input/output
- Two functional modes: indirect mode and memory-mapped mode
- Fully programmable opcode and frame format
- Integrated RX FIFO and TX FIFO
- Supports 8, 16, and 32-bit data accesses

4.17 UART Interfaces (UART)

The BK7259 includes six Universal Asynchronous Receiver/Transmitter (UART) interfaces, which support full-duplex, asynchronous serial communication at a baud rate up to 10 Mbps.

UART0 supports flash download. When UART0 is used for flash download, UART0_RX and UART0_TX are mapped to GPIO10 as DL_UART_RX and GPIO11 as DL_UART_TX, respectively.

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The UART interface offers the features below:

- Configurable data length (5, 6, 7, or 8 bits)
- Even, odd, or none parity check
- Programmable stop bits (1 or 2 bits)
- Each UART embeds a 128-byte TX FIFO and a 128-byte RX FIFO. FIFO mode is disabled by default and can be enabled by software.
- Hardware flow control with RTS and CTS signals (UART0)
- Flash download (UART0)
- Programmable digital filter

4.18 Smart Card Controller (SC)

The BK7259 features a Smart Card controller (SC). The Smart Card controller is a communication controller that transmits data between the superior system and the Smart Card. It can perform a complete smart card session, including card activation, card deactivation, cold/warm reset, Answer to Reset (ATR) response reception, data transfers, etc.

The features of the Smart Card controller (SC) are listed below:

- Supports the ISO/IEC 7816-3:2006 and EMV 4.3 specifications
- Performs functions needed for complete smart card sessions, including:
  - Card activation and deactivation
  - Cold/warm reset
  - Answer to Reset (ATR) response reception
  - Data transfers to and from the card
- Extensive interrupt support system
- Adjustable clock rate and bit (baud) rate
- Configurable automatic byte repetition
- Handles commonly used communication protocols:
  - T=0 for asynchronous half-duplex character transmission, and
  - T=1 for asynchronous half-duplex block transmission
- Automatic convention detection
- Adjustable FIFOs for Receive and Transmit buffers (64 bytes) with threshold
- Configurable timing functions:
  - Smart card activation time

- Smart card reset time
- Guard time
- Timeout timers

4.19 SD and eMMC Host Controllers (SDMMC)

Two SD/SDIO and eMMC host controllers (SDMMC) are available on the BK7259. The SDMMC can be used as a host to communicate with external SD cards, SDIO cards, and eMMC devices.

The features of the SDMMC include the following:

- Embedded MultiMediaCard System Specification version 5.1 compliant
  - Card support for three data bus modes: 1-bit (default), 4-bit, and 8-bit
- SD memory card specifications version 4.20 compliant
  - SPI mode and UHS-II mode not supported
- SDIO card specification version 4.10 compliant
  - Card support for two data bus modes: 1-bit (default) and 4-bit
  - SPI mode and UHS-II mode not supported
- Supports SDMA/ADMA2/ADMA3 transfers, allowing high-speed transfer without CPU load
- Supports Command Queuing Engine (CQE) and compliant with eMMC CQ HCI
  - Programmable scheduler algorithm selection of task execution
  - Supports data prefetch for back-to-back WRITE operations
- Supports fully software driven tuning operations

4.20 I2C Interfaces (I2C)

I2C is a popular inter-IC interface that requires only two bus lines, the serial data line (SDA) and the serial clock line (SCL). The BK7259 embeds two I2C interfaces, which can operate in master or slave mode.

The features of the I2C interface are listed below:

- Master and slave modes
- Standard-mode (up to 100 kbps)
- Fast-mode (up to 400 kbps)
- Fast-mode Plus (up to 1 Mbps)
- 7-bit and 10-bit addressing
- Bus idle and SCL low timeout condition detection

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- Embedded 16-byte TX FIFO and 16-byte RX FIFO

4.21 I3C Interface (I3C)

Improved Inter-Integrated Circuit (I3C) is a two-wire, shared (multidrop), serial communication interface developed by the MIPI Alliance to enhance the legacy I2C interface. The BK7259 includes an I3C interface.

The I3C interface supports the following key features:

- I3C transmission modes:
  - Single data rate messaging (SDR):
    - Fm/Fm+
    - 11 MHz to 12.9 MHz (mixed fast bus)
    - Up to 12.9 MHz (pure bus)
  - High data rate messaging (HDR):
    - HDR-DDR
- Dynamic address assignment
- I3C slave generated requests:
  - In-band interrupt
  - Hot-join request
  - Mastership request
- I3C common command codes
- Time control:
  - Asynchronous Mode 0
- Legacy I2C device co-existence on the same bus
- Retaining registers
- Command queue
- Immediate commands
- Separate TX, RX, and IBI storage elements
- MCTP flow control
- General-purpose inputs/outputs accessible tough I3C I/F
- Multi-role (main master, secondary master, slave)
- Target reset
- Group addressing

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4.22 USB Full-speed Controller (USB_FS)

The BK7259 embeds a USB full-speed controller with an integrated transceiver. It can operate as a host or a device.

The USB full-speed controller features are the following:

- Compliant with the Universal Serial Bus Specification Rev 1.1 and 2.0
- Full-speed (FS) operation (up to 12 Mbps)
- One bidirectional control endpoint0
- Seven IN/OUT endpoints configurable to support bulk, interrupt or isochronous data transfer
- A FIFO of 2 Kbytes configurable to be allocated to 8 endpoints

4.23 USB High-speed Controller (USB_HS)

The BK7259 embeds a USB high-speed (up to 480 Mbps) controller with an integrated transceiver. It can operate as a host or a device.

The USB high-speed controller features are the following:

- Compliant with the Universal Serial Bus Specification Rev 1.1 and 2.0
- Full-speed (FS) operation (up to 12 Mbps) and high-speed operation (up to 480 Mbps)
- One bidirectional control endpoint0
- Seven IN/OUT endpoints configurable to support bulk, interrupt or isochronous data transfer
- A FIFO of 8 Kbytes configurable to be allocated to 8 endpoints
- USB 2.0 Link Power Management (LPM) support
- The application processor embeds 64 KB USB RAM dedicated to high-speed USB:
  - The RISC cores can store high-speed USB related instructions and data in it.
  - The Cortex-M52 and Cortex-M55 cores can access it.
  - The GDMA and HPMA can access it.

4.24 CAN Controllers (CAN)

The BK7259 features two Controller Area Network (CAN) controllers. The CAN controller uses the basic CAN principle and meets all constraints of the CAN-specification 2.0B active. Furthermore, the CAN controller can be configured to meet the specification of CAN with flexible data rate CAN FD. CAN 2.0 carries a data payload up to 8 bytes and CAN FD up to 64 bytes.

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The CAN controller supports two operating modes, normal and standby, which can be selected via the CAN_STBY pin. If a high level is applied to the CAN_STBY pin, the external transceiver enters the standby mode.

The features of the CAN controller include:

- Supports CAN specification
  - CAN 2.0B (payload up to 8 bytes, verified by Bosch reference model)
  - CAN FD (payload up to 64 bytes, ISO 11898-1:2015 or non-ISO Bosch)
- Programmable data rate:
  - CAN 2.0B: up to 1 Mbit/s
  - CAN FD: up to 10 Mbit/s
- Two transmit buffers
  - One primary transmit buffer (PTB)
  - Optional configurable secondary transmit buffer (STB)
  - Number of buffer slots selectable by a generic parameter
  - Operation in FIFO or priority decision mode
- Independent and programmable internal 29-bit acceptance filters
- Extended features
  - Single-shot transmission mode (for PTB and/or for STB)
  - Listen only mode
  - Loop back mode (internal and external)
  - Transceiver standby mode
- Configurable interrupt sources
- Time-stamping:
  - ISO 11898-4 time-triggered CAN (TTCAN) with partial hardware support
  - CiA 603 time-stamping
- Compatible with AUTOSAR
- Optimized for SAE J1939

4.25 LIN Controller (LIN)

The BK7259 has a Local Interconnect Network (LIN) controller. The LIN controller is a communication controller that performs serial communication. It implements the data link layer of the LIN Protocol Specification. The LIN protocol uses a single master/multiple slave concept for the frame transfer between nodes of the LIN network.

The LIN controller supports Sleep mode. If a low level is applied to the LIN_SLEEP pin, the external transceiver enters the Sleep mode.

The features of the LIN controller are listed here:

- Support of LIN specification 2.2A
- Backward compatibility to LIN 1.3
- Configurable for support of master or slave functionality
- Programmable data rate between 1 kbit/s and 20 kbit/s (for master)
- Automatic bit rate detection (for slave)
- 8-byte data buffer
- 8-bit host controller interface

4.26 GDMA Controller (GDMA)

The BK7259 has a general-purpose DMA controller (GDMA) with 8 DMA channels to unload CPU activity. The 8 channels are shared by peripherals that have GDMA capabilities.

The GDMA controller has the features below:

- Supports transfer of the following types:
  - Single block transfer
  - Repeated block transfer
  - Linked-list transfer
- Memory-mapped data transfers from a source to a destination:
  - Peripheral-to-memory
  - Memory-to-peripheral
  - Memory-to-memory
  - Peripheral-to-peripheral
- Programmable data width for destination and source:
  - 8 bits (byte)
  - 16 bits (half-word)
  - 32 bits (word)
- Eight-grade priority for transfers arbitration at a channel level
- Per channel event generation on the following events:
  - Transfer complete

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  - Half transfer complete
  - Bus error
- 8 concurrent DMA channels:
  - Each channel has a FIFO for queuing source and destination transfers.
- Per linked-list item within a channel:
  - Separately programmed source and destination transfers
  - Number of data bytes to be transferred from the source can be programmed for each block.
  - Linear source and destination addressing: either fixed or contiguously incremented addressing, programmed at a block level, between successive single transfers
  - Event generation on DMA half-transfer complete and transfer complete
  - Pointer to the next linked-list item and its data structure in memory, with automatic update of the DMA linked-list control registers
- Secure/non-secure support:
  - A non-secure channel performs non-secure DMA transfers.
  - A secure channel can perform secure or non-secure DMA transfers.
  - TrustZone-aware DMA AHB master port
- Privileged/unprivileged support:
  - An unprivileged channel performs unprivileged DMA transfers.
  - A privileged channel performs privileged DMA transfers.
  - Privileged-aware AHB master port
  - Privileged-aware AHB slave port

A selection of peripherals on the BK7259 have GDMA capabilities, including UART0, UART1, UART2, UART3, SPI0, SPI1, SPI2, DMIC0, DMIC1, audio ADCs, audio DACs, I2S0, I2S1, I2S2, I2S3, SPDIF0, SPDIF1, AUX ADC, SPI3, UART4, I2S4, AUX DAC0, and AUX DAC1.

4.27 High-performance DMA Controller (HPDMA)

The BK7259 has a high-performance DMA controller (HPDMA) with 4 DMA channels to offload CPU activity. The HPDMA controller performs programmable data transfers between memory-mapped peripherals and/or memories.

The HPDMA controller has the features below:

- Supports transfer of the following types:
  - Single block transfer

  - Repeated block transfer
  - Linked-list transfer
- Dual bidirectional AXI master
- Each channel supports two-dimensional array transfers with a configurable step size.
- Memory-mapped data transfers from a source to a destination:
  - Peripheral-to-memory
  - Memory-to-peripheral
  - Memory-to-memory
  - Peripheral-to-peripheral
- Programmable data width for destination and source:
  - 8 bits (byte)
  - 16 bits (half-word)
  - 32 bits (word)
  - 64 bits
  - 128 bits
- Four-grade priority for transfers arbitration at a channel level
- Per channel event generation on the following events:
  - Transfer complete
  - Half transfer complete
  - Internal FIFO error
  - Bus error
- Each channel can be independently configured to generate interrupts for specific events.
- 4 concurrent DMA channels:
  - Each channel has a FIFO for queuing source and destination transfers.
  - Intra-channel DMA transfers chaining using programmable linked-list into memory, supporting two execution modes: run-to-completion and link step mode
- Per linked-list item within a channel:
  - Separately programmed source and destination transfers
  - Number of data bytes to be transferred from the source can be programmed for each block.
  - Linear source and destination addressing: either fixed or contiguously incremented addressing, programmed at a block level, between successive single transfers
  - Event generation on DMA half-transfer complete and transfer complete

  - Pointer to the next linked-list item and its data structure in memory, with automatic update of the DMA linked-list control registers
- Debug:
  - Support for channel suspend and resume
  - Channel status reporting, including FIFO level and event flags
- Secure/non-secure support:
  - Support for secure and non-secure DMA transfers, independently at a channel level
  - TrustZone-aware AXI master port
  - TrustZone-aware AHB slave port
- Privileged/unprivileged support:
  - Support for privileged and unprivileged DMA transfers, independently at a channel level
  - Privileged-aware AXI master port
  - Privileged-aware AHB slave port

UART5 on the BK7259 can utilize these HPDMA channels.

4.28 Ethernet MAC Interface (ENET)

The BK7259 provides a media access controller (MAC) for Ethernet LAN communications through a reduced medium-independent interface (RMII) or a reduced gigabit media independent interface (RGMII). The Ethernet MAC interface (ENET) is compliant with the IEEE 802.3-2015 specification and can be used in applications such as network interface cards, and data center bridges and nodes. The BK7259 requires an external physical interface device (PHY) to connect to the physical LAN bus. The PHY is connected to the device RMII port using 9 signals or RGMII port using 14 signals, and can be clocked using a 25 MHz clock from the BK7259 or an external crystal oscillator.

The ENET includes the following features:

- 10, 100, and 1000 Mbps data transfer rates
- Half-duplex (CSMA/CD) and full-duplex operation
- 32-bit CRC generation and removal
- Tagged MAC frame support (VLAN support)
- Dedicated DMA controller allowing high-speed transfers between the system memory and the internal FIFOs
- Embedded TX FIFO and RX FIFO to buffer transmit and receive frames. Both FIFOs are 2 Kbytes.
- MAC control sublayer (control frames) support
- Different types of address filtering for physical and multicast address (multicast and group addresses)

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- 32-bit status code for each transmitted or received frame
- Supports hardware PTP (precision time protocol) in accordance with IEEE 1588-2008 (version 2)
- Triggers interrupt when system time becomes greater than target time

4.29 PWM

The BK7259 has an advanced-control PWM module (PWM). The PWM module consists of 12 32-bit auto-reload counters driven by 12 programmable prescalers. The PWM module has 12 independent channels for input capture, pulse edge counting, or generation of output waveforms (output compare).

The 12 independent channels can be allocated to any counter comparator output. Each of the 12 channels can generate independent waveforms or complementary waveforms when coupled with any other channels.

The features of the PWM module are listed here:

- 12 32-bit up, down, up-and-down auto-reload counters
- 12 8-bit programmable prescalers capable of dividing the clock frequency of each counter by any factor between 1 and 256
- 12 independent channels for:
  - Input capture
  - Pulse edge counting
  - PWM generation (edge or center-aligned mode)
- Complementary outputs with programmable dead-time and configurable dead-time mode
- Synchronization circuit to control the counter with external signals and to interconnect several counters together
- Repetition counter to update the registers only after a given number of cycles of the counter
- Dedicated hardware support for smooth and precise LED dimming/fading
- Interrupt generation on the following events:
  - Update: counter overflow or underflow, counter initialization (by software or internal/external trigger)
  - Counter start
  - Input capture
  - Output compare
  - LED fading end
- Change of polarity, duty cycle, and base frequency on every PWM period
- Supports incremental (quadrature) encoder and Hall-sensor circuitry for positioning purposes

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4.30 Video Peripherals

4.30.1 Display Controller (DISPLAY)

The TFT LCD display controller (DISPLAY) provides a 24-bit parallel digital RGB (Red, Green, Blue) and outputs all signals to interfaces of various LCD and TFT panels.

The display controller has the following features:

- Up to 24-bit RGB parallel pixel output
- Input from the GPU, input color format: tiled HV sampled ARGB8888 1 bpp
- Three output color formats:
  - RBG888
  - RBG666
  - RBG565
- Programmable clocks for different display panels
- Configurable window position and size
- AXI master interface with burst of 64 words

4.30.2 Display Serial Interface (DSI)

The display serial interface (DSI) provides an interface for communicating with MIPI DSI compliant displays. It supports for up to four D-PHY data lanes, data rate up to 1.5 Gbps per lane.

4.30.3 Digital Video Port Interface (DVP)

The digital video port (DVP) interface provides a 10-bit parallel interface to sensors. The input from the sensor is fed to the video processors, and the output of the video processors is written to the data memory.

The DVP interface features include:

- 10-bit parallel interface
- Programmable polarity for pixel clock and synchronization signals

4.30.4 Camera Serial Interface (CSI)

The camera serial interface (CSI) provides an interface for communicating with MIPI CSI compliant cameras. It supports for up to two D-PHY data lanes, data rate up to 1.5 Gbps per lane.

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4.30.5 Image Signal Processor (ISP)

The image signal processor (ISP) provides a wide range of image processing features with less power consumption. It offers image processing features with support for DVP/MIPI CSI sensors and DMA data.

The ISP provides the following features:

- Input format: RAW8/RAW10/RAW12, 8/10-bit YUV422
- Output format: RAW8/RAW10/RAW12, 8-bit YUV420/422/444 planar/semi planar/interleaved, RGB888 planar/interleaved
- Input resolution: maximum 2560*2560, multi-sensor throughput up to 120 MBps @ 120 MHz, 1440p @ 20 fps + 1080p @ 20 fps @ 240 MHz
- Functional feature
  - Test Pattern (TP)
  - Black Level Subtraction (BLS)
  - Sensor linear compensation
  - Gamma-in/out in RAW/RGB
  - Lens Shade Correction (LSC)
  - Digital gain
  - Wide Dynamic Range (WDR) tone mapping for local and global
  - Green Equalization (GE)
  - Defect Pixel Correction (DPC)
  - 2D NR (noise reduction) for RAW domain
  - Demosaic interpolation (Debayer)
  - Sharpen
  - Color Correction (CC)
  - Color adjustment (hue, brightness, contrast, saturation)
  - 3A (AE/AWB/AF) and statistics output
  - Anti-flicker
  - Color Aberration Correction (CAC)
- Output: 2 paths (YUV420/422, RGB888, RAW8/10/12)
- Output-processor: crop, downscaling

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4.30.6 Video Processing Unit (VPU)

The video processing unit (VPU) is designed specifically for video codecs and video processing. It includes a hardware video pre-processor, a hardware video decoder, a hardware video encoder, and a hardware video post-processor.

The VPU offers the following features:

- H.264 encoder
  - Input size: maximum 16384*16384 (before cropping)
  - Input format: RGB, YUV420/422
  - OSD/mosaic: 8 rectangles, YUV420/RGB, alpha blending
  - Encoding resolution: maximum 2560*2560, multi-source throughput up to 180 MBps, 1440p @ 30 fps or 1440p @ 20 fps + 1080p @ 20 fps
  - Profile: BP/MP/HP, up to level 6
  - Macroblock size: 16*16, 8*8, 4*4, 8*16, 16*8
  - Search range: H: ±123, V: ±21
  - All modes for prediction are supported
  - ROI for 2 rectangles with separate QP
  - Rate control: CBR/VBR/ABR/CVBR/CQP
- H.264 decoder
  - Input size: maximum 1920*1920 @ 60 fps, minimum 48*48
  - Input format: H.264 byte/NAL/SVC/MVC
  - Error detection, concealment, and error resilience tools
  - Profile from level 1-5.1 BP/MP/HP
  - Format conversion to YUV420/422/ARGB
  - Up/down scaling to 3x
  - Alpha blending with 2 rectangular areas in YUV/RGB with programmable alpha
  - De-blocking filter
  - Rotation for 90/180/270 and flip
  - Contrast/brightness/saturation adjustment
- JPEG encoder
  - Input size: maximum 32768*32768
  - Input format: RGB, YUV420/422
  - OSD/mosaic: 8 rectangles, YUV420/RGB, alpha blending

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  - Encoding resolution: maximum 32768*32768, maximum throughput 360 MBps
  - Output: JFIF/MJPEG
  - Rate control: CBR/VBR/CQP
- JPEG decoder
  - Input size: 48*48 to 32768*32768
  - Input format: JFIF/YUV400/420/422/444/440/411
  - Error detection
  - Format conversion to RGB
  - Up/down scaling
  - Programmable alpha blending
  - Dithering
  - Cropping for digital zoom
  - Rotation for 90/180/270 and flip
  - Contrast/brightness/saturation adjustment

4.30.7 Graphics Processing Unit (GPU)

The graphics processing unit (GPU) provides hardware acceleration for graphics processing, to bring high-end graphic capabilities to a variety of IoT devices.

The GPU supports the following features:

- Input format
  - INDEX1/2/4/8
  - A4/A8/L8
  - YUY2/NV12/NV16/YV16
  - ARGB1555/4444/8565/8888, RGB565/888
- Tessellation: lines, quadratics, and cubic Bezier curves
- LVGL support for 2D and 2.5D applications
- Compression: visual lossless from 1.6 to 3.0
- Blending: 8 porter duff modes
- Rotation: any angle rotation
- Linear and curved paint paths
- Filter and interpolation: point/linear/bi-linear
- Coordinate systems and transformations

4.30.8 Display Processor Unit (DPU)

The display processor unit (DPU) outputs video and graphic streams from a single output panel with up to 1080p display resolution. The single output panel can connect to a Display Pixel Interface (DPI) or enhanced Display Pixel Interface (eDPI) device for RGB output.

The features of the DPU are as follows:

- Input format
  - RGB888/565 (linear and tiled) with compressed or uncompressed
  - ARGB8565/8888 (linear and tiled) with compressed or uncompressed
  - YUY2/NV12
- Pre-processor
  - Rotation (90, 180, 270)/flip
  - Swizzle
  - YUV to RGB
  - Color keying
- Post-processor
  - Alpha blending with video layer
  - Gamma correction
  - Dithering
  - RCD (round corner display) blending
- Output interface
  - MIPI DPI
  - Maximum 1920*1080 @ 60 fps for linear output
  - Maximum 1024*768 @ 60 fps for tiled output

4.31 Audio Peripherals

The BK7259 comes with a rich set of audio peripherals to enhance the listening experience.

4.31.1 10-band Digital Equalizer (EQ)

A dedicated 10-band digital equalizer is implemented prior to digital-to-analog conversion, allowing the user to customize the frequency response of the audio output. The equalizer is implemented in hardware to reduce overall chip power consumption.

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4.31.2 Audio ADCs

The BK7259 contains three 24-bit audio ADCs with programmable sampling rates of 32 kHz, 48 kHz, 64 kHz, 96 kHz, 128 kHz, 192 kHz, or 384 kHz.

The audio ADC supports the following features:

- Programmable sampling rates from 32 kHz to 384 kHz
- Supports differential and single-ended microphone input
- Programmable microphone bias (1.8 to 2.4 V)
- Supports programmable input analog gains
- Digital volume control
- Supports up to 3 digital microphones

4.31.3 Audio DACs

The BK7259 also integrates two 24-bit audio DACs with programmable sampling rates of 8 kHz, 16 kHz, 44.1 kHz, 48 kHz, 88.2 kHz, or 96 kHz.

The audio DAC supports the following features:

- Programmable sampling rates from 8 kHz to 96 kHz
- Minimum 600 Ω load
- Differential output or single-ended output
- De-pop function in stereo output
- -80 dB crosstalk between channels
- Digital volume control
- Configurable 10-band EQ

4.31.4 Microphone Input Amplifiers and Bias Generator

The BK7259 contains three fully differential analog microphone input amplifiers and a low-noise microphone bias generator, allowing the microphone to interface with passive resistors and capacitors.

The microphone signal can be amplified with the amplifier over a 0 to 30 dB gain range with a 2 dB step size.

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4.31.5 Audio Amplifiers

The BK7259 provides two audio amplifiers, each capable of driving a 600 Ω speaker with load capacitance up to 30 pF.

4.31.6 Digital Microphone Interfaces (DMIC)

The BK7259 has two digital microphone interfaces. Each DMIC interface supports up to two digital microphones. The DMIC interface captures PDM stream from digital microphones and converts it to PCM data. The PCM sampling rate can be up to 384 kHz when the PDM clock frequency is 3.072 MHz.

4.31.7 Voice Activity Detector (VAD)

The voice activity detector (VAD) is a voice-triggered system wake-up mechanism. The VAD consists of a front-end programmable gain amplifier (PGA) and a 10-bit ADC. It generates an interrupt on detection of voice activity to wake up the system.

The VAD features:

- Supports microphone data from analog microphone
- Supports single microphone voice detection
- Programmable sampling rates from 100 Hz to 4 kHz

4.31.8 SPDIF Interfaces (SPDIF)

The BK7259 integrates two SPDIF Interfaces. The SPDIF interface is a digital audio interface used to carry uncompressed PCM audio or compressed surround sound.

4.31.9 HDMI CEC Controller (HDMI CEC)

The BK7259 provides a high-definition multimedia interface (HDMI) consumer electronics control (CEC) controller. The HDMI CEC controller is designed to control HDMI connected audiovisual devices in an environment.

4.31.10 I2S Interfaces (I2S)

The BK7259 integrates five I2S interfaces that support master and slave modes with a sampling rate from 8 kHz to 384 kHz. The I2S interface supports both PCM mono channel mode and I2S stereo channel mode.

Listed here are the I2S features:

- Master or slave mode
- Full duplex or half-duplex communication
- Various sampling rates
- 12-bit programmable prescaler
- Multiple I2S protocols supported:
  - I2S Philips standard
  - MSB-Justified standard (Left-Justified)
  - LSB-Justified standard (Right-Justified)
  - PCM standard
- Programmable data order with LSB first or MSB first
- Programmable data width between 1 and 32 bits
- Programmable clock polarity
- Integrated 32-bit RX FIFO and 32-bit TX FIFO, both with a depth of 32 x 3 channels
- Master clock can be output to drive external audio devices.

4.32 Auxiliary ADC (AUX ADC)

The auxiliary ADC (AUX ADC) is a 14-bit successive approximation analog-to-digital converter. The AUX ADC has multiple external analog input channels and internal dedicated channels. The AUX ADC supports A/D conversion performed in one-shot or continuous mode.

The AUX ADC module has the following features:

- Programmable sampling rates
- 14-bit resolution
- Up to 10 external analog input channels: ADC1/2/3/4/5/6/10/12/13/15
- Five internal dedicated channels:
  - VBAT monitoring channel (VBAT/4), connected to ADC0
  - Internal temperature sensor (VTEMP), connected to ADC7
  - Two internal calibration channels, connected to ADC8/11
  - Internal touch sensor (VOUT_TH), connected to ADC9
- Conversion modes:
  - One-shot mode
  - Continuous mode

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4.33 Auxiliary DACs (AUX DAC)

The BK7259 embeds two auxiliary DACs (AUX DAC). The AUX DAC is a 12-bit, voltage output digital-to-analog converter. Each AUX DAC has one output channel. In dual DAC channel mode, conversions can be performed independently or simultaneously by grouping both channels for synchronous updates. The AUX DAC features built-in rail-to-rail output buffers capable of sourcing or sinking up to 20 mA. Individual gain/offset calibration can be applied to each AUX DAC output channel.

The AUX DAC has the following features:

- One output channel for each AUX DAC
- High current drive capability
- Supports dual-channel simultaneous conversion
- Synchronized update capability
- Noise-wave generation
- Triangular-wave generation
- External triggers for conversion
- Per channel gain/offset calibration
- Internal reference
- Each AUX DAC output can be disconnected from the corresponding output pin.

4.34 Analog Comparators (ACOMP)

The BK7259 integrates two low-power rail-to-rail analog comparators (ACOMP), each with 10 input channels. The ACOMP utilizes an 8-bit DAC as an internal reference. The ACOMP can generate an interrupt or redirect outputs to PWM inputs for input capture.

The ACOMP features are as follows:

- Up to 10 non-inverting input channels per ACOMP
- Programmable internal reference voltage
- Programmable hysteresis
- Programmable speed/power consumption
- Redirection of outputs to PWM inputs

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4.35 Timer Groups (TIMG)

The BK7259 includes six general-purpose timer groups (TIMG). Each group has three 32-bit timers. Each group consists of three 32-bit counters driven by a 4-bit prescaler.

Each TIMG module has the following features:

- Three timers (Timer0/1/2)
- Three 32-bit up counters
- 4-bit prescaler, factor between 1 and 16
- Capable of reading the real-time value of the counter

4.36 Watchdog Timers (WDT)

The BK7259 has one watchdog timer in the AON power domain, one window watchdog timer in each CPU core, and one AHB window watchdog timer dedicated to the application processor subsystem. The purpose of the watchdog timer is to detect and recover from failures or malfunctions.

4.36.1 AON Watchdog Timer (AON WDT)

The AON power domain watchdog timer (AON WDT) is based on a 24-bit up-counter. It runs on the 32 kHz CLK_ROSC clock and has a maximum programmable period of up to 4.66 (2^24/1 kHz) hours. The AON WDT triggers a reset on expiry of a programmed time period.

4.36.2 CPU Window Watchdog Timers (CPU WWDT)

Each CPU core has a window watchdog timer (CPU WWDT). Each CPU window watchdog timer is based on a 16-bit up-counter. All CPU window watchdog timers are clocked from nominal 32 kHz clock sources. The CPU WWDT triggers a reset on expiry of a specified time period. A reset is also generated if the up-counter is reloaded outside the window.

4.36.3 AHB Window Watchdog Timer (AHB WWDT)

The AHB window watchdog timer is dedicated to the application processor subsystem. It is based on a 16-bit up-counter and is clocked from the CLK_X32K clock. The AHB WWDT triggers a reset on expiry of a specified time period. A reset is also generated if the up-counter is reloaded outside the window.

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4.37 Real-time Counters (RTC)

4.37.1 High-voltage RTC (HV RTC)

The high-voltage RTC is supplied by VBAT. It features a 36-bit counter and a tick event generator. The HV RTC clock source can be selected from CLK_X32K or CLK_ROSC. It is used for low-power timing, and it keeps running even when the system is in deep sleep0 mode.

4.37.2 AON RTC

The AON RTC in the AON power domain features a 64-bit up-counter and a tick event generator. The AON RTC runs on the CLK_LPO0 clock. It is used for low-power timing, and it keeps running even when the system is in deep sleep1 mode.

4.38 IrDA Interfaces (IrDA)

The BK7259 embeds four hardware IrDA interfaces that support waveform analysis and waveform generation. The IrDA interface monitors the start of infrared signals, records the sequence of infrared waveforms, stores the waveforms in the RX FIFO for software analysis, and writes the waveforms to be sent to the TX FIFO when sending, thereby enabling the analysis and transmission of any infrared protocol.

The IrDA interface has the following features:

- Single-duplex mode
- Carrier modulation for transmission
- Integrated 512-byte RX FIFO and 512-byte TX FIFO

4.39 Temperature Sensor (TEMP)

The BK7259 integrates an on-chip temperature sensor that can measure on-chip temperature over -40 to +125 °C with an accuracy of ±5 °C. The digital results can be read from the AUX ADC.

Usually, the software initiates the calibration of a specific module based on the temperature value, narrowing the difference in chip performance at different temperatures. The host can also read the on-chip temperature and decide whether to reduce transmit power or suspend operation at high temperatures.

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4.40 Touch Sensor (TOUCH)

The BK7259 has one capacitive sensing I/O, which immediately detects capacitance changes induced by touch or proximity of objects.

4.41 Security

The BK7259 provides state-of-the-art security based on a powerful security architecture. It integrates a Secure Element (SE) for cryptograph and system security control. The SE embeds all-round and robust security features to set up a top-secret execution environment for IoT devices.

The key features of the SE include:

- Secure boot
- Secure debug
- Secure connection
- Firmware Over-The-Air (FOTA)
- Provisioning
- TEE_M
- TrustEngine

System Architecture

The SE software implements an all-round device-to-cloud security architecture, including:

- Provisioning mechanism in the manufacturing stage.
- Secure boot, secure debug, TRNG calibration, and FOTA update in the device boot stage.
- Secure connection and FOTA download in the device run-time stage.

The SE also introduces a remote management solution to enable users to access, monitor and manage IoT devices in a secure manner. The remote management solution includes FOTA SaaS, Secure Debug SaaS, and Provisioning SaaS.

4.41.1 Secure Boot

The secure boot solution provides the legitimacy and trustworthiness for images running on devices. It ensures that only official images which are published from OEMs are executable.

The secure boot functions include following aspects:

- Image verification

- Image encryption
- Extended program execution
- Image anti-rollback protection

4.41.2 Secure Debug

Secure debug provides a reliable mechanism for enabling the debug feature of a device after the device deployment.

Debug is one of the most commonly-used features in the device. With the debug feature, you can access all the device data, including the device firmware and the device root key.

4.41.3 FOTA

The FOTA solution implements a series of light-weight and trustworthy firmware upgrade interaction between IoT devices and servers, which includes downloading, verification, and installation.

4.41.4 Provisioning

Provisioning is the process to initialize secure credentials onto devices. It is required to happen in a secure environment such as the product line in the manufacturing stage.

The provisioning solution provides a full secure mechanism from the cloud server to devices. It keeps the security and integrity of provisioning materials, such as the device ID, model key, and the secure boot/secure debug public key hash, also keeps the device top secret such as the device root key that occurs only on the device side.

4.41.5 Secure Connection

The secure connection feature uses Mbed TLS to provide connection security in the transport layer over UDP. The key exchange method is Pre-Shared Key (PSK). The PSK’s deployment happens in the manufacturing stage.

Mbed TLS provides cryptograph and SSL/TLS capabilities with TLS 1.1 and TLS 1.2.

4.41.6 BootROM

The SE includes the BootROM reference code which is integrated with the secure boot solution.

There are primary boot and recovery boot paths in the BootROM.

- Primary boot is the main boot path. Normally devices should boot to the primary path.
- Recovery boot is the secondary boot path.

4.41.7 Bootloader

The SE includes the Bootloader reference code which is integrated with the secure debug, provisioning, TRNG calibration, and FOTA solution.

In the primary bootloader, the secure debug agent, provisioning agent, and TRNG agent are integrated. The primary bootloader boots to the next image—TEE_M.

In the recovery bootloader, both the secure debug agent, TRNG agent, and the FOTA update agent are integrated. The recovery bootloader does not boot to any images.

4.41.8 TEE_M

The SE introduces TEE_M. TEE_M provides an implementation of secure world software for Armv8-M, which follows Arm Platform Security Architecture (PSA) PSA_Firmware_Framework_1.0-bet0. TEE_M provides the following services by default:

- Secure Storage: The Secure Storage (Only supports Protected Storage) service supports PSA_Storage_API-1.0-bet2 without extended functions.
- Crypto: The Crypto service supports PSA_Cryptography_API_Reference_1.0_bet1.
- DTLS: DTLS (leverages Mbed TLS) is integrated in TEE_M.

4.41.9 TrustEngine

The SE introduces TrustEngine as a secure component in the system. It provides the following features:

- High security assurance. The crypto engine supports key ladders, lifecycle management and True Random Number Generator (TRNG) which enhance the system security.
- High performance and low power for encryption/decryption operation. This is achieved by TrustEngine internal cryptography engines.
- Reduction of software complicity in security. Some security functions are implemented in TrustEngine hardware, which can reduce the risk of sensitive information leakage to non-secure hosts.

4.41.9.1 Features

TrustEngine includes the following features:

- Symmetric schemes, AES-ECB/CBC/CTR/CBC-MAC/CMAC/CCM/GCM (key size 128-bit, 192-bit and 256-bit)
- Symmetric schemes, SM4-ECB/CBC/CTR/CBC-MAC/CMAC/CCM/GCM
- Digest schemes, SHA1/224/256
- Digest scheme, SM3

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- Asymmetric schemes, RSA 1024/2048/3072/4096 and ECCP 192/224/256/384/512/521
- Asymmetric scheme, SM2
- Key ladder for key management
- Lifecycle management
- True random number generator

4.41.9.2 Supported Standards and Specifications

TrustEngine is compliant with the following standards:

- FIPS PUB 180-4: Secure Hash Standard (SHS)
- FIPS PUB 197: Advanced Encryption Standard (AES)
- NIST SP 800-38A Recommendation for Block Cipher Modes of Operation-Methods and Techniques
- NIST SP 800-38B Recommendation for Block Cipher Modes of Operation: The CMAC Mode for Authentication
- NIST SP 800-38C Recommendation for Block Cipher Modes of Operation-the CCM Mode for Authentication and Confidentiality
- NIST SP 800-38D, Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC
- NIST SP 800-90B Recommendation for the Entropy Sources Used for Random Bit Generation
- SM2 Public Key Cryptographic Algorithm Based on Elliptic Curves (GB/T 32918-2016)
- SM3 Cryptographic Hash Algorithm (GB/T 32905-2016)
- SM4 Block Cipher Algorithm (GB/T 32907-2016)

4.41.9.3 Components

TrustEngine consists of five major function blocks.

- Symmetric Cryptography Accelerator (SCA)
- Asymmetric Cryptography Accelerator (ACA)
- HASH engine
- One-time programmable storage access controller
- True random number generator

Symmetric Cryptography Accelerator (SCA)

SCA in TrustEngine is responsible for data encryption/decryption using the symmetric cryptography algorithm. The data encryption/decryption operations are performed through the SCA engine by sending special commands to TrustEngine.

Asymmetric Cryptography Accelerator (ACA)

The ACA engine in TrustEngine is responsible for accelerating the asymmetric cryptography, such as:

- Asymmetric encryption and decryption: RSA and ECC
- Digital signature and verification: RSA signatures and ECDSA
- Key exchange: DH (Diffie-Hellman) and ECDH

HASH Accelerator

The HASH engine in TrustEngine is responsible for digest calculation. The digest of certain data can be calculated through the HASH engine by sending special commands to TrustEngine.

True Random Number Generator

The True Random Number Generator (TRNG) generates the random bits using the internal entropy source (ring oscillator inverter chain) or the external entropy source. The random bits are required by both secure and non-secure hosts.

4.42 Serial-wire Debug Port (SW-DP)

The serial-wire debug port (SW-DP) enables a debugger to be connected to the BK7259 device. The SW-DP interface uses the two following pins:

- SWCLK: clock from the debugger to target (BK7259 device)
- SWDIO: bi-directional serial data

The SWCLK and SWDIO lines both have an internal pull-down resistor.

5 Electrical Characteristics

Note: Values currently listed in this section are preliminary measurements and are subject to change.

5.1 Absolute Maximum Ratings

Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. Exposure to absolute-maximum-rated conditions for prolonged periods may affect the reliability of the device.

Parameter	Description	Min.	Max.	Unit
VBAT	Chip power supply voltage	-0.3	3.6	V
VCCPA	Supply voltage for PA	-0.3	3.6	V
VCCPAD	Supply voltage for PA driver	-0.3	3.6	V
VCCCB	Supply voltage for bandgap and analog peripherals	-0.3	3.6	V
VCCAUD	Supply voltage for audio ADCs/DACs	-0.3	3.6	V
MICBIAS	Microphone bias output voltage	-0.3	3.6	V
VDDAON	AON LDO output voltage	-0.3	1.05	V
SWA	Analog buck switch output voltage	-0.3	3.6	V
VDDA	Analog LDO output voltage	-0.3	1.8	V
	Analog buck output voltage	-0.3	1.8	V
VCCA	Supply voltage for Analog	-0.3	1.8	V
VCCPLL	Supply voltage for RF PLL	-0.3	1.8	V
VCCRXFE	Supply voltage for RX	-0.3	1.8	V
VCCIF	Supply voltage for IF	-0.3	1.8	V
SWD	Digital buck switch output voltage	-0.3	3.6	V
VDDD	Digital LDO output voltage	-0.3	1.2	V
	Digital buck output voltage	-0.3	1.2	V
VDDDIG	Digital core LDO output voltage	-0.3	1.07	V
SWH	High-power digital buck switch output voltage	-0.3	3.6	V
VDDH	High-power digital buck output voltage	-0.3	1.3	V

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Parameter	Description	Min.	Max.	Unit
VDDCORE_HS	High-power digital core LDO output voltage	-0.3	1.07	V
VDDRAM	EXMEM LDO output voltage	-0.3	2.1	V
VDD1V8_HSUSB	Supply voltage for high-speed USB	-0.3	2.1	V
VDD3V3_HSUSB	Supply voltage for high-speed USB	-0.3	4.0	V
VOUT1V2	1.2 V LDO output voltage	-0.3	1.87	V
VOUT1V8	1.8 V LDO output voltage	-0.3	2.86	V
VOUT2V8	2.8 V LDO output voltage	-0.3	4.0	V
VOUT3V0	3.0 V LDO output voltage	-0.3	4.0	V
VCCGPIO	Supply voltage for GPIOs	-0.3	3.36	V
CSI_VDDH	Supply voltage for MIPI CSI	-0.36	3.63	V
DSI_VDDH	Supply voltage for MIPI DSI	-0.66	3.66	V
P_{RX}	RX input power	-	10	dBm
T_{STR}	Storage temperature range	-55	150	°C

5.2 ESD Ratings

Parameter	Description	Test Conditions	Value	Unit
ESD HBM	Electrostatic discharge voltage (human body model), per ANSI/ESDA/JEDEC JS-001	GPIO pins	TBD	V
		RF/analog pins	TBD	V
ESD CDM	Electrostatic discharge voltage (charge device model), per ANSI/ESDA/JEDEC JS-002	GPIO pins	TBD	V
		RF/analog pins	TBD	V
Latch-up	Latch-up sensitivity, per JESD78F.02	25 °C	TBD	mA

5.3 Recommended Operating Conditions

Parameter	Description	Min.(1)	Typ.	Max.	Unit
VBAT(2)	Chip power supply voltage	2.3	3.3	3.6	V
VCCPA	Supply voltage for PA	2.3	3.3	3.6	V
VCCPAD	Supply voltage for PA driver	2.3	3.3	3.6	V

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Parameter	Description	Min.(1)	Typ.	Max.	Unit
VCCCB	Supply voltage for bandgap and analog peripherals	2.3	3.3	3.6	V
VCCAUD	Supply voltage for audio ADCs/DACs	2.3	3.3	3.6	V
MICBIAS	Microphone bias output voltage	1.8	-	2.4	V
VDDAON	AON LDO output voltage	0.6	TBD	0.95	V
VDDA	Analog LDO output voltage	1.3	TBD	1.65	V
	Analog buck output voltage	1.3	TBD	1.65	V
VCCA	Supply voltage for Analog	1.3	TBD	1.65	V
VCCPLL	Supply voltage for RF PLL	1.3	TBD	1.65	V
VCCRXFE	Supply voltage for RX	1.3	TBD	1.65	V
VCCIF	Supply voltage for IF	1.3	TBD	1.65	V
VDDD	Digital LDO output voltage	0.75	TBD	1.1	V
	Digital buck output voltage	0.75	TBD	1.1	V
VDDDIG	Digital core LDO output voltage	0.45	TBD	0.975	V
VDDH	High-power digital buck output voltage	0.85	TBD	1.2	V
VDDCORE_HS	High-power digital core LDO output voltage	0.6	TBD	0.975	V
VDDRAM	EXMEM LDO output voltage	1.8	-	1.95	V
VDD1V8_HSUSB	Supply voltage for high-speed USB	1.8	-	1.95	V
VDD3V3_HSUSB	Supply voltage for high-speed USB	2.3	-	3.6	V
VOUT1V2	1.2 V LDO output voltage	1.0	1.2	1.7	V
VOUT1V8	1.8 V LDO output voltage	1.1	1.8	2.6	V
VOUT2V8	2.8 V LDO output voltage	2.1	2.8	3.6	V
VOUT3V0	3.0 V LDO output voltage	2.1	3.0	3.6	V
VCCGPIO	Supply voltage for GPIOs	1.8	-	3.05	V
CSI_VDDH	Supply voltage for MIPI CSI	2.97	3.3	3.63	V
DSI_VDDH	Supply voltage for MIPI DSI	2.97	3.3	3.63	V
T_{OPR}	Operating temperature range	-40	-	85	°C

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Parameter	Description	Min.(1)	Typ.	Max.	Unit

(1) The minimum voltage specified includes the ripple on the supply voltage and all other transient dips. Care must be taken when operating at the minimum specified voltage.

(2) To ensure RF performance, the ripple on the supply must be less than Vpp = 100 mV.

5.4 DC Electrical Characteristics

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
R_{PU}	Weak pull-up resistor	-	-	48	-	kΩ
R_{PD}	Weak pull-down resistor	-	-	48	-	kΩ
V_{IH_CEN}	CEN input high voltage (chip enable threshold)	-	TBD	-	TBD	V
V_{IL_CEN}	CEN input low voltage (chip disable threshold)	-	TBD	-	TBD	V
GPIO0–GPIO55						
VIH	High-level input voltage	-	0.7 VBAT	-	VBAT + 0.3	V
VIL	Low-level input voltage	-	-0.3	-	0.3 VBAT	V
VOH	High-level output voltage	-	0.8 VBAT	-	-	V
VOL	Low-level output voltage	-	-	-	0.1 VBAT	V
V_{T+}	Schmitt-trigger high-level input voltage	VBAT = 3.3 V	-	TBD	-	V
		VBAT = 1.8 V	-	TBD	-	V
V_{T-}	Schmitt-trigger low-level input voltage	VBAT = 3.3 V	-	TBD	-	V
		VBAT = 1.8 V	-	TBD	-	V
I_{OH}	High-level source current	VOH = 0.8 VBAT, GPIO_CAPACITY = 3	-	TBD	-	mA
		VOH = 0.8 VBAT, GPIO_CAPACITY = 2	-	TBD	-	mA
		VOH = 0.8 VBAT, GPIO_CAPACITY = 1	-	TBD	-	mA
		VOH = 0.8 VBAT, GPIO_CAPACITY = 0	-	TBD	-	mA
I_{OL}	Low-level sink current	VOL = 0.15 VBAT, GPIO_CAPACITY = 3	-	TBD	-	mA

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Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
		VOL = 0.15 VBAT, GPIO_CAPACITY = 2	-	TBD	-	mA
		VOL = 0.15 VBAT, GPIO_CAPACITY = 1	-	TBD	-	mA
		VOL = 0.15 VBAT, GPIO_CAPACITY = 0	-	TBD	-	mA
GPIO56–GPIO71						
VIH	High-level input voltage	-	0.7 VCCGPIO	-	VCCGPIO + 0.3	V
VIL	Low-level input voltage	-	-0.3	-	0.3 VCCGPIO	V
VOH	High-level output voltage	-	0.8 VCCGPIO	-	-	V
VOL	Low-level output voltage	-	-	-	0.1 VCCGPIO	V
V_{T+}	Schmitt-trigger high-level input voltage	VCCGPIO = 3.3 V	-	TBD	-	V
		VCCGPIO = 1.8 V	-	TBD	-	V
V_{T-}	Schmitt-trigger low-level input voltage	VCCGPIO = 3.3 V	-	TBD	-	V
		VCCGPIO = 1.8 V	-	TBD	-	V
I_{OH}	High-level source current	VOH = 0.8 VCCGPIO, GPIO_CAPACITY = 3	-	TBD	-	mA
		VOH = 0.8 VCCGPIO, GPIO_CAPACITY = 2	-	TBD	-	mA
		VOH = 0.8 VCCGPIO, GPIO_CAPACITY = 1	-	TBD	-	mA
		VOH = 0.8 VCCGPIO, GPIO_CAPACITY = 0	-	TBD	-	mA
I_{OL}	Low-level sink current	VOL = 0.15 VCCGPIO, GPIO_CAPACITY = 3	-	TBD	-	mA
		VOL = 0.15 VCCGPIO, GPIO_CAPACITY = 2	-	TBD	-	mA
		VOL = 0.15 VCCGPIO, GPIO_CAPACITY = 1	-	TBD	-	mA
		VOL = 0.15 VCCGPIO, GPIO_CAPACITY = 0	-	TBD	-	mA

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5.5 AON LDO

Parameter	Description	Min.	Typ.	Max.	Unit
VDDAON	AON LDO output voltage	0.6	TBD	0.95	V
Load current	-	-	-	TBD	mA

5.6 Analog LDO

Parameter	Description	Min.	Typ.	Max.	Unit
VDDA	Analog LDO output voltage	1.3	TBD	1.65	V
Load current	-	-	-	TBD	mA

5.7 Digital LDO

Parameter	Description	Min.	Typ.	Max.	Unit
VDDD	Digital LDO output voltage	0.75	TBD	1.1	V
Load current	-	-	-	TBD	mA

5.8 Digital Core LDO

Parameter	Description	Min.	Typ.	Max.	Unit
VDDDIG	Digital core LDO output voltage	0.45	TBD	0.975	V
Load current	-	-	-	TBD	mA

5.9 High-power Digital Core LDO

Parameter	Description	Min.	Typ.	Max.	Unit
VDDCORE_HS	High-power digital core LDO output voltage	0.6	TBD	0.975	V
Load current	-	-	-	TBD	mA

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5.10 EXMEM LDO

Parameter	Description	Min.	Typ.	Max.	Unit
VDDRAM	EXMEM LDO output voltage	1.8	-	1.95	V
Load current	-	-	-	TBD	mA

5.11 1V2 LDO

Parameter	Description	Min.	Typ.	Max.	Unit
VOUT1V2	1.2 V LDO output voltage	1.0	1.2	1.7	V
Load current	-	-	-	TBD	mA

5.12 1V8 LDO

Parameter	Description	Min.	Typ.	Max.	Unit
VOUT1V8	1.8 V LDO output voltage	1.1	1.8	2.6	V
Load current	-	-	-	TBD	mA

5.13 2V8 LDO

Parameter	Description	Min.	Typ.	Max.	Unit
VOUT2V8	2.8 V LDO output voltage	2.1	2.8	3.6	V
Load current	-	-	-	TBD	mA

5.14 3V0 LDO

Parameter	Description	Min.	Typ.	Max.	Unit
VOUT3V0	3.0 V LDO output voltage	2.1	3.0	3.6	V
Load current	-	-	-	TBD	mA

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5.15 Analog Buck

Parameter	Description	Min.	Typ.	Max.	Unit
VDDA	Analog buck output voltage	1.3	TBD	1.65	V
Load current	-	-	-	TBD	mA
Switching frequency	Buck modulation frequency	TBD	TBD	TBD	MHz
Output filter capacitor capacitance	-	-	TBD	-	μF
Inductor inductance	-	-	TBD	-	μH
Inductor DC resistance	-	-	-	TBD	mΩ
Inductor saturation current	-	TBD	-	-	mA

5.16 Digital Buck

Parameter	Description	Min.	Typ.	Max.	Unit
VDDD	Digital buck output voltage	0.75	TBD	1.1	V
Load current	-	-	-	TBD	mA
Switching frequency	Buck modulation frequency	TBD	TBD	TBD	MHz
Output filter capacitor capacitance	-	-	TBD	-	μF
Inductor inductance	-	-	TBD	-	μH
Inductor DC resistance	-	-	-	TBD	mΩ
Inductor saturation current	-	TBD	-	-	mA

5.17 High-power Digital Buck

Parameter	Description	Min.	Typ.	Max.	Unit
VDDH	High-power digital buck output voltage	0.85	TBD	1.2	V
Load current	-	-	-	TBD	mA
Switching frequency	Buck modulation frequency	TBD	TBD	TBD	MHz

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Parameter	Description	Min.	Typ.	Max.	Unit
Output filter capacitor capacitance	-	-	TBD	-	μF
Inductor inductance	-	-	TBD	-	μH
Inductor DC resistance	-	-	-	TBD	mΩ
Inductor saturation current	-	TBD	-	-	mA

5.18 26 MHz Crystal Characteristics

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
F0	Nominal frequency	-	-	26	-	MHz
ΔF/F0	Frequency tolerance	25 °C	-10	-	+10	ppm
TC	Frequency stability over operating temperature	-40 to 105 °C crystal	-20	-	+20	ppm
		-30 to 85 °C crystal	-10	-	+10	ppm
CL	Load capacitance	-	7	7.3	12	pF
TS	Trim sensitivity	3225 package	-	32	-	ppm/pF
		2016 package	-	17	-	ppm/pF

5.19 32.768 kHz Crystal Characteristics

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
F0	Nominal frequency	-	-	32.768	-	kHz
ΔF/F0	Frequency tolerance	25 °C	-20	-	+20	ppm
T_{OPR}	Operating temperature range	-	-40	-	85	°C
CL	Load capacitance	-	-	12.5	-	pF
ESR	Equivalent series resistance	25 °C	-	-	70	kΩ

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5.20 Current Consumption

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Conditions	Min.	Typ.	Max.	Unit
Wi-Fi Active Mode					
RX current	11b: 11 Mbps DSSS	-	TBD	-	mA
	11g: 54 Mbps OFDM	-	TBD	-	mA
	11n: MCS7, HT20	-	TBD	-	mA
	11n: MCS7, HT40	-	TBD	-	mA
	11ax: MCS7, HE20	-	TBD	-	mA
	11ax: MCS9, HE20	-	TBD	-	mA
	11ax: MCS7, HE40	-	TBD	-	mA
	11ax: MCS9, HE40	-	TBD	-	mA
TX current	11b: 11 Mbps DSSS @ TBD dBm	-	TBD	-	mA
	11g: 54 Mbps OFDM @ TBD dBm	-	TBD	-	mA
	11n: MCS7, HT20 @ TBD dBm	-	TBD	-	mA
	11n: MCS7, HT40 @ TBD dBm	-	TBD	-	mA
	11ax: MCS7, HE20 @ TBD dBm	-	TBD	-	mA
	11ax: MCS9, HE20 @ TBD dBm	-	TBD	-	mA
	11ax: MCS7, HE40 @ TBD dBm	-	TBD	-	mA
	11ax: MCS9, HE40 @ TBD dBm	-	TBD	-	mA
Wi-Fi IEEE Power Save Mode					
Standby associated, DTIM = 1	Without TCP keep-alive	-	TBD	-	µA
Standby associated, DTIM = 3	Without TCP keep-alive	-	TBD	-	µA
Standby associated, DTIM = 10	Without TCP keep-alive	-	TBD	-	µA
Bluetooth Active Mode Dedicated Bluetooth transceiver enabled					
Bluetooth LE RX current	-	-	TBD	-	mA

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Parameter	Conditions	Min.	Typ.	Max.	Unit
Bluetooth LE TX current	2 dBm	-	TBD	-	mA
Bluetooth LE advertising	Interval: 2 s	-	TBD	-	µA
Bluetooth LE connection slave	Interval: 1 s	-	TBD	-	µA
Classic Bluetooth A2DP music streaming	-	-	TBD	-	mA
Bluetooth Sniff Mode Dedicated Bluetooth transceiver enabled					
Classic Bluetooth sniff	Interval: 500 ms	-	TBD	-	µA
Sleep Mode					
Deep sleep0	Only HV RTC is active	-	2.5	-	µA
Deep sleep1	128 KB SRAM retention	-	25	-	µA
Sleep	-	-	70	-	µA
Shutdown Mode					
Shutdown	-	-	1	-	µA

5.21 RF Characteristics

Unless otherwise noted, the RF data is measured at the chip port, where RF cable is connected, including the front-end loss. The front-end circuit is a matching network designed to ensure that the product complies with wireless regulations in various countries.

5.21.1 WLAN RF Characteristics

5.21.1.1 WLAN Receiver Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Conditions	Min.	Typ.	Max.	Unit
General					
Frequency range	-	2412	-	2484	MHz
Sensitivity					
Sensitivity - IEEE 802.11b	1 Mbps DSSS	-	TBD	-	dBm

Parameter	Conditions	Min.	Typ.	Max.	Unit
(8% PER)	2 Mbps DSSS	-	TBD	-	dBm
	5.5 Mbps DSSS	-	TBD	-	dBm
	11 Mbps DSSS	-	TBD	-	dBm
Sensitivity - IEEE 802.11g (10% PER)	6 Mbps OFDM	-	TBD	-	dBm
	9 Mbps OFDM	-	TBD	-	dBm
	12 Mbps OFDM	-	TBD	-	dBm
	18 Mbps OFDM	-	TBD	-	dBm
	24 Mbps OFDM	-	TBD	-	dBm
	36 Mbps OFDM	-	TBD	-	dBm
	48 Mbps OFDM	-	TBD	-	dBm
	54 Mbps OFDM	-	TBD	-	dBm
Sensitivity - IEEE 802.11n, 20 MHz (10% PER)	HT20, MCS0	-	TBD	-	dBm
	HT20, MCS1	-	TBD	-	dBm
	HT20, MCS2	-	TBD	-	dBm
	HT20, MCS3	-	TBD	-	dBm
	HT20, MCS4	-	TBD	-	dBm
	HT20, MCS5	-	TBD	-	dBm
	HT20, MCS6	-	TBD	-	dBm
	HT20, MCS7	-	TBD	-	dBm
Sensitivity - IEEE 802.11n, 40 MHz (10% PER)	HT40, MCS0	-	TBD	-	dBm
	HT40, MCS1	-	TBD	-	dBm
	HT40, MCS2	-	TBD	-	dBm
	HT40, MCS3	-	TBD	-	dBm
	HT40, MCS4	-	TBD	-	dBm
	HT40, MCS5	-	TBD	-	dBm
	HT40, MCS6	-	TBD	-	dBm
	HT40, MCS7	-	TBD	-	dBm
	HE20, MCS0	-	TBD	-	dBm

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Parameter	Conditions	Min.	Typ.	Max.	Unit
Sensitivity - IEEE 802.11ax, 20 MHz (10% PER)	HE20, MCS1	-	TBD	-	dBm
	HE20, MCS2	-	TBD	-	dBm
	HE20, MCS3	-	TBD	-	dBm
	HE20, MCS4	-	TBD	-	dBm
	HE20, MCS5	-	TBD	-	dBm
	HE20, MCS6	-	TBD	-	dBm
	HE20, MCS7	-	TBD	-	dBm
	HE20, MCS8	-	TBD	-	dBm
	HE20, MCS9	-	TBD	-	dBm
Sensitivity - IEEE 802.11ax, 40 MHz (10% PER)	HE40, MCS0	-	TBD	-	dBm
	HE40, MCS1	-	TBD	-	dBm
	HE20, MCS2	-	TBD	-	dBm
	HE40, MCS3	-	TBD	-	dBm
	HE40, MCS4	-	TBD	-	dBm
	HE40, MCS5	-	TBD	-	dBm
	HE40, MCS6	-	TBD	-	dBm
	HE40, MCS7	-	TBD	-	dBm
	HE40, MCS8	-	TBD	-	dBm
	HE40, MCS9	-	TBD	-	dBm
Maximum Receive Level					
Maximum receive level @ 2.4 GHz	11b: 1 Mbps (8% PER, 1024 octets)	-	TBD	-	dBm
	11b: 11 Mbps (8% PER, 1024 octets)	-	TBD	-	dBm
	11g: 6 Mbps (10% PER, 1000 octets)	-	TBD	-	dBm
	11g: 54 Mbps (10% PER, 1000 octets)	-	TBD	-	dBm
	11n: HT20 MCS0 (10% PER, 4096 octets)	-	TBD	-	dBm

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Parameter	Conditions		Min.	Typ.	Max.	Unit
	11n: HT20 MCS7 (10% PER, 4096 octets)		-	TBD	-	dBm
	11n: HT40 MCS0 (10% PER, 4096 octets)		-	TBD	-	dBm
	11n: HT40 MCS7 (10% PER, 4096 octets)		-	TBD	-	dBm
	11ax: HE20 MCS0 (10% PER, 4096 octets)		-	TBD	-	dBm
	11ax: HE20 MCS9 (10% PER, 4096 octets)		-	TBD	-	dBm
	11ax: HE40 MCS0 (10% PER, 4096 octets)		-	TBD	-	dBm
	11ax: HE40 MCS9 (10% PER, 4096 octets)		-	TBD	-	dBm
Adjacent Channel Rejection						
Adjacent channel (±30 MHz) rejection - IEEE 802.11b (8% PER with desired signal level as specified in Conditions)	1 Mbps DSSS	-74 dBm	-	TBD	-	dB
	2 Mbps DSSS	-74 dBm	-	TBD	-	dB
Adjacent channel (±25 MHz) rejection - IEEE 802.11b (8% PER with desired signal level as specified in Conditions)	5.5 Mbps DSSS	-70 dBm	-	TBD	-	dB
	11 Mbps DSSS	-70 dBm	-	TBD	-	dB
Adjacent channel (±25 MHz) rejection - IEEE 802.11g (10% PER with desired signal level as specified in Conditions)	6 Mbps OFDM	-79 dBm	-	TBD	-	dB
	54 Mbps OFDM	-62 dBm	-	TBD	-	dB
Adjacent channel (±25 MHz) rejection - IEEE 802.11n (10% PER with desired signal level as specified in Conditions)	HT20, MCS0	-79 dBm	-	TBD	-	dB
	HT20, MCS7	-61 dBm	-	TBD	-	dB
Adjacent channel (±40 MHz) rejection - IEEE 802.11n	HT40, MCS0	-76 dBm	-	TBD	-	dB

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Parameter	Conditions		Min.	Typ.	Max.	Unit
(10% PER with desired signal level as specified in Conditions)	HT40, MCS7	-58 dBm	-	TBD	-	dB
Adjacent channel (±20 MHz) rejection - IEEE 802.11ax (10% PER with desired signal level as specified in Conditions)	HE20, MCS0	-79 dBm	-	TBD	-	dB
	HE20, MCS9	-54 dBm	-	TBD	-	dB
Adjacent channel (±40 MHz) rejection - IEEE 802.11ax (10% PER with desired signal level as specified in Conditions)	HE40, MCS0	-76 dBm	-	TBD	-	dB
	HE40, MCS9	-51 dBm	-	TBD	-	dB
Spurious Emissions						
Spurious emissions	< 1 GHz		-	TBD	-	dBm
	> 1 GHz		-	TBD	-	dBm

5.21.1.2 WLAN Transmitter Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Conditions	Min.	Typ.	Max.	Unit
General					
Frequency range	-	2412	-	2484	MHz
TX Power					
TX power - IEEE 802.11b (SEM compliant)	1 Mbps DSSS	-	TBD	-	dBm
	11 Mbps DSSS	-	TBD	-	dBm
TX power - IEEE 802.11g (EVM compliant)	6 Mbps OFDM	-	TBD	-	dBm
	54 Mbps OFDM	-	TBD	-	dBm
TX power - IEEE 802.11n (EVM compliant)	HT20, MCS0	-	TBD	-	dBm
	HT20, MCS7	-	TBD	-	dBm
	HT40, MCS0	-	TBD	-	dBm
	HT40, MCS7	-	TBD	-	dBm
TX power - IEEE 802.11ax (EVM compliant)	HE20, MCS0	-	TBD	-	dBm
	HE20, MCS9	-	TBD	-	dBm

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Parameter	Conditions	Min.	Typ.	Max.	Unit
	HE40, MCS0	-	TBD	-	dBm
	HE40, MCS9	-	TBD	-	dBm
Spurious Emissions					
Spurious emissions (at maximum output power)	< 1 GHz	-	TBD	-	dBm
	> 1 GHz	-	TBD	-	dBm

5.21.2 Bluetooth RF Characteristics

5.21.2.1 Bluetooth BR/EDR Receiver Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Conditions	Min.	Typ.	Max.	Unit
General					
Frequency range	-	2402	-	2480	MHz
Bluetooth Basic Rate					
Sensitivity	GFSK, 0.1% BER	-	TBD	-	dBm
Maximum input level	GFSK, 0.1% BER	-	TBD	-	dBm
C/I co-channel	GFSK, 0.1% BER	-	TBD	-	dB
C/I 1 MHz adjacent channel	GFSK, 0.1% BER	-	TBD	-	dB
C/I 2 MHz adjacent channel	GFSK, 0.1% BER	-	TBD	-	dB
C/I ≥ 3 MHz adjacent channel	GFSK, 0.1% BER	-	TBD	-	dB
C/I image channel	GFSK, 0.1% BER	-	TBD	-	dB
C/I 1 MHz adjacent to image channel	GFSK, 0.1% BER	-	TBD	-	dB
Out-of-band blocking	30–2000 MHz	-	TBD	-	dBm
	2000–2400 MHz	-	TBD	-	dBm
	2500–3000 MHz	-	TBD	-	dBm
	3000 MHz–12.75 GHz	-	TBD	-	dBm
Intermodulation	-	-	TBD	-	dBm

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Parameter	Conditions	Min.	Typ.	Max.	Unit
Bluetooth Enhanced Data Rate					
Sensitivity	π/4-DQPSK, 0.01% BER, 2 Mbps	-	TBD	-	dBm
	8-DPSK, 0.01% BER, 3 Mbps	-	TBD	-	dBm
EDR BER floor performance	π/4-DQPSK	-	-	TBD	-
	8-DPSK	-	-	TBD	-
Maximum input level	π/4-DQPSK, 0.1% BER	-	TBD	-	dBm
	8-DPSK, 0.1% BER	-	TBD	-	dBm
C/I co-channel	π/4-DQPSK, 0.1% BER	-	TBD	-	dB
	8-DPSK, 0.1% BER	-	TBD	-	dB
C/I 1 MHz adjacent channel	π/4-DQPSK, 0.1% BER	-	TBD	-	dB
	8-DPSK, 0.1% BER	-	TBD	-	dB
C/I 2 MHz adjacent channel	π/4-DQPSK, 0.1% BER	-	TBD	-	dB
	8-DPSK, 0.1% BER	-	TBD	-	dB
C/I ≥ 3 MHz adjacent channel	π/4-DQPSK, 0.1% BER	-	TBD	-	dB
	8-DPSK, 0.1% BER	-	TBD	-	dB
C/I image channel	π/4-DQPSK, 0.1% BER	-	TBD	-	dB
	8-DPSK, 0.1% BER	-	TBD	-	dB
C/I 1 MHz adjacent to image channel	π/4-DQPSK, 0.1% BER	-	TBD	-	dB
	8-DPSK, 0.1% BER	-	TBD	-	dB

5.21.2.2 Bluetooth BR/EDR Transmitter Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Min.	Typ.	Max.	Unit
General				
Frequency range	2402	-	2480	MHz
Bluetooth Basic Rate				
TX power	-	TBD	-	dBm

Parameter		Min.	Typ.	Max.	Unit
TX power control step		-	TBD	-	dB
20 dB bandwidth		-	TBD	-	MHz
Adjacent channel power	|M-N| = 2	-	TBD	-	dBm
	|M-N| >= 3	-	TBD	-	dBm
Modulation characteristics	Δf1avg	-	TBD	-	kHz
	Δf2max	-	TBD	-	kHz
	Δf2avg/Δf1avg	-	TBD	-	-
Initial carrier frequency tolerance		-	TBD	-	kHz
Carrier frequency drift	DH1 packet	-	TBD	-	kHz
	DH3 packet	-	TBD	-	kHz
	DH5 packet	-	TBD	-	kHz
	Drift rate	-	TBD	-	kHz/50 μs
Bluetooth Enhanced Data Rate					
TX power	π/4-DQPSK	-	TBD	-	dBm
	8-DPSK	-	TBD	-	dBm
Relative TX power	π/4-DQPSK	-	TBD	-	dB
	8-DPSK	-	TBD	-	dB
Carrier frequency stability	ω0, π/4-DQPSK	-	TBD	-	kHz
	ω0, 8-DPSK	-	TBD	-	kHz
	ωi, π/4-DQPSK	-	TBD	-	kHz
	ωi, 8-DPSK	-	TBD	-	kHz
	ωi + ω0, π/4-DQPSK	-	TBD	-	kHz
	ωi + ω0, 8-DPSK	-	TBD	-	kHz
Modulation accuracy	RMS DEVM, π/4-DQPSK	-	TBD	-	-
	RMS DEVM, 8-DPSK	-	TBD	-	-
	Peak DEVM, π/4-DQPSK	-	TBD	-	-
	Peak DEVM, 8-DPSK	-	TBD	-	-
	99% DEVM, π/4-DQPSK	-	TBD	-	-

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Parameter		Min.	Typ.	Max.	Unit
	99% DEVM, 8-DPSK	-	TBD	-	-
EDR differential phase encoding	-	-	TBD	-	%

5.21.2.3 Bluetooth LE Receiver Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Conditions	Min.	Typ.	Max.	Unit
General					
Frequency range	-	2402	-	2480	MHz
Bluetooth LE 1 Mbps					
Sensitivity	30.8% PER	-	TBD	-	dBm
Maximum input level	30.8% PER	-	TBD	-	dBm
C/I co-channel	-	-	TBD	-	dB
C/I 1 MHz adjacent channel	-	-	TBD	-	dB
C/I -1 MHz adjacent channel	-	-	TBD	-	dB
C/I 2 MHz adjacent channel	-	-	TBD	-	dB
C/I -2 MHz adjacent channel	-	-	TBD	-	dB
C/I 3 MHz adjacent channel	-	-	TBD	-	dB
C/I -3 MHz adjacent channel	-	-	TBD	-	dB
C/I > 3 MHz adjacent channel	-	-	TBD	-	dB
C/I < -3 MHz adjacent channel	-	-	TBD	-	dB
C/I image channel	-	-	TBD	-	dB
C/I 1 MHz adjacent to image channel	-	-	TBD	-	dB
C/I -1 MHz adjacent to image channel	-	-	TBD	-	dB
Out-of-band blocking	30–2000 MHz	-	TBD	-	dBm
	2003–2399 MHz	-	TBD	-	dBm
	2484–2997 MHz	-	TBD	-	dBm

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Parameter	Conditions	Min.	Typ.	Max.	Unit
	3000 MHz–12.75 GHz	-	TBD	-	dBm
Intermodulation	-	-	TBD	-	dBm
Bluetooth LE 2 Mbps					
Sensitivity	30.8% PER	-	TBD	-	dBm
Maximum input level	30.8% PER	-	TBD	-	dBm
C/I co-channel	-	-	TBD	-	dB
C/I 2 MHz adjacent channel	-	-	TBD	-	dB
C/I -2 MHz adjacent channel	-	-	TBD	-	dB
C/I 4 MHz adjacent channel	-	-	TBD	-	dB
C/I -4 MHz adjacent channel	-	-	TBD	-	dB
C/I 6 MHz adjacent channel	-	-	TBD	-	dB
C/I -6 MHz adjacent channel	-	-	TBD	-	dB
C/I > 6 MHz adjacent channel	-	-	TBD	-	dB
C/I < -6 MHz adjacent channel	-	-	TBD	-	dB
C/I image channel	-	-	TBD	-	dB
C/I 2 MHz adjacent to image channel	-	-	TBD	-	dB
C/I -2 MHz adjacent to image channel	-	-	TBD	-	dB
Out-of-band blocking	30–2000 MHz	-	TBD	-	dBm
	2003–2399 MHz	-	TBD	-	dBm
	2484–2997 MHz	-	TBD	-	dBm
	3000 MHz–12.75 GHz	-	TBD	-	dBm
Intermodulation	-	-	TBD	-	dBm
Bluetooth LE 125 kbps					
Sensitivity	30.8% PER	-	TBD	-	dBm
Maximum input level	30.8% PER	-	TBD	-	dBm
C/I co-channel	-	-	TBD	-	dB
C/I 1 MHz adjacent channel	-	-	TBD	-	dB

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Parameter	Conditions	Min.	Typ.	Max.	Unit
C/I -1 MHz adjacent channel	-	-	TBD	-	dB
C/I 2 MHz adjacent channel	-	-	TBD	-	dB
C/I -2 MHz adjacent channel	-	-	TBD	-	dB
C/I 3 MHz adjacent channel	-	-	TBD	-	dB
C/I -3 MHz adjacent channel	-	-	TBD	-	dB
C/I > 3 MHz adjacent channel	-	-	TBD	-	dB
C/I < -3 MHz adjacent channel	-	-	TBD	-	dB
C/I image channel	-	-	TBD	-	dB
C/I 1 MHz adjacent to image channel	-	-	TBD	-	dB
C/I -1 MHz adjacent to image channel	-	-	TBD	-	dB
Bluetooth LE 500 kbps					
Sensitivity	30.8% PER	-	TBD	-	dBm
Maximum input level	30.8% PER	-	TBD	-	dBm
C/I co-channel	-	-	TBD	-	dB
C/I 1 MHz adjacent channel	-	-	TBD	-	dB
C/I -1 MHz adjacent channel	-	-	TBD	-	dB
C/I 2 MHz adjacent channel	-	-	TBD	-	dB
C/I -2 MHz adjacent channel	-	-	TBD	-	dB
C/I 3 MHz adjacent channel	-	-	TBD	-	dB
C/I -3 MHz adjacent channel	-	-	TBD	-	dB
C/I > 3 MHz adjacent channel	-	-	TBD	-	dB
C/I < -3 MHz adjacent channel	-	-	TBD	-	dB
C/I image channel	-	-	TBD	-	dB
C/I 1 MHz adjacent to image channel	-	-	TBD	-	dB
C/I -1 MHz adjacent to image channel	-	-	TBD	-	dB

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5.21.2.4 Bluetooth LE Transmitter Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter		Conditions	Min.	Typ.	Max.	Unit
General						
Frequency range		-	2402	-	2480	MHz
TX power		-	TBD	TBD	TBD	dBm
Bluetooth LE 1 Mbps						
In-band emissions	±2 MHz offset	-	-	TBD	-	dBm
	±3 MHz offset	-	-	TBD	-	dBm
	>±3 MHz offset	-	-	TBD	-	dBm
Modulation characteristics	Δf1avg	-	-	TBD	-	kHz
	Δf2max	-	-	TBD	-	kHz
	Δf2avg/Δf1avg	-	-	TBD	-	-
Carrier frequency offset and drift	Max $\|f_n\|$ n = 0, 1, 2, 3…k	-	-	TBD	-	kHz
	Max $\|f_0 - f_n\|$ n = 2, 3, 4…k	-	-	TBD	-	kHz
	$\|f_1 - f_0\|$	-	-	TBD	-	kHz
	Max $\|f_n - f_{n-5}\|$ n = 6, 7, 8…k	-	-	TBD	-	kHz/50 μs
Bluetooth LE 2 Mbps						
In-band emissions	±4 MHz offset	-	-	TBD	-	dBm
	±5 MHz offset	-	-	TBD	-	dBm
	>±5 MHz offset	-	-	TBD	-	dBm
Modulation characteristics	Δf1avg	-	-	TBD	-	kHz
	Δf2max	-	-	TBD	-	kHz
	Δf2avg/Δf1avg	-	-	TBD	-	-
Carrier frequency offset and drift	Max $\|f_n\|$ n = 0, 1, 2, 3…k	-	-	TBD	-	kHz
	Max $\|f_0 - f_n\|$ n = 2, 3, 4…k	-	-	TBD	-	kHz
	$\|f_1 - f_0\|$	-	-	TBD	-	kHz
	Max $\|f_n - f_{n-5}\|$ n = 6, 7, 8…k	-	-	TBD	-	kHz/50 μs

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Parameter		Conditions	Min.	Typ.	Max.	Unit
Bluetooth LE 125 kbps						
In-band emissions	±2 MHz offset	-	-	TBD	-	dBm
	±3 MHz offset	-	-	TBD	-	dBm
	>±3 MHz offset	-	-	TBD	-	dBm
Modulation characteristics	Δf1avg	-	-	TBD	-	kHz
	Δf1max	-	-	TBD	-	kHz
Carrier frequency offset and drift	Max $\|f_n\|$ n = 0, 1, 2, 3…k	-	-	TBD	-	kHz
	Max $\|f_0 - f_n\|$ n = 1, 2, 3…k	-	-	TBD	-	kHz
	$\|f_0 - f_3\|$	-	-	TBD	-	kHz
	$\|f_n - f_{n-3}\|$ n = 7, 8, 9…k	-	-	TBD	-	kHz/48 μs
Bluetooth LE 500 kbps						
In-band emissions	±2 MHz offset	-	-	TBD	-	dBm
	±3 MHz offset	-	-	TBD	-	dBm
	>±3 MHz offset	-	-	TBD	-	dBm

5.21.3 IEEE 802.15.4 RF Characteristics

5.21.3.1 IEEE 802.15.4 Receiver Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Conditions	Min.	Typ.	Max.	Unit
General					
Frequency range	-	2400	-	2483.5	MHz
O-QPSK 250 kbps					
Sensitivity	1 % PER	-	TBD	-	dBm
Maximum input level	1 % PER	-	TBD	-	dBm
Co-channel rejection	1 % PER, desired signal @ -82 dBm	-	TBD	-	dB
Adjacent channel (+5 MHz) rejection	1 % PER, desired signal @ -82 dBm	-	TBD	-	dB

Parameter	Conditions	Min.	Typ.	Max.	Unit
Adjacent channel (-5 MHz) rejection	1 % PER, desired signal @ -82 dBm	-	TBD	-	dB
Alternate channel (+10 MHz) rejection	1 % PER, desired signal @ -82 dBm	-	TBD	-	dB
Alternate channel (-10 MHz) rejection	1 % PER, desired signal @ -82 dBm	-	TBD	-	dB
Channel (+15 MHz) rejection	1 % PER, desired signal @ -82 dBm	-	TBD	-	dB
Channel (-15 MHz) rejection	1 % PER, desired signal @ -82 dBm	-	TBD	-	dB
Spurious emissions	< 1 GHz	-	TBD	-	dBm
	> 1 GHz	-	TBD	-	dBm

5.21.3.2 IEEE 802.15.4 Transmitter Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Conditions	Min.	Typ.	Max.	Unit
General					
Frequency range	-	2400	-	2483.5	MHz
O-QPSK 250 kbps					
Maximum output power	-	-	TBD	-	dBm
Error-vector magnitude (EVM)	-	-	TBD	-	%
Center frequency tolerance	-	-	TBD	-	ppm
Power spectral density (PSD)	Relative, at greater than 3.5 MHz offset	-	TBD	-	dB
	Absolute, at greater than 3.5 MHz offset	-	TBD	-	dBm
Spurious emissions	30–1000 MHz	-	TBD	-	dBm
	1–12.75 GHz	-	TBD	-	dBm

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5.21.4 2.4 GHz Proprietary RF Characteristics

5.21.4.1 2.4 GHz Proprietary Receiver Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Conditions	Min.	Typ.	Max.	Unit
General					
Frequency range	-	2400	-	2483.5	MHz
2.4 GHz Proprietary 1 Mbps					
Sensitivity	0.1% BER	-	TBD	-	dBm
Maximum input level	0.1% BER	-	TBD	-	dBm
C/I co-channel	-	-	TBD	-	dB
C/I 1 MHz adjacent channel	-	-	TBD	-	dB
C/I -1 MHz adjacent channel	-	-	TBD	-	dB
C/I 2 MHz adjacent channel	-	-	TBD	-	dB
C/I -2 MHz adjacent channel	-	-	TBD	-	dB
C/I 3 MHz adjacent channel	-	-	TBD	-	dB
C/I -3 MHz adjacent channel	-	-	TBD	-	dB
C/I > 3 MHz adjacent channel	-	-	TBD	-	dB
C/I < -3 MHz adjacent channel	-	-	TBD	-	dB
C/I image channel	-	-	TBD	-	dB
C/I 1 MHz adjacent to image channel	-	-	TBD	-	dB
C/I -1 MHz adjacent to image channel	-	-	TBD	-	dB
Out-of-band blocking	30–2000 MHz	-	TBD	-	dBm
	2003–2399 MHz	-	TBD	-	dBm
	2484–2997 MHz	-	TBD	-	dBm
	3000 MHz–12.75 GHz	-	TBD	-	dBm

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5.21.4.2 2.4 GHz Proprietary Transmitter Characteristics

Measured with T = 25 °C, VBAT = 3.3 V unless otherwise stated.

Parameter	Conditions	Min.	Typ.	Max.	Unit
General					
Frequency range	-	2400	-	2483.5	MHz
TX power	-	TBD	TBD	TBD	dBm
Modulation 20 dB Bandwidth					
Modulation 20 dB bandwidth – 1 Mbps	-	-	TBD	-	MHz

5.22 Audio Characteristics

5.22.1 Audio ADC and DAC characteristics

Parameter	Conditions	Min.	Typ.	Max.	Unit
Audio ADC					
ADC SNR	1 kHz sine wave	-	106	-	dB
ADC THD+N	-	-	TBD	-	dB
ADC sampling rate	-	32	-	384	kHz
Audio DAC					
DAC single-end output	Load = 600 Ω	-	TBD	-	Vrms
DAC single-end output THD+N	1.2 Vrms @ 600 Ω load	-	TBD	-	dB
	1.5 Vrms @ 600 Ω load	-	TBD	-	dB
DAC differential output	Load = 600 Ω	-	TBD	-	Vrms
DAC differential output THD+N	1.2 Vrms @ 600 Ω load	-	TBD	-	dB
	1.5 Vrms @ 600 Ω load	-	TBD	-	dB
DAC output SNR	1 kHz sine wave	-	104	-	dB
DAC load capacitance	-	600	-	-	Ω
DAC sampling rate	-	8	-	96	kHz

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5.22.2 Voice Activity Detector Characteristics

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
VCCCB	Supply voltage	-	2.3	-	3.6	V
V_{IN_PP}	Input signal amplitude	-	-	TBD	-	V_{pp}
F_S	Sampling rate	-	0.1	-	4	kHz
SNR_{TH}	Wake-up SNR threshold	-	3	-	-	dB
t_{LAT}	Wake-up latency	-	-	-	10	ms
I_{VAD}	Supply current	-	-	TBD	-	μA

5.23 AUX ADC Characteristics

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
VCCCB	Supply voltage	-	2.3	-	3.6	V
F_{CLK}	Conversion clock	-	0.86	-	TBD	MHz
C_{CONV}	Conversion time	-	-	26	-	Cycle
F_S	Sampling rate	-	0.03	-	TBD	MHz
V_{REF}	Reference voltage	Internal	-	1.5 or 1.8	-	V
		External	-	VBAT/2	-	V
V_{IN}	Input voltage range	ADC0	-	VBAT/4	-	V
		Other channels	0	-	V_{REF} * 2	V
R_{IN}	Input impedance	-	-	TBD	-	MΩ
C_{IN}	Input capacitance	-	-	4	-	pF
DNL	Differential nonlinearity	-	-	±0.5	-	LSB
INL	Integral nonlinearity	-	-	±1	-	LSB
ENOB	Effective number of bits	-	-	12	-	Bit
SNDR	Signal-to-noise-and-distortion ratio	-	-	74	-	dB
SFDR	Spurious-free dynamic range	-	-	84	-	dB

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Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
$T_{STARTUP}$	Startup time	-	-	5	-	μs
I_{ADC}	Supply current	-	-	1	-	mA

5.24 AUX DAC Characteristics

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
VCCCB	Supply voltage	-	2.3	-	3.6	V
V_{REF}	Reference voltage	-	0.9	-	VBAT	V
C_L	Output load capacitance	-	-	50	100	pF
I_L	Output load current	-	-1	-	20	mA
f_{UPDATE}	DAC conversion rate	-	-	-	500	kSPS
I_{DAC}	Supply current	Operating mode	-	TBD	-	μA
		Disabled	-	TBD	-	μA
t_{DAC}	Full-scale settling time	-	-	TBD	-	μs
$V_{DACOUTL}$	DAC output voltage range low	High-speed mode, no load, DAC set to 0x000	-	TBD	-	mV
$V_{DACOUTH}$	DAC output voltage range high	High-speed mode, no load, DAC set to 0xFFF	-	TBD	-	mV
INL	Integral nonlinearity	-	-	TBD	-	LSB
DNL	Differential nonlinearity	-	-	TBD	-	LSB
E_{OFFSET}	Offset error	-	-	TBD	-	%FSR
E_G	Gain error	-	-	TBD	-	%FSR
PSRR	Power supply rejection ratio	-	-	TBD	-	dB
T_{CO}	Temperature coefficient offset voltage at middle scale	-	-	TBD	-	μV/°C
T_{EO}	Temperature coefficient offset error	-	-	TBD	-	μV/°C
T_{GE}	Temperature coefficient gain error	-	-	TBD	-	ppm/°C
R_{OUT}	Output resistance	Load = 10 kΩ	-	TBD	-	Ω

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Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
SR	Slew rate	-	-	TBD	-	V/μs
CT	DAC to DAC crosstalk	-	-	TBD	-	dB
T_{PU}	Power-up time	-	-	TBD	-	μs

5.25 Analog Comparator Characteristics

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
VBAT	Supply voltage	-	2.3	-	3.6	V
V_{REFH}	8-bit DAC reference voltage high	-	TBD	-	TBD	V
I_{ACMP}	Supply current	High-speed mode	-	TBD	-	μA
		Normal mode	-	TBD	-	μA
		Low-power mode	-	TBD	-	μA
V_{AIN}	Analog input voltage	-	TBD	-	TBD	V
V_{AIO}	Analog input offset voltage	High-speed mode	-	TBD	-	mV
		Normal mode	-	TBD	-	mV
		Low-power mode	-	TBD	-	mV
V_{HYS}	Analog comparator hysteresis	Hysteresis select bits = 00	-	TBD	-	mV
		Hysteresis select bits = 01	-	TBD	-	mV
		Hysteresis select bits = 10	-	TBD	-	mV
		Hysteresis select bits = 11	-	TBD	-	mV
V_{ACMPOH}	Output high	-	-	TBD	-	V
V_{ACMPOL}	Output low	-	-	TBD	-	V
t_D	Propagation delay	High-speed mode, 100 mV overdrive	-	TBD	-	ns
		High-speed mode, 30 mV overdrive	-	TBD	-	ns
		Normal mode, 30 mV overdrive	-	TBD	-	ns
		Low-power mode, 30 mV overdrive	-	TBD	-	ns

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Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
t_{init}	Analog comparator initialization delay	-	-	TBD	-	μs
I_{8BDAC}	8-bit DAC current adder (enabled)	High-power mode	-	TBD	-	μA
		Low-power mode	-	TBD	-	μA
INL	8-bit DAC integral nonlinearity	Low/high-power mode	TBD	-	TBD	LSB
		Low-power mode	TBD	-	TBD	LSB
DNL	8-bit DAC differential nonlinearity	Low/high-power mode	TBD	-	TBD	LSB
		Low-power mode	TBD	-	TBD	LSB

5.26 Touch Sensor Characteristics

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
VCCCB	Supply voltage	-	2.3	-	3.6	V
f_{CLK}	Clock frequency	-	-	32	-	kHz
I_{TH}	Supply current	High-power mode	-	TBD	-	μA
		Low-power mode	-	TBD	-	μA
ΔC_{MIN}	Minimum detectable capacitance (sensitivity)	-	-	50	-	fF
C_{RANGE}	Maximum input capacitance range	-	-	-	23	pF

5.27 BK7259 Inference Performance

Model	Conditions	Performance (FPS)
MobileNet V1 (224x224x3)	Dedicated 448 KB SRAM	79
MobileNet V2 (224x224x3)	Dedicated 448 KB SRAM	84
Tiny Yolo V2 (224x224x3)	Dedicated 448 KB SRAM	57

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6 Package Information

Figure 6-1 QFN128 12.3 x 12.3 x 0.85 mm Package Outline

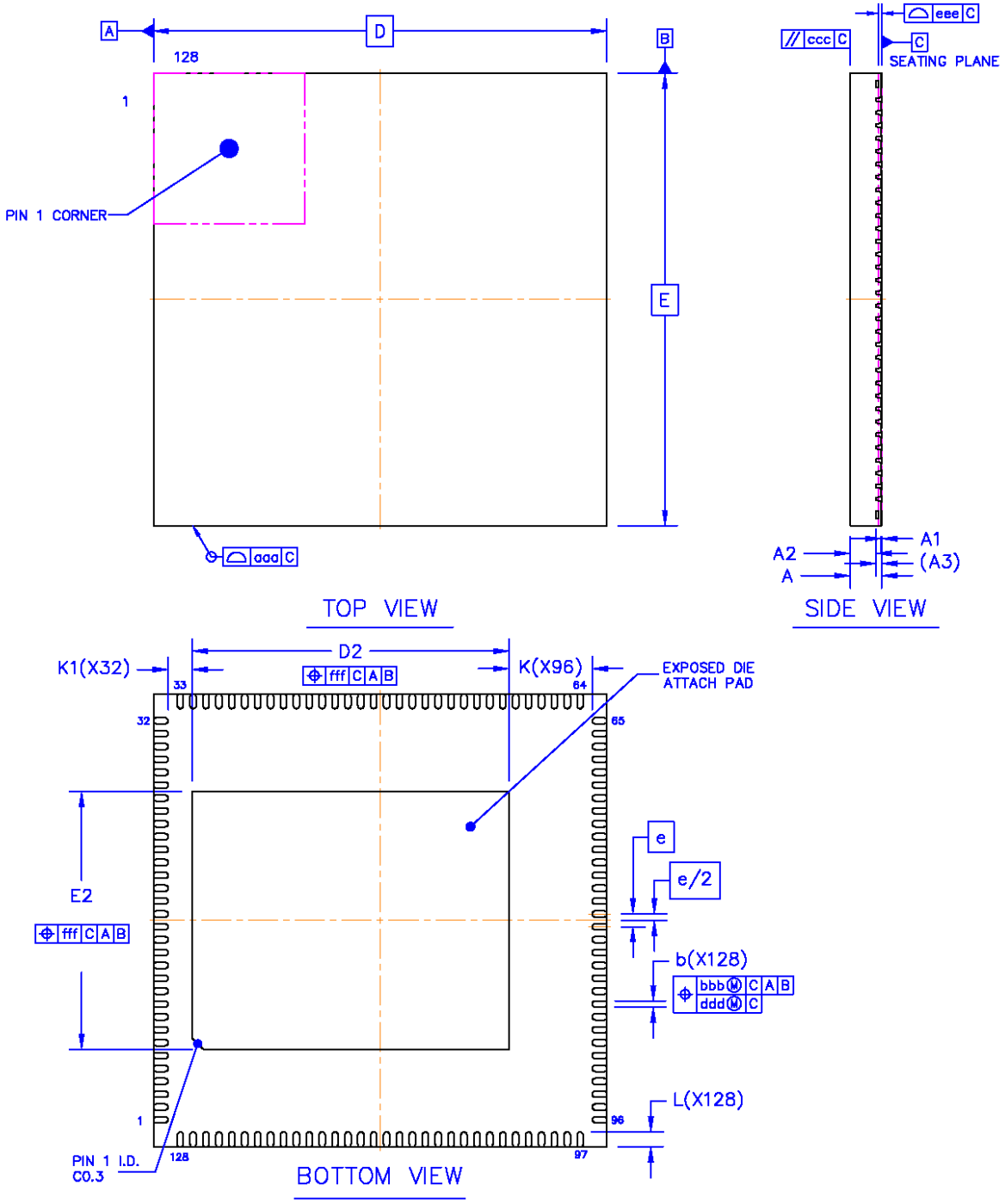


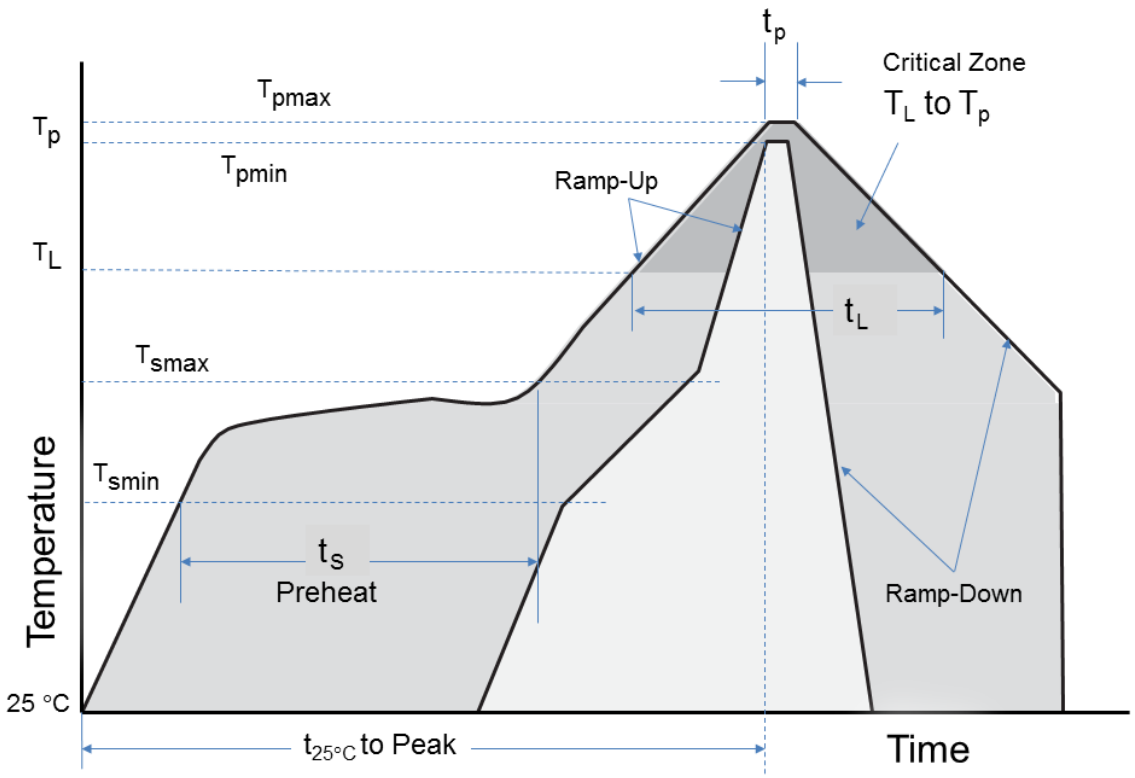
Table 6-1 QFN128 Package Dimensions

Symbol	Dimensions in Millimeters		
	Min.	Nom.	Max.
A	0.80	0.85	0.90
A1	0.00	0.02	0.05
A2	-	0.70	-
A3	0.152 REF		
b	0.12	0.17	0.22
D	12.30 BSC		
E	12.30 BSC		
e	0.35 BSC		
D2	8.50	8.60	8.70
E2	6.90	7.00	7.10
L	0.30	0.40	0.50
K	2.25 REF		
K1	0.65 REF		
aaa	0.10		
ccc	0.10		
eee	0.08		
bbb	0.07		
ddd	0.05		
fff	0.10		

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7 Reflow Soldering Profile

Figure 7-1 Reflow Soldering Profile



Profile Feature		Specification
Average ramp-up rate (T_{smax} to T_p)		3 °C/s max.
Preheat	Temperature min. (T_{smin})	150 °C
	Temperature max. (T_{smax})	200 °C
	Time (t_s)	60 s to 180 s
Time maintained above	Temperature (T_L)	217 °C
	Time (t_L)	60 s to 150 s
Peak/classification temperature (T_p)		260 °C
Time within 5 °C of actual peak temperature (t_p)		20 s to 40 s
Ramp-down rate		6 °C/s max.

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Profile Feature	Specification
Time 25 °C to peak temperature	8 minutes max.

RoHS Compliant

The product does not contain lead, mercury, cadmium, hexavalent chromium, PBB, PBDE, DEHP, BBP, DBP, or DIBP content in accordance with EU RoHS Directive (EU) 2015/863 amending Annex II to Directive 2011/65/EU.

ESD Sensitivity

Integrated circuits are ESD sensitive and can be damaged by static electricity. Proper ESD techniques should be used when handling these devices.



Moisture Sensitivity Level

The product is qualified to moisture sensitivity level MSL3 in accordance with IPC/JEDEC J-STD-020.

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8 Ordering Information

Figure 8-1 Ordering Code Scheme

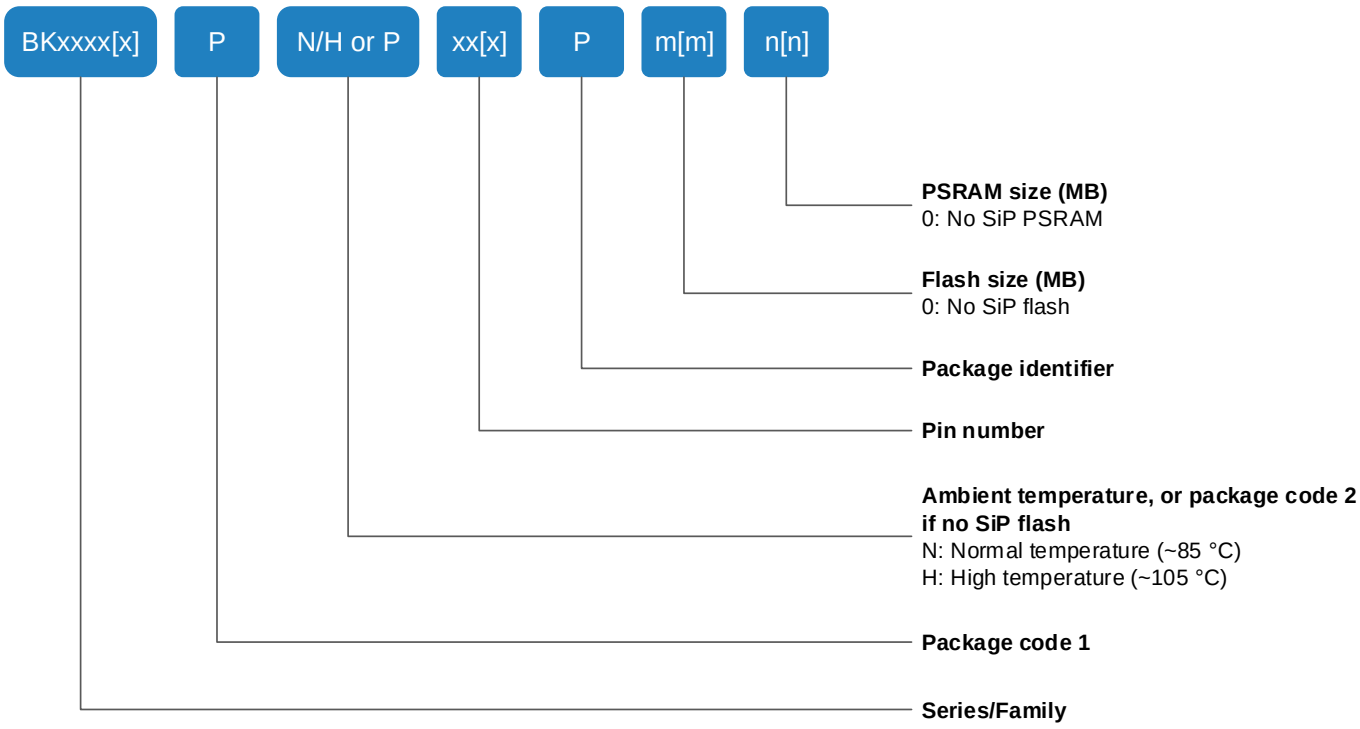


Table 8-1 Ordering Information

Ordering Code	Package	SiP(1) Flash	SiP PSRAM	Packing	Minimum Ordering Qty (MOQ)(2)
BK7259QN128A832	12.3 mm x 12.3 mm x 0.85 mm QFN128	8 MB	32 MB (2 x 16 MB)	Tape and Reel	1500
BK7259QN128B832	12.3 mm x 12.3 mm x 0.85 mm QFN128	8 MB	32 MB (2 x 16 MB)	Tape and Reel	1500

(1) A system in a package (SiP) refers to flash/PSRAM enclosed in the package.

(2) The MOQ for customized or exclusive ordering codes is one wafer lot. Please consult your sales representative for the exact quantity.

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Revision History

Version	Date	Description
1.0	2024/11/8	Initial release
1.1	2025/2/14	• Changed Bluetooth LE support to dual-mode Bluetooth support • Updated the maximum resolution of the ISP in Section 4.30.5 Image Signal Processor (ISP) • Updated the ordering code scheme in Section 8 Ordering Information: 0 means no SiP flash
1.2	2025/12/30	Overall update
1.3	2026/6/26	• Added information for BK7259QN128B832 QFN128 package • Updated the maximum operating temperature • Section 1 Features – Added UL MU-MIMO feature – Corrected the number of virtual Wi-Fi interfaces – Modified Wi-Fi operating modes – Modified flash/PSRAM capacity • Clarified that the actual flash and PSRAM sizes are design-dependent and added a note regarding AUDIO in Figure 4-1 in Section 4.3 Memory Organization • Added a note regarding AUDIO in Figure 4-3 in Section 4.10 Clock Management • Added parameters SR_{VBAT_RISE} to Table 4-2 and SR_{VBAT_FALL} to Table 4-3 in Section 4.12.1 Power Scheme • Updated the per-channel event generation list for GDMA in Section 4.26 GDMA Controller (GDMA) • Updated the description for the ISP noise reduction feature in Section 4.30.5 Image Signal Processor (ISP) • Section 5.4 DC Electrical Characteristics – Updated section heading – Added values for R_{PU} and R_{PD} – Added V_{IH_CEN} and V_{IL_CEN} parameters • Revised Figure 8-1 Ordering Code Scheme and updated Table 8-1 Ordering Information in Section 8 Ordering Information

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